

SOA Exam P Study Primer

Topic 1: General Probability (23–30% of the exam)

September 2026 syllabus

How to use this primer

Topic 1 has seven learning outcomes, (a) through (g). Each section below covers one outcome: what the outcome asks, the definitions and formulas needed, worked examples in the style of the exam’s five-choice multiple-choice questions, the traps that cost points, and practice problems with answers. The exam is 30 questions in three hours, so a typical question must be solved in about six minutes. Every section ends with a one-page summary of what to memorise.

Contents

1 Outcome (a): Sets, sample spaces, events, and the axioms of probability	4
1.1 What the outcome asks	4
1.2 Definitions	4
1.3 Venn diagram method	6
1.4 Countable additivity and geometric series	8
1.5 Testing whether two described events are equal	8
1.6 Worked examples	9
1.7 Traps	15
1.8 Practice problems	16
1.9 Answers to practice problems	18
1.10 Sample exam cross-reference	19
1.11 One-page summary	19
2 Outcome (b): Combinatorics	21
2.1 What the outcome asks	21
2.2 Counting tools	21
2.3 Classical probability with counting	23
2.4 Lattice counting of ordered pairs	24
2.5 Selections restricted by rows and columns	24
2.6 Pooled populations and unknown counts	25
2.7 Worked examples	26
2.8 Traps	31
2.9 Practice problems	32
2.10 Answers to practice problems	34
2.11 Sample exam cross-reference	34
2.12 One-page summary	35

3	Outcome (c): Independence	36
3.1	What the outcome asks	36
3.2	Definitions	36
3.3	How to recognise independence in a word problem	39
3.4	Worked examples	39
3.5	Traps	46
3.6	Practice problems	47
3.7	Answers to practice problems	49
3.8	Sample exam cross-reference	50
3.9	One-page summary	50
4	Outcome (d): Mutually exclusive events	52
4.1	What the outcome asks	52
4.2	Definitions and rules	52
4.3	Disjoint decomposition as a technique	53
4.4	Partitions pinned down by linear constraints	54
4.5	Worked examples	56
4.6	Traps	59
4.7	Practice problems	61
4.8	Answers to practice problems	62
4.9	Sample exam cross-reference	63
4.10	One-page summary	63
5	Outcome (e): Addition and multiplication rules	65
5.1	What the outcome asks	65
5.2	Addition rule	65
5.3	Multiplication rule	67
5.4	Combining the rules	68
5.5	Worked examples	69
5.6	Traps	74
5.7	Practice problems	75
5.8	Answers to practice problems	77
5.9	Sample exam cross-reference	78
5.10	One-page summary	78
6	Outcome (f): Conditional probability	80
6.1	What the outcome asks	80
6.2	Definition and properties	80
6.3	Three computational settings	81
6.4	Worked examples	83
6.5	Traps	89
6.6	Practice problems	90
6.7	Answers to practice problems	92
6.8	Sample exam cross-reference	93
6.9	One-page summary	94

7	Outcome (g): Law of total probability and Bayes' theorem	95
7.1	What the outcome asks	95
7.2	Law of total probability	95
7.3	Bayes' theorem	96
7.4	The table method	96
7.5	Worked examples	97
7.6	Variants of the likelihood and the denominator	102
7.7	Traps	107
7.8	Practice problems	108
7.9	Answers to practice problems	111
7.10	Sample exam cross-reference	112
7.11	One-page summary	112

1 Outcome (a): Sets, sample spaces, events, and the axioms of probability

1.1 What the outcome asks

The exam does not ask you to recite definitions. It tests whether you can convert a word problem into set notation, fill a two-event or three-event Venn diagram from partial information, and apply the axioms and their direct consequences (complement rule, inclusion-exclusion, monotonicity) to produce a number. The most common question gives four to seven proportions of policyholders holding various coverages and asks for the proportion in one region of the diagram or in a union of regions (“none”, “exactly one”, “at most one”, “auto but not home”). A second type gives two probabilities with no other information and asks for the largest or smallest value a third probability can take; that is solved with the axioms alone. The arithmetic is short; points are lost in the translation from English to sets.

1.2 Definitions

Sample space, outcomes, events. The *sample space* S of a random experiment (select a policyholder at random, observe next year’s claims) is the set of all possible results; each element of S is an *outcome*. An *event* is a subset of S . A *occurs* when the observed outcome lies in A . S is the certain event and $\emptyset$ the impossible event.

Set operations. For events $A, B \subseteq S$:

- Union $A \cup B$: outcomes in A or B or both. English: “ A or B ”, “at least one of A, B ”.
- Intersection $A \cap B$ (also written AB): outcomes in both. English: “ A and B ”, “both”.
- Complement $A^c = S \setminus A$: outcomes not in A . English: “not A ”.
- Difference $A \setminus B = A \cap B^c$: outcomes in A but not in B . English: “ A but not B ”, “ A only” (when only two events are in play).
- A and B are *disjoint* (mutually exclusive) when $A \cap B = \emptyset$; $A \subseteq B$ when every outcome of A is in B , so A occurring forces B .

Union and intersection distribute over each other: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. Any event A splits as the disjoint union $A = (A \cap B) \cup (A \cap B^c)$; this identity is used constantly.

Key fact: De Morgan’s laws

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

In words: “neither A nor B ” is the complement of “at least one”; “not both” is the complement of “both”. The same pattern holds for any number of events: the complement of a union is the intersection of the complements, and the complement of an intersection is the union of the complements.

Set functions and the collection of events. A *set function* is a function whose inputs are sets. The number of elements $|A|$, the length of an interval, and probability are all set functions. The domain of a probability is a collection $\mathcal{F}$ of subsets of S , the *events*. $\mathcal{F}$ must contain S , must contain A^c whenever it contains A , and must contain $A_1 \cup A_2 \cup \dots$ whenever it contains each A_i ; a collection with these closure properties is a *sigma-field* (or sigma-algebra). When S is finite, $\mathcal{F}$ is every subset of S ; on the exam, every set you can describe is an event.

Probability as a set function. A *probability* is a set function $\mathbb{P} : \mathcal{F} \rightarrow \mathbb{R}$ satisfying three axioms (Kolmogorov, 1933).

Key fact: The three axioms of probability

- A1.** Nonnegativity: $\mathbb{P}(A) \geq 0$ for every event A .
- A2.** Normalisation: $\mathbb{P}(S) = 1$.
- A3.** Countable additivity: if $A_1, A_2, \dots$ are pairwise disjoint ($A_i \cap A_j = \emptyset$ for $i \neq j$), then

$$\mathbb{P}(A_1 \cup A_2 \cup \dots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots$$

Everything else in this outcome is a consequence of the three axioms.

Key fact: Consequences of the axioms

- C1.** $\mathbb{P}(\emptyset) = 0$.
- C2.** Finite additivity: for pairwise disjoint $A_1, \dots, A_n$, $\mathbb{P}(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mathbb{P}(A_i)$.
- C3.** Complement rule: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- C4.** Monotonicity: if $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.
- C5.** $0 \leq \mathbb{P}(A) \leq 1$.
- C6.** Difference rule: $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$.
- C7.** Inclusion-exclusion, two events:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

- C8.** Inclusion-exclusion, three events:

$$\begin{aligned} \mathbb{P}(A \cup B \cup C) &= \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ &\quad - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) \\ &\quad + \mathbb{P}(A \cap B \cap C). \end{aligned}$$

- C9.** Boole's inequality: $\mathbb{P}(A_1 \cup \dots \cup A_n) \leq \sum_{i=1}^n \mathbb{P}(A_i)$.
- C10.** Bonferroni's inequality: $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$.

Each consequence comes from writing a set as a disjoint union and applying A3: $S = A \cup A^c$ gives C3; $B = A \cup (B \setminus A)$ gives C4 and C5; $A = (A \cap B) \cup (A \cap B^c)$ gives C6; $A \cup B = A \cup (B \cap A^c)$ gives C7, and C7 applied twice gives C8. When the exam asks which statements “must be true”, the answer is one of C1–C10 or an axiom.

Identities from adding two inclusion-exclusion equations. Some questions give probabilities of unions that involve complements, such as $\mathbb{P}(A \cup B)$ and $\mathbb{P}(A \cup B^c)$, and ask for $\mathbb{P}(A)$. Write inclusion-exclusion for each union and add. Since $\mathbb{P}(B) + \mathbb{P}(B^c) = 1$ and $\mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) = \mathbb{P}(A)$ (C6), the sum collapses.

Key fact: Union-with-complement identities

$$\begin{aligned}
\mathbb{P}(A \cup B) + \mathbb{P}(A \cup B^c) &= 1 + \mathbb{P}(A), \\
\mathbb{P}(A \cup B) + \mathbb{P}(A^c \cup B) &= 1 + \mathbb{P}(B), \\
\mathbb{P}(A \cup B^c) &= 1 - \mathbb{P}(A^c \cap B), \\
\mathbb{P}(A^c \cup B^c) &= 1 - \mathbb{P}(A \cap B), \\
\mathbb{P}(A \cup B) + \mathbb{P}(A^c \cup B^c) &= 1 + \mathbb{P}(\text{exactly one of } A, B), \\
\mathbb{P}(A \cup B^c) + \mathbb{P}(A^c \cup B) &= 2 - \mathbb{P}(\text{exactly one of } A, B).
\end{aligned}$$

The third and fourth lines are De Morgan plus the complement rule: $A \cup B^c$ is the complement of $A^c \cap B$, so its probability is one minus the “ B but not A ” region. Every identity in the box is also a statement about the 2×2 table: $A \cup B^c$ is three of the four cells (all but $A^c \cap B$), so $\mathbb{P}(A \cup B) + \mathbb{P}(A \cup B^c)$ counts the two A cells twice and the two A^c cells once, which is $\mathbb{P}(A) + 1$. When an identity is not remembered, fill the table with unknowns and add cells.

1.3 Venn diagram method

A Venn diagram partitions S into disjoint regions, so the probability of any event is the sum of the regions it contains (C2). Name the events, fill the regions from the innermost intersection outward, then read off the requested set as a sum of regions.

Two events (four regions). The four regions are $A \cap B$, $A \cap B^c$, $A^c \cap B$, and $A^c \cap B^c$. A 2×2 table is the fastest bookkeeping; row and column totals are the marginal probabilities, and the grand total is 1.

	B	B^c	Total
A	$\mathbb{P}(A \cap B)$	$\mathbb{P}(A \cap B^c)$	$\mathbb{P}(A)$
A^c	$\mathbb{P}(A^c \cap B)$	$\mathbb{P}(A^c \cap B^c)$	$\mathbb{P}(A^c)$
Total	$\mathbb{P}(B)$	$\mathbb{P}(B^c)$	1

Any three independent pieces of information determine the whole table.

Three events (eight regions). Fill in the order shown. Each region’s probability is obtained by subtracting regions already known from a given total.

Order	Region	Description	How to compute
1	$A \cap B \cap C$	all three	given, or found last from the total
2	$A \cap B \cap C^c$	A and B only	$\mathbb{P}(A \cap B) - \mathbb{P}(A \cap B \cap C)$
3	$A \cap B^c \cap C$	A and C only	$\mathbb{P}(A \cap C) - \mathbb{P}(A \cap B \cap C)$
4	$A^c \cap B \cap C$	B and C only	$\mathbb{P}(B \cap C) - \mathbb{P}(A \cap B \cap C)$
5	$A \cap B^c \cap C^c$	A only	$\mathbb{P}(A) - (\text{regions 1, 2, 3})$
6	$A^c \cap B \cap C^c$	B only	$\mathbb{P}(B) - (\text{regions 1, 2, 4})$
7	$A^c \cap B^c \cap C$	C only	$\mathbb{P}(C) - (\text{regions 1, 3, 4})$
8	$A^c \cap B^c \cap C^c$	none	$1 - (\text{regions 1–7})$

When the problem gives “none” instead of “all three”, the same table is solved in reverse: the sum of regions 1–7 equals $1 - \mathbb{P}(\text{none})$, which with inclusion-exclusion determines $\mathbb{P}(A \cap B \cap C)$.

Binary classifications. A problem that classifies each policyholder as young or old, male or female, and married or single is a three-event problem in disguise: pick one label from each pair as the event (Y, M, R for young, male, married) and the other label is its complement. “Young, female, single” is the region $Y \cap M^c \cap R^c$, which is “ Y only” in the eight-region table. Counts work exactly like probabilities; the grand total replaces 1.

A disjoint pair inside a three-event problem. When the problem states that two of the three events are mutually exclusive, say $B \cap C = \emptyset$, two regions vanish: “ B and C only” and “all three”. Six regions remain: A only, B only, C only, $A \cap B$, $A \cap C$, and none. Name them with letters, translate every sentence of the problem into a linear equation in those letters, and solve. Useful translations:

- “exactly two of the three” is the sum $(A \cap B) + (A \cap C)$;
- “exactly one” is $(A \text{ only}) + (B \text{ only}) + (C \text{ only})$;
- “ B but not A ” is B only (because $B \cap C$ is empty);
- “not A ” is $(B \text{ only}) + (C \text{ only}) + (\text{none})$;
- “twice as many have B as have C ” is $(B \text{ only}) + (A \cap B) = 2[(C \text{ only}) + (A \cap C)]$.

The total of the six regions is 1 (or the population size). Five independent facts plus the total determine all six regions; the exam usually gives exactly enough, and the order of solving is: everything outside A first, then the “exactly two” total, then the ratio.

A fourth set disjoint from the other three. A survey may add a fourth category D with the statement that no member of D belongs to A, B , or C . Then D is disjoint from $A \cup B \cup C$, and by finite additivity

$$\mathbb{P}(A \cup B \cup C \cup D) = \mathbb{P}(A \cup B \cup C) + \mathbb{P}(D),$$

so “none of the four” is $1 - \mathbb{P}(A \cup B \cup C) - \mathbb{P}(D)$. Run the three-event inclusion-exclusion for A, B, C , then add D as a ninth region. Since D overlaps nothing, “exactly one of the four” is “exactly one of the three” plus $\mathbb{P}(D)$, and “exactly two of the four” equals “exactly two of the three”.

From English to sets. The two tables below are the translation dictionary; every phrase in them appears in exam questions. Write $p_A = \mathbb{P}(A)$, $p_B = \mathbb{P}(B)$, $p_{AB} = \mathbb{P}(A \cap B)$.

English phrase (two events)	Set expression	Probability
A or B ; at least one	$A \cup B$	$p_A + p_B - p_{AB}$
A and B ; both	$A \cap B$	p_{AB}
not A	A^c	$1 - p_A$
neither A nor B	$A^c \cap B^c = (A \cup B)^c$	$1 - p_A - p_B + p_{AB}$
not both; at most one	$A^c \cup B^c = (A \cap B)^c$	$1 - p_{AB}$
A but not B ; A only	$A \cap B^c$	$p_A - p_{AB}$
exactly one of A, B	$(A \cap B^c) \cup (A^c \cap B)$	$p_A + p_B - 2p_{AB}$

For three events write $\Sigma_1 = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C)$, $\Sigma_2 = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C)$, and $T = \mathbb{P}(A \cap B \cap C)$. Region numbers refer to the eight-region table.

English phrase (three events)	Set expression	Probability
at least one	$A \cup B \cup C$	$\Sigma_1 - \Sigma_2 + T$
none	$A^c \cap B^c \cap C^c$	$1 - \Sigma_1 + \Sigma_2 - T$
all three	$A \cap B \cap C$	T
A only	$A \cap B^c \cap C^c$	$\mathbb{P}(A) - \mathbb{P}(AB) - \mathbb{P}(AC) + T$
exactly one	regions 5, 6, 7	$\Sigma_1 - 2\Sigma_2 + 3T$
exactly two	regions 2, 3, 4	$\Sigma_2 - 3T$
at least two	regions 1, 2, 3, 4	$\Sigma_2 - 2T$
at most one	complement of “at least two”	$1 - \Sigma_2 + 2T$

Every three-event formula is a sum of regions from the eight-region table; when in doubt, fill the regions and add.

1.4 Countable additivity and geometric series

Axiom A3 covers infinite sequences of disjoint events, and the exam tests it with regions whose areas or probabilities form a geometric progression: the i th region has probability r^i (or kr^i) for $i = 1, 2, \dots$, with $0 < r < 1$. The regions are disjoint, so the probability of hitting one of the first N is the sum of the first N terms, and the probability of hitting any of infinitely many is the sum of the series.

Key fact: Geometric sums for disjoint regions

For $0 < r < 1$,

$$\sum_{i=1}^N r^i = \frac{r(1-r^N)}{1-r}, \quad \sum_{i=1}^{\infty} r^i = \frac{r}{1-r}, \quad \sum_{i=0}^{\infty} r^i = \frac{1}{1-r}.$$

The infinite sum must not exceed 1 (A3 with A2), so a design with areas r^i , $i \geq 1$, is admissible only when $r \leq 1/2$. If the union of the regions is the whole space, the constant k in $\mathbb{P}(A_i) = kr^i$ is fixed by $kr/(1-r) = 1$.

The probability of missing all N regions is $1 - r(1 - r^N)/(1 - r)$, which decreases to the floor $1 - r/(1 - r) = (1 - 2r)/(1 - r)$ as $N \rightarrow \infty$. A “minimum N such that the miss probability is below p ” question is solved by setting the closed form below p , isolating r^N , and taking logarithms:

$$r^N < c \iff N > \frac{\ln c}{\ln r} \quad (\text{the inequality flips because } \ln r < 0),$$

then rounding up to the next integer. If p is at or below the floor, no finite N works, and no infinite design works either.

1.5 Testing whether two described events are equal

A question may define F and G in words, list five events in words, and ask how many of the five equal $F \cap G$ (or $F \cup G$). Reading alone is unreliable; the descriptions are chosen to differ by one boundary case. The systematic method:

1. Choose coordinates for an outcome. For claims in two years, an outcome is the pair (N_1, N_2) of claim counts.

2. Coarsen each coordinate to the fewest classes that every description in the list can still distinguish. If the descriptions mention “none”, “exactly one”, and “two or more”, use the classes 0, 1, 2^+ .
3. Draw the grid of classes and mark the cells that belong to the target event.
4. For each listed description, mark its cells and compare. Equal events have identical cell sets; one differing cell settles the case.

The listed descriptions are usually algebraic rewrites of the target, and each rewrite is right or wrong by one of the set laws:

- De Morgan: “not (F^c and G^c)” equals $F \cup G$; “not (F^c or G^c)” equals $F \cap G$.
- Disjoint decomposition: $F \cup G = F \cup (F^c \cap G)$.
- Absorption: $F \cup (F \cap H) = F$ and $F \cap (F \cup H) = F$; adding a subset of F to a union with F changes nothing.
- Distribution: $F \cap (G \cup H) = (F \cap G) \cup (F \cap H)$.
- Given one coordinate, a condition on a total becomes a condition on the other coordinate: with $N_1 = 1$ fixed, “ $N_1 + N_2 \geq 2$ ” is “ $N_2 \geq 1$ ”.

A description that is a proper subset of the target (it adds a condition) or a proper superset (it drops one) is not equal to it, even when it is “almost” the same. Watch for “more than one” (at least two) versus “one or more” (at least one), and for a condition on the two-year total that silently allows a year with zero.

1.6 Worked examples

Example 1.1: Three-event coverage survey, none of the three

An insurance company surveys its policyholders about three types of coverage: auto, homeowners, and life. The survey finds that 40% have auto coverage, 35% have homeowners coverage, and 25% have life coverage. It also finds that 15% have both auto and homeowners, 10% have both auto and life, 8% have both homeowners and life, and 5% have all three coverages.

Calculate the percentage of policyholders with none of the three coverages.

(A) 22% (B) 28% (C) 33% (D) 38% (E) 72%

Solution

Let A , H , L be the events that a randomly chosen policyholder has auto, homeowners, and life coverage. “None of the three” is $(A \cup H \cup L)^c$, so by the complement rule and three-event inclusion-exclusion,

$$\mathbb{P}(A \cup H \cup L) = 0.40 + 0.35 + 0.25 - 0.15 - 0.10 - 0.08 + 0.05 = 0.72,$$

and $\mathbb{P}(\text{none}) = 1 - 0.72 = 0.28$.

Answer: (B).

Example 1.2: Exactly one of three

An insurance agency’s records show that, for a randomly selected customer,

- the probability of having auto coverage is 0.50, homeowners coverage 0.40, and life coverage 0.30;

- the probability of having both auto and homeowners is 0.20, both auto and life is 0.15, and both homeowners and life is 0.10;
- the probability of having all three is 0.05.

Calculate the probability that a randomly selected customer has exactly one of the three coverages.

(A) 0.35 (B) 0.40 (C) 0.45 (D) 0.50 (E) 0.55

Solution

Fill the eight regions in the standard order. All three: 0.05. Pairs only: auto–homeowners $0.20 - 0.05 = 0.15$; auto–life $0.15 - 0.05 = 0.10$; homeowners–life $0.10 - 0.05 = 0.05$. Singles only: auto $0.50 - 0.15 - 0.10 - 0.05 = 0.20$; homeowners $0.40 - 0.15 - 0.05 - 0.05 = 0.15$; life $0.30 - 0.10 - 0.05 - 0.05 = 0.10$. None: $1 - 0.80 = 0.20$.

Exactly one is the sum of the three “only” regions: $0.20 + 0.15 + 0.10 = 0.45$.

Formula check: $\Sigma_1 - 2\Sigma_2 + 3T = 1.20 - 2(0.45) + 3(0.05) = 1.20 - 0.90 + 0.15 = 0.45$.

Answer: (C).

Example 1.3: Bounding a probability from the axioms alone

For a randomly selected claim from an auto insurer’s files, the probability that the claim involves bodily injury is 0.60 and the probability that it involves property damage is 0.70. No other information about the claims is available.

Determine the smallest possible value of the probability that a claim involves both bodily injury and property damage.

(A) 0 (B) 0.10 (C) 0.30 (D) 0.42 (E) 0.60

Solution

Let B be bodily injury and D be property damage. Inclusion-exclusion gives $\mathbb{P}(B \cap D) = \mathbb{P}(B) + \mathbb{P}(D) - \mathbb{P}(B \cup D) = 1.30 - \mathbb{P}(B \cup D)$. Since $\mathbb{P}(B \cup D) \leq 1$ (C5), $\mathbb{P}(B \cap D) \geq 1.30 - 1 = 0.30$. The value 0.30 is attained when $B \cup D = S$, so it is the minimum. (This is Bonferroni’s inequality, C10.)

The largest possible value is $\min(0.60, 0.70) = 0.60$, attained when $B \subseteq D$ (C4). The choice $0.42 = (0.60)(0.70)$ assumes independence, which is not given.

Answer: (C).

Example 1.4: Complements and De Morgan

A property insurer’s records show that, for a randomly selected policyholder, the probability of not having auto coverage is 0.65, the probability of not having homeowners coverage is 0.55, and the probability of having neither coverage is 0.35.

Calculate the probability that a randomly selected policyholder has both auto and homeowners coverage.

(A) 0.10 (B) 0.15 (C) 0.20 (D) 0.35 (E) 0.45

Solution

Let A and H be auto and homeowners coverage. Translate each statement. $\mathbb{P}(A^c) = 0.65$ so $\mathbb{P}(A) = 0.35$. $\mathbb{P}(H^c) = 0.55$ so $\mathbb{P}(H) = 0.45$. “Neither” is $A^c \cap H^c = (A \cup H)^c$ by De Morgan, so $\mathbb{P}(A \cup H) = 1 - 0.35 = 0.65$. Then

$$\mathbb{P}(A \cap H) = \mathbb{P}(A) + \mathbb{P}(H) - \mathbb{P}(A \cup H) = 0.35 + 0.45 - 0.65 = 0.15.$$

Table check: $A \cap H^c = 0.35 - 0.15 = 0.20$, $A^c \cap H = 0.45 - 0.15 = 0.30$, neither = 0.35; the four regions sum to $0.15 + 0.20 + 0.30 + 0.35 = 1.00$.

Answer: (B).

Example 1.5: Translating “only”, “none”, “no customer has both”, “at most one”

An insurance agency has 200 customers. Each customer has some combination of auto, homeowners, and renters coverage, or none of them. The agency’s records show:

- 120 customers have auto coverage;
- 60 customers have homeowners coverage and 50 have renters coverage; no customer has both homeowners and renters coverage;
- 65 customers have auto coverage only;
- 25 customers have none of the three coverages.

Calculate the number of customers who have at most one of the three coverages.

(A) 55 (B) 65 (C) 120 (D) 145 (E) 175

Solution

Let A , H , R be the three coverages and work with counts out of 200. “No customer has both homeowners and renters” means $H \cap R = \emptyset$, so $A \cap H \cap R$ is also empty and every two-coverage customer has auto plus one other. “Auto only” is 65, so $120 - 65 = 55$ customers have auto plus another coverage; these are exactly the two-coverage customers. Since H and R are disjoint, $|H \cup R| = 60 + 50 = 110$, of whom 55 also have auto, leaving $110 - 55 = 55$ with homeowners only or renters only. Exactly one coverage: $65 + 55 = 120$. Check: $120 + 55 + 25 = 200$.

“At most one” is none or exactly one: $25 + 120 = 145$.

Answer: (D).

Example 1.6: Unions with complements

For a randomly selected policy in a commercial lines portfolio, let A be the event that the policy carries general liability coverage and B the event that it carries property coverage. You are given

$$\mathbb{P}(A \cup B) = 0.78, \quad \mathbb{P}(A \cup B^c) = 0.67, \quad \mathbb{P}(B) = 0.55.$$

Calculate the probability that the policy carries both coverages.

(A) 0.22 (B) 0.33 (C) 0.45 (D) 0.55 (E) 0.67

Solution

Write inclusion-exclusion for each union and add:

$$\begin{aligned}\mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B), \\ \mathbb{P}(A \cup B^c) &= \mathbb{P}(A) + \mathbb{P}(B^c) - \mathbb{P}(A \cap B^c), \\ 0.78 + 0.67 &= 2\mathbb{P}(A) + 1 - [\mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)] = 2\mathbb{P}(A) + 1 - \mathbb{P}(A),\end{aligned}$$

so $\mathbb{P}(A) = 1.45 - 1 = 0.45$. Then $\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) = 0.45 + 0.55 - 0.78 = 0.22$. Shorter route: $A \cup B^c$ is the complement of $A^c \cap B$ (De Morgan), so $\mathbb{P}(A^c \cap B) = 1 - 0.67 = 0.33$, and $\mathbb{P}(A \cap B) = \mathbb{P}(B) - \mathbb{P}(A^c \cap B) = 0.55 - 0.33 = 0.22$.

Table check: $A \cap B = 0.22$, $A \cap B^c = 0.45 - 0.22 = 0.23$, $A^c \cap B = 0.33$, neither = $1 - 0.78 = 0.22$; the sum is 1.00.

Answer: (A).

Example 1.7: Disjoint pair inside a three-product profile

An agent sells disability income insurance, term life insurance, and whole life insurance. No client holds both term life and whole life. The agent's client profile is:

- (i) 20% of clients hold none of the three products;
- (ii) 63% of clients hold disability income insurance;
- (iii) three times as many clients hold term life as hold whole life;
- (iv) 23% of clients hold exactly two of the three products;
- (v) 12% of clients hold term life but not disability income insurance.

Calculate the percentage of clients who hold both disability income and whole life insurance.

- (A) 5% (B) 8% (C) 12% (D) 15% (E) 18%

Solution

Let D, T, W be the three products. Since $T \cap W = \emptyset$, the regions $T \cap W$ only and $D \cap T \cap W$ are empty, leaving six regions: d (D only), t (T only), w (W only), dt , dw , and n (none). Translate:

$$n = 0.20, \quad d + dt + dw = 0.63, \quad t + dt = 3(w + dw), \quad dt + dw = 0.23, \quad t = 0.12.$$

Fact (v) gives t directly because T but not D cannot include any W . Outside D lie t, w , and n , with total $1 - 0.63 = 0.37$, so $w = 0.37 - 0.20 - 0.12 = 0.05$. Substitute into the ratio: $0.12 + dt = 3(0.05 + dw) = 0.15 + 3dw$, so $dt = 0.03 + 3dw$. With $dt + dw = 0.23$: $0.03 + 4dw = 0.23$, so $dw = 0.05$ and $dt = 0.18$.

Check: $d = 0.63 - 0.18 - 0.05 = 0.40$, and $0.40 + 0.12 + 0.05 + 0.18 + 0.05 + 0.20 = 1.00$. Term life totals $0.12 + 0.18 = 0.30$, whole life $0.05 + 0.05 = 0.10$, ratio 3.

Answer: (A).

Example 1.8: Three overlapping benefits and a fourth disjoint one

A company offers its 250 employees four voluntary benefits: dental, vision, legal, and a commuter plan. Enrollment records show:

- (i) 96 are enrolled in dental, 58 in vision, and 40 in legal;

- (ii) 27 are enrolled in dental and vision, 18 in dental and legal, and 12 in vision and legal;
- (iii) 7 are enrolled in all three of dental, vision, and legal;
- (iv) 35 are enrolled in the commuter plan, and none of these is enrolled in dental, vision, or legal.

Calculate the number of employees enrolled in exactly one of the four benefits.

- (A) 71 (B) 101 (C) 114 (D) 136 (E) 179

Solution

Let D, V, L, K be the four benefits. For the three overlapping ones, $\Sigma_1 = 96 + 58 + 40 = 194$, $\Sigma_2 = 27 + 18 + 12 = 57$, $T = 7$. Exactly one of D, V, L is $\Sigma_1 - 2\Sigma_2 + 3T = 194 - 114 + 21 = 101$. Region check: D only = $96 - 20 - 11 - 7 = 58$, V only = $58 - 20 - 5 - 7 = 26$, L only = $40 - 11 - 5 - 7 = 17$, total 101.

K is disjoint from $D \cup V \cup L$, so every one of its 35 members has exactly one benefit, and none of them changes the count for D, V, L . Exactly one of the four: $101 + 35 = 136$.

For reference, $|D \cup V \cup L| = 194 - 57 + 7 = 144$, $|D \cup V \cup L \cup K| = 144 + 35 = 179$ by finite additivity, and none of the four = $250 - 179 = 71$.

Answer: (D).

Example 1.9: Countable additivity with geometric areas

A carnival game uses a board of area 1 painted in red and blue regions. A blindfolded player marks a point on the board and wins if the point lies in a blue region. The board can be painted with any number of red regions, and the red regions are designed so that the i th red region has area $(2/5)^i$, $i = 1, 2, \dots$

Calculate the minimum number of red regions that makes the probability of winning less than 34%.

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Solution

With N red regions, the red regions are disjoint, so by additivity the losing probability is the sum of their areas and the winning probability is

$$w_N = 1 - \sum_{i=1}^N \left(\frac{2}{5}\right)^i = 1 - \frac{(2/5)(1 - (2/5)^N)}{1 - 2/5} = 1 - \frac{2}{3} \left(1 - \left(\frac{2}{5}\right)^N\right) = \frac{1}{3} + \frac{2}{3} \left(\frac{2}{5}\right)^N.$$

Require $w_N < 0.34$:

$$\frac{2}{3} \left(\frac{2}{5}\right)^N < 0.34 - \frac{1}{3} = \frac{1}{150} \iff \left(\frac{2}{5}\right)^N < \frac{1}{100} \iff N > \frac{\ln(0.01)}{\ln(0.4)} = 5.03.$$

The smallest integer is $N = 6$. Check: $w_5 = 1/3 + (2/3)(0.01024) = 0.3402$, not below 0.34; $w_6 = 1/3 + (2/3)(0.004096) = 0.3361$.

The winning probability never drops below the floor $1/3$: the infinite series of red areas sums to $(2/5)/(3/5) = 2/3$. A target of “less than 33%” would have no answer, with any finite or infinite number of red regions.

Answer: (E).

Example 1.10: Which described events equal $F \cup G$

A policyholder holds an auto policy for two years. Let F be the event that the policyholder has at most one accident in year one, and G the event that the policyholder has no accidents in year two. Consider the following events:

- (i) The policyholder has at most one accident in the two-year period.
- (ii) It is not the case that the policyholder has two or more accidents in year one and at least one accident in year two.
- (iii) The policyholder has at most one accident in year one, or has two or more accidents in year one and no accidents in year two.
- (iv) The policyholder has at most one accident in year one, or has no accidents in either year.
- (v) The policyholder has at most one accident in year one, or has more accidents in year one than in year two and no accidents in year two.

Determine the number of events from the list that are the same as $F \cup G$.

- (A) None (B) Exactly one (C) Exactly two (D) Exactly three (E) All

Solution

Let N_1 and N_2 be the accident counts in the two years. Every description distinguishes only the classes 0, 1, and 2^+ for each year, so the outcome grid has nine cells. F is the rows $N_1 = 0$ and $N_1 = 1$; G is the column $N_2 = 0$; $F \cup G$ is those two rows plus the cell $(2^+, 0)$, seven cells in all. A bullet marks membership.

(N_1, N_2)	$F \cup G$	(i)	(ii)	(iii)	(iv)	(v)
$(0, 0)$	•	•	•	•	•	•
$(0, 1)$	•	•	•	•	•	•
$(0, 2^+)$	•		•	•	•	•
$(1, 0)$	•	•	•	•	•	•
$(1, 1)$	•		•	•	•	•
$(1, 2^+)$	•		•	•	•	•
$(2^+, 0)$	•		•	•		•
$(2^+, 1)$						
$(2^+, 2^+)$						

(i) is the event $N_1 + N_2 \leq 1$, three cells; it omits $(0, 2^+)$, which is in F . Not equal. (ii) is the complement of $F^c \cap G^c$, which by De Morgan is $F \cup G$. Equal. (iii) is $F \cup (F^c \cap G)$, the disjoint decomposition of $F \cup G$. Equal. (iv) is $F \cup \{(0, 0)\}$; the added cell is already in F , so this is just F , which lacks $(2^+, 0)$. Not equal. (v) is $F \cup \{N_1 > N_2, N_2 = 0\} = F \cup \{N_1 \geq 1, N_2 = 0\}$. The added cells are $(1, 0)$, already in F , and $(2^+, 0)$; the result is $F \cup G$. Equal.

Three of the five events equal $F \cup G$.

Answer: (D).

1.7 Traps

Trap: “At least one” versus “exactly one” versus “at most one”

“At least one of A, B ” is $A \cup B$ and includes the intersection. “Exactly one” excludes the intersection: $\mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B)$. “At most one” is the complement of “both”: $1 - \mathbb{P}(A \cap B)$. For three events, “at most one” is not $1 - T$; it is $1 - (\Sigma_2 - 2T)$, the complement of “at least two”. Likewise “neither A nor B ” is $(A \cup B)^c$ while “not both” is $(A \cap B)^c$; De Morgan’s laws tell you which is which.

Trap: $\mathbb{P}(A \cap B^c)$ is not $\mathbb{P}(A) - \mathbb{P}(B)$

“ A but not B ” has probability $\mathbb{P}(A) - \mathbb{P}(A \cap B)$. Subtracting $\mathbb{P}(B)$ instead removes outcomes that were never in A . The identity $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$ holds only when $A \subseteq B$.

Trap: Assuming disjoint or independent when the problem does not say so

If a question gives $\mathbb{P}(A)$ and $\mathbb{P}(B)$ and asks for $\mathbb{P}(A \cup B)$ with no other information, the answer is not determined; it lies between $\max(\mathbb{P}(A), \mathbb{P}(B))$ and $\min(1, \mathbb{P}(A) + \mathbb{P}(B))$. Adding the probabilities assumes disjointness; multiplying them assumes independence. Use either only when the problem states it or the structure forces it (a policyholder cannot be both under 25 and over 65).

Trap: “ A and B ” includes those who also have C

In a three-event problem, “15% have both auto and homeowners” is $\mathbb{P}(A \cap H) = 0.15$, which includes the policyholders who also have life. The region “auto and homeowners only” is $\mathbb{P}(A \cap H) - \mathbb{P}(A \cap H \cap L)$. Conversely, when the problem says “15% have auto and homeowners but not life”, it is giving the region directly. Confusing the two shifts the answer by a multiple of T .

Trap: $\mathbb{P}(A \cup B^c)$ is not $1 - \mathbb{P}(A \cap B)$

The complement of $A \cup B$ is $A^c \cap B^c$, not $A \cup B^c$. The event $A \cup B^c$ is three of the four cells of the 2×2 table (everything except $A^c \cap B$), so its probability is $1 - \mathbb{P}(A^c \cap B)$, and it is typically large. When two unions involving complements are given, add the two inclusion-exclusion equations rather than guessing a relation between them.

Trap: An infinite union of disjoint regions has a floor, and a ceiling

With region probabilities r^i , $i \geq 1$, the probability of missing every region decreases toward $(1 - 2r)/(1 - r)$, not toward 0; a question that asks for a miss probability below that floor has no answer. In the other direction, the series must sum to at most 1, so r^i areas on a board of area 1 require $r \leq 1/2$. When solving $r^N < c$, dividing by $\ln r$ (negative) reverses the inequality, and the answer is the next integer above the bound, not the nearest integer.

Trap: Coarsening the outcome grid too far

The classes 0, 1, 2^+ are enough when every description compares a count to a fixed number. A description that compares the two years to each other (“more accidents in year one than in year two”) can split the 2^+ class: the outcomes (3, 2) and (2, 3) both sit in the cell $(2^+, 2^+)$ but fall on opposite sides of the comparison. Either refine the classes or test the ambiguous cell with explicit outcomes before declaring two events equal.

1.8 Practice problems

Practice 1.1

A survey of homeowners insurance policyholders finds that 50% have a security system, 35% have a fire sprinkler system, and 25% have a backup generator. Further, 15% have both a security system and a sprinkler system, 10% have both a security system and a generator, 5% have both a sprinkler system and a generator, and 2% have all three.

Calculate the percentage of policyholders with none of the three.

- (A) 12% (B) 18% (C) 20% (D) 28% (E) 32%

Practice 1.2

For a randomly selected policyholder, let A be the event that the policyholder files a claim in the first policy year and B the event that the policyholder files a claim in the second policy year. You are given $\mathbb{P}(A^c) = 0.60$, $\mathbb{P}(B) = 0.35$, and $\mathbb{P}(A \cup B) = 0.60$.

Calculate the probability that the policyholder files a claim in exactly one of the two years.

- (A) 0.30 (B) 0.45 (C) 0.55 (D) 0.60 (E) 0.75

Practice 1.3

For a randomly selected claim, the probability that the claim involves a rental vehicle is 0.75 and the probability that it involves a towing charge is 0.65. No other information is available.

Determine the largest possible value of the probability that the claim involves neither a rental vehicle nor a towing charge.

- (A) 0 (B) 0.0875 (C) 0.25 (D) 0.35 (E) 0.40

Practice 1.4

An insurer classifies each policy by whether it has a young driver (Y), a sports car (S), and an accident in the past three years (C). For a randomly selected policy,

$$\mathbb{P}(Y) = 0.30, \quad \mathbb{P}(S) = 0.25, \quad \mathbb{P}(C) = 0.20,$$

$$\mathbb{P}(Y \cap S) = 0.10, \quad \mathbb{P}(Y \cap C) = 0.08, \quad \mathbb{P}(S \cap C) = 0.06, \quad \mathbb{P}(Y \cap S \cap C) = 0.04.$$

Calculate the probability that a randomly selected policy has at most one of the three characteristics.

- (A) 0.16 (B) 0.24 (C) 0.76 (D) 0.84 (E) 0.90

Practice 1.5

The probability that a randomly selected auto policyholder lacks at least one of collision coverage and comprehensive coverage is 0.90. The probability of having collision coverage is 0.45 and the probability of having comprehensive coverage is 0.30.

Calculate the probability that the policyholder has at least one of the two coverages.

- (A) 0.10 (B) 0.35 (C) 0.55 (D) 0.65 (E) 0.75

Practice 1.6

Of 500 auto policyholders, 280 have collision coverage, 340 have comprehensive coverage, and 60 have neither.

Calculate the number of policyholders who have collision coverage only.

- (A) 80 (B) 100 (C) 160 (D) 180 (E) 220

Practice 1.7

For a randomly selected policyholder, let A be the event that the policyholder files a claim this year and B the event that the policyholder renews at year end. You are given

$$\mathbb{P}(A \cup B) = 0.85, \quad \mathbb{P}(A \cup B^c) = 0.75, \quad \mathbb{P}(A^c \cup B) = 0.90.$$

Calculate the probability that at most one of the events A and B occurs.

- (A) 0.15 (B) 0.25 (C) 0.35 (D) 0.40 (E) 0.50

Practice 1.8

An insurance agent sells auto, homeowners, and renters insurance. No client holds both homeowners and renters insurance. The agent's client profile is:

- (i) 12% of clients hold none of the three products;
- (ii) 43% of clients hold exactly one of the three products;
- (iii) 73% of clients hold auto insurance;
- (iv) 5% of clients hold renters insurance but not auto insurance;
- (v) twice as many clients hold both auto and homeowners insurance as hold both auto and renters insurance.

Calculate the percentage of clients who hold homeowners insurance.

- (A) 20% (B) 28% (C) 30% (D) 38% (E) 40%

Practice 1.9

A bank surveys 400 customers about four products: a checking account, a savings account, a brokerage account, and a certificate of deposit. The survey finds:

- (i) 130 hold a checking account, 100 a savings account, and 70 a brokerage account;
- (ii) 40 hold checking and savings, 30 hold checking and brokerage, and 20 hold savings and brokerage;
- (iii) 60 hold a certificate of deposit, and none of these holds a checking, savings, or brokerage account;
- (iv) 115 hold none of the four products.

Calculate the number of customers who hold all three of the checking, savings, and brokerage accounts.

- (A) 15 (B) 25 (C) 30 (D) 45 (E) 60

Practice 1.10

Each incoming claim is assigned to exactly one adjuster from the sequence $A_1, A_2, A_3, \dots$, and the probability that a claim is assigned to adjuster A_i is $k(3/4)^i$ for $i = 1, 2, \dots$, where k is a constant.

Calculate the smallest n such that the probability that a claim is assigned to one of the first n adjusters is at least 0.95.

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Practice 1.11

A policyholder holds a policy for two years. Let F be the event that the policyholder has no claims in year one and G the event that the policyholder has at least two claims in year two. Consider the following events:

- (i) The policyholder has at least one claim in year one and at most one claim in year two.
- (ii) It is not the case that the policyholder has no claims in year one or has at least two claims in year two.
- (iii) The policyholder has at least one claim in year one, and has either no claims or exactly one claim in year two.
- (iv) The policyholder has exactly one claim in year one and at most one claim in year two, or has two or more claims in year one and at most one claim in year two.
- (v) The policyholder has at least one claim in year one and at most one claim in year two, or has exactly one claim in year one and no claims in year two.

Determine the number of events from the list that are the same as $(F \cup G)^c$.

- (A) None (B) Exactly one (C) Exactly two (D) Exactly three (E) All

Practice 1.12

An auto insurer has 5,000 policies. Each policy is classified as urban or rural, as covering a sedan or a truck, and as commuting use or pleasure use. Of these policies, 2,100 are urban, 2,800 cover sedans, and 3,200 are for commuting use. Further, 1,150 are urban sedans, 1,900 are commuting sedans, and 1,350 are urban commuting policies. Finally, 600 policies are rural trucks for pleasure use.

Calculate the number of policies that are urban trucks for pleasure use.

- (A) 300 (B) 450 (C) 650 (D) 700 (E) 950

1.9 Answers to practice problems

1. **(B)**. $\mathbb{P}(\text{at least one}) = 1.10 - 0.30 + 0.02 = 0.82$, so none is 0.18.
2. **(B)**. $\mathbb{P}(A) = 0.40$; $\mathbb{P}(A \cap B) = 0.40 + 0.35 - 0.60 = 0.15$; exactly one $= 0.40 + 0.35 - 2(0.15) = 0.45$ (equivalently $0.60 - 0.15$).
3. **(C)**. Neither $= 1 - \mathbb{P}(R \cup T)$, and $\mathbb{P}(R \cup T) \geq \max(0.75, 0.65) = 0.75$ by monotonicity, so neither ≤ 0.25 , attained when $T \subseteq R$. (0.0875 assumes independence, which is not given.)

4. (D). At least two = $\Sigma_2 - 2T = 0.24 - 0.08 = 0.16$; at most one = 0.84.
5. (D). “Lacks at least one” is $(C \cap M)^c$, so $\mathbb{P}(C \cap M) = 0.10$; $\mathbb{P}(C \cup M) = 0.45 + 0.30 - 0.10 = 0.65$.
6. (B). At least one = 440; both = $280 + 340 - 440 = 180$; collision only = $280 - 180 = 100$.
7. (E). $\mathbb{P}(A) = 0.85 + 0.75 - 1 = 0.60$; $\mathbb{P}(B) = 0.85 + 0.90 - 1 = 0.75$; $\mathbb{P}(A \cap B) = 0.60 + 0.75 - 0.85 = 0.50$; at most one = $1 - 0.50 = 0.50$. (Neither = 0.15; exactly one = 0.35.)
8. (E). Exactly two = $1 - 0.12 - 0.43 = 0.45$; auto only = $0.73 - 0.45 = 0.28$; homeowners only + renters only = $0.43 - 0.28 = 0.15$, so homeowners only = 0.10; auto-renters = 0.15 and auto-homeowners = 0.30 from the ratio; homeowners = $0.10 + 0.30 = 0.40$.
9. (A). The certificate holders are disjoint from the other three, so $|C \cup S \cup B| = 400 - 115 - 60 = 225 = 300 - 90 + T$, giving $T = 15$.
10. (C). The union of all A_i is S , so $k(3/4)/(1/4) = 3k = 1$ and $k = 1/3$. $\mathbb{P}(A_1 \cup \dots \cup A_n) = 1 - (3/4)^n \geq 0.95$ requires $n \geq \ln(0.05)/\ln(0.75) = 10.41$, so $n = 11$.
11. (E). The target is $F^c \cap G^c = \{N_1 \geq 1, N_2 \leq 1\}$. (i) is the direct statement; (ii) is De Morgan; (iii) splits $N_2 \leq 1$ into $N_2 = 0$ or $N_2 = 1$; (iv) splits $N_1 \geq 1$ into $N_1 = 1$ or $N_1 \geq 2$; (v) adds the cell $(1, 0)$, already in the target, so absorption leaves it unchanged.
12. (A). Union of urban, sedan, commuting = $5000 - 600 = 4400 = 8100 - 4400 + T$, so $T = 700$; urban only = $2100 - 1150 - 1350 + 700 = 300$.

1.10 Sample exam cross-reference

After studying this section, test on the following questions from the sample exam.

- Bank 1, Q1: three-event inclusion-exclusion, none of the three (Example 1, Practice 1).
- Bank 1, Q3: add $\mathbb{P}(A \cup B)$ and $\mathbb{P}(A \cup B^c)$ to isolate $\mathbb{P}(A)$ (Example 6, Practice 7).
- Bank 1, Q5: three binary classifications as a three-event count; “young, female, single” is a single region (Example 2, Practice 12).
- Bank 1, Q87: disjoint pair inside three products, six-region linear system with an “exactly two” constraint and a ratio (Example 7, Practice 8).
- Bank 1, Q100: three overlapping sets plus a fourth disjoint from them; finite additivity for the union (Example 8, Practice 9).
- Bank 1, Q108: countable additivity over disjoint regions with geometric areas; minimum N by logarithms (Example 9, Practice 10).
- Bank 1, Q153: event equality tested on an outcome grid; De Morgan, decomposition, absorption (Example 10, Practice 11).

1.11 One-page summary

- Sample space S : all outcomes. Event: a subset of S . Probability: a set function on the events with $\mathbb{P}(A) \geq 0$, $\mathbb{P}(S) = 1$, and countable additivity over pairwise disjoint events.
- Consequences: $\mathbb{P}(\emptyset) = 0$; $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$; $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$; $0 \leq \mathbb{P}(A) \leq 1$; $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$.
- De Morgan: $(A \cup B)^c = A^c \cap B^c$ (“neither”); $(A \cap B)^c = A^c \cup B^c$ (“not both”).
- Inclusion-exclusion: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$; $\mathbb{P}(A \cup B \cup C) = \Sigma_1 - \Sigma_2 + T$.
- Exactly one of two: $\mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B)$. Exactly one of three: $\Sigma_1 - 2\Sigma_2 + 3T$. Exactly two of three: $\Sigma_2 - 3T$. At least two of three: $\Sigma_2 - 2T$.
- Bounds with no other information: $\max(\mathbb{P}(A), \mathbb{P}(B)) \leq \mathbb{P}(A \cup B) \leq \min(1, \mathbb{P}(A) + \mathbb{P}(B))$ and $\max(0, \mathbb{P}(A) + \mathbb{P}(B) - 1) \leq \mathbb{P}(A \cap B) \leq \min(\mathbb{P}(A), \mathbb{P}(B))$. Boole: $\mathbb{P}(\bigcup A_i) \leq \sum \mathbb{P}(A_i)$.
- Venn bookkeeping: fill regions from the triple intersection outward (all three; each pair only; each single only; none); every answer is a sum of regions.

- Translation: “or” / “at least one” is union; “and” / “both” is intersection; “not” is complement; “but not” and “only” subtract the intersection; “every policyholder has at least one” means the union is S ; “no one has both” means disjoint.
- Unions with complements: $\mathbb{P}(A \cup B) + \mathbb{P}(A \cup B^c) = 1 + \mathbb{P}(A)$; $\mathbb{P}(A \cup B^c) = 1 - \mathbb{P}(A^c \cap B)$; $\mathbb{P}(A^c \cup B^c) = 1 - \mathbb{P}(A \cap B)$. Derive any of these by adding two inclusion-exclusion equations.
- Disjoint pair inside three events: two regions vanish; name the six that remain, translate each fact into a linear equation, solve outside-in. A fourth set disjoint from the other three adds by finite additivity: $\mathbb{P}(A \cup B \cup C \cup D) = \mathbb{P}(A \cup B \cup C) + \mathbb{P}(D)$.
- Geometric regions: $\sum_{i=1}^N r^i = r(1 - r^N)/(1 - r)$, $\sum_{i=1}^{\infty} r^i = r/(1 - r) \leq 1$. Miss probability with N regions is $1 - r(1 - r^N)/(1 - r)$, floor $(1 - 2r)/(1 - r)$; solve $r^N < c$ as $N > \ln c / \ln r$ and round up.
- Event equality: draw the outcome grid (classes 0, 1, 2^+ per coordinate), mark the target, mark each description; one differing cell breaks equality. Rewrites that preserve equality: De Morgan, $F \cup G = F \cup (F^c \cap G)$, absorption $F \cup (F \cap H) = F$.
- Never assume disjointness or independence unless stated; check that the regions sum to 1 (or to the stated count).

2 Outcome (b): Combinatorics

2.1 What the outcome asks

The exam tests classical probability: when every outcome of an experiment is equally likely,

$$\mathbb{P}(A) = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}},$$

so the whole problem reduces to counting two sets correctly and consistently. The questions ask you to decide whether order matters (permutation versus combination), whether selection is with or without replacement, and how to split a group into several labelled subgroups (multinomial partitions). The typical setting is selecting policies from a portfolio, forming a committee from a group of policyholders, assigning claims to adjusters, or arranging files in a queue. “At least one” questions are almost always fastest through the complement. Exam arithmetic is done on a calculator, so answers are usually given to three decimals.

2.2 Counting tools

Multiplication principle. If a procedure consists of k stages and stage i can be done in n_i ways regardless of what happened at the earlier stages, the procedure can be done in $n_1 n_2 \cdots n_k$ ways. Every formula below is a consequence of this principle.

Key fact: Permutations: order matters, no replacement

The number of ordered arrangements of all n distinct objects is

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1, \quad 0! = 1.$$

The number of ordered arrangements of k objects chosen from n distinct objects is

$${}_n P_k = \frac{n!}{(n-k)!} = n(n-1) \cdots (n-k+1).$$

Key fact: Combinations: order does not matter, no replacement

The number of unordered subsets of size k from n distinct objects is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{{}_n P_k}{k!}.$$

Useful identities:

$$\binom{n}{k} = \binom{n}{n-k}, \quad \binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2},$$

and Pascal's rule

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Pascal's rule has a counting interpretation that recurs in exam problems: fix one particular object; the k -subsets that contain it number $\binom{n-1}{k-1}$ and the k -subsets that do not contain it number

$\binom{n-1}{k}$. Dividing, the probability that a random k -subset contains a specific object is $\binom{n-1}{k-1}/\binom{n}{k} = k/n$.

Key fact: Binomial theorem

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}, \quad \sum_{k=0}^n \binom{n}{k} = 2^n, \quad \sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

The identity $\sum_k \binom{n}{k} = 2^n$ counts all subsets of an n -set. It is also why the binomial probabilities in Topic 2 sum to one.

Key fact: Permutations with repeated objects (multinomial coefficient)

The number of distinguishable arrangements of n objects of which n_1 are identical of type 1, n_2 identical of type 2, $\dots$, n_r identical of type r (with $n_1 + \dots + n_r = n$) is

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \dots n_r!}.$$

The same number counts the ways to partition n distinct objects into r labelled groups of sizes $n_1, \dots, n_r$. It equals the product of successive combinations $\binom{n}{n_1} \binom{n-n_1}{n_2} \dots \binom{n_r}{n_r}$.

For example, the word MISSISSIPPI has $11!/(1! 4! 4! 2!) = 34650$ distinguishable arrangements, and 12 distinct claims can be divided among three adjusters, four claims each, in $12!/(4! 4! 4!) = 34650$ ways.

Key fact: With replacement

Choosing k times from n objects with replacement, order recorded: n^k outcomes. Choosing k times from n objects with replacement, order ignored (a multiset): $\binom{n+k-1}{k}$ outcomes. The second count is rarely needed on the exam; the first is needed constantly, in particular as the denominator in “at least one match” problems.

Order matters?	Replacement?	Count	Typical wording
Yes	No	${}_nP_k = \frac{n!}{(n-k)!}$	arrange, rank, sequence, fill distinct positions
Yes	Yes	n^k	each of k items independently takes one of n values
No	No	$\binom{n}{k}$	select, choose, committee, sample, hand
No	Yes	$\binom{n+k-1}{k}$	multiset: only how many of each type is recorded

Decision table for choosing k from n distinct objects.

Circular arrangements. n distinct objects around a circle, where only relative position matters, can be arranged in $(n-1)!$ ways: fix one object to remove the rotational symmetry and arrange

the remaining $n - 1$ in a row. If the circle can also be flipped over (a necklace), divide again by 2.

Stars and bars. The number of ways to write n as an ordered sum of r nonnegative integers $x_1 + x_2 + \cdots + x_r = n$, $x_i \geq 0$, is

$$\binom{n+r-1}{r-1}.$$

Picture n identical stars and $r - 1$ bars; each arrangement of the $n + r - 1$ symbols is one solution. If every $x_i \geq 1$ is required, give each variable one unit first and count solutions of $y_1 + \cdots + y_r = n - r$: $\binom{n-1}{r-1}$. For example, 10 identical bonus units can be distributed among 4 agents in $\binom{13}{3} = 286$ ways, and in $\binom{9}{3} = 84$ ways if every agent gets at least one. If all 286 distributions are equally likely, the probability that every agent gets at least one unit is $84/286 = 0.294$.

Restrictions. Two standard devices handle restrictions on arrangements.

- **Objects that must be together:** glue them into one block, arrange the blocks, then multiply by the number of internal orderings of the block. Three files that must be adjacent among 7 files in a row: $5! \cdot 3! = 720$ arrangements out of $7! = 5040$.
- **Objects that must be separated:** arrange the unrestricted objects first, then place the restricted objects in the gaps (including the two ends). No two of 3 special files adjacent among 7: arrange the 4 others ($4!$), choose 3 of the 5 gaps ($\binom{5}{3}$), order the special files in those gaps ($3!$): $24 \cdot 10 \cdot 6 = 1440$.

2.3 Classical probability with counting

With equally likely outcomes, $\mathbb{P}(A) = |A|/|S|$. The numerator and denominator must be counted in the same way: both as ordered lists, or both as unordered sets. Either convention gives the same probability if applied consistently.

Key fact: The hypergeometric counting pattern

A group contains a objects of type I and b objects of type II. A sample of n objects is drawn at random without replacement. The probability that the sample contains exactly x of type I is

$$\mathbb{P}(X = x) = \frac{\binom{a}{x} \binom{b}{n-x}}{\binom{a+b}{n}}.$$

With more than two types, the numerator becomes a product of one binomial coefficient per type, and the denominator is $\binom{N}{n}$ with N the total. Topic 2 names this the hypergeometric distribution; on this outcome it is a counting pattern.

Same problem, two methods. A portfolio holds 10 policies: 4 auto and 6 homeowners. Three policies are selected at random without replacement. Find the probability that exactly 2 of the 3 are auto.

Counting (unordered). Total $\binom{10}{3} = 120$. Favourable $\binom{4}{2} \binom{6}{1} = 6 \cdot 6 = 36$. Probability $36/120 = 0.300$.

Sequential conditional probabilities (ordered). The sequence auto, auto, home has probability

$$\frac{4}{10} \cdot \frac{3}{9} \cdot \frac{6}{8} = \frac{72}{720} = 0.100.$$

The sequences auto, home, auto and home, auto, auto each have the same probability (the numerators 4, 3, 6 appear in a different order, the denominators are 10, 9, 8 regardless). There are $\binom{3}{2} = 3$ positions for the home policy, so the probability is $3 \times 0.100 = 0.300$.

The two methods agree. The counting method is faster when the sample is unordered; the sequential method is faster when the question involves only two or three draws, or asks about a specific position (“the third policy selected is auto”).

2.4 Lattice counting of ordered pairs

Two dice, or two people each choosing an integer at random, produce an ordered pair (x, y) with x and y each taken from $\{1, 2, \dots, n\}$. The n^2 pairs are equally likely and form an n -by- n grid of lattice points. An event described by the difference $x - y$ is a set of diagonals of that grid, and the diagonals have a simple length pattern: the main diagonal $x = y$ has n points, and for each $d \geq 1$ the two diagonals $x - y = d$ and $y - x = d$ have $n - d$ points each.

Key fact: Band around the diagonal

For x, y chosen independently and uniformly from $\{1, \dots, n\}$ and $0 \leq k \leq n - 1$,

$$\#\{|x - y| \leq k\} = n + 2 \sum_{d=1}^k (n - d) = n + 2kn - k(k + 1),$$

$$\#\{|x - y| > k\} = (n - k)(n - k - 1) = 2 \binom{n - k}{2}.$$

The two counts add to n^2 . The second form has a direct reading: the pairs with $x - y > k$ are in one-to-one correspondence, through $x' = x - k$, with the pairs $x' > y$ drawn from $\{1, \dots, n - k\}$, of which there are $\binom{n - k}{2}$; double for $y - x > k$.

For two fair six-sided dice, the pairs with $|x - y| \leq 2$ number $6 + 2(5 + 4) = 24$, so $\mathbb{P}(|X - Y| \leq 2) = 24/36 = 2/3$, and the pairs with $|x - y| > 2$ number $4 \cdot 3 = 12$. When the two ranges are different (x from 1 to m , y from 1 to n) the band is clipped unevenly at the edges; count row by row, one value of x at a time, and add. The same row-by-row count is the fallback whenever the closed form does not obviously apply.

2.5 Selections restricted by rows and columns

Items laid out in an array with r rows and c columns are sometimes selected under the restriction that no two selected items share a row or a column. Selecting k such items ($k \leq \min(r, c)$) is a two-stage procedure.

Key fact: No two in the same row or column

The number of unordered sets of k items from an r -by- c array with no two in the same row or column is

$$\binom{c}{k} \cdot \frac{r!}{(r-k)!} = \binom{r}{k} \cdot \frac{c!}{(c-k)!}.$$

Choose the k columns as a set, then assign a distinct row to each chosen column in order (or the same with rows and columns exchanged). Sequentially: the first item can be any of rc ; the second must avoid one row and one column, leaving $rc - r - c + 1 = (r-1)(c-1)$; the third leaves $(r-2)(c-2)$; and so on. The product $rc(r-1)(c-1)\cdots$ counts ordered selections and must be divided by $k!$. For a probability, divide the unordered count by $\binom{rc}{k}$.

2.6 Pooled populations and unknown counts

Two further variations of the hypergeometric counting pattern appear on the exam.

Key fact: Defectives pooled from several sources

When a stock of N items is assembled from sources of sizes $N_1, \dots, N_m$ with defect rates $p_1, \dots, p_m$, the number of defective items in the stock is the integer $D = p_1N_1 + \cdots + p_mN_m$. A sample drawn without replacement from the pooled stock follows the hypergeometric pattern with $a = D$ and $b = N - D$; the source of each item is irrelevant once D is known. The rates are used only to form the count D ; they are not probabilities of the sampled items.

Key fact: Ratio of two hypergeometric probabilities

With r objects of type I among N and a sample of n , the ratio of consecutive probabilities is

$$\frac{\mathbb{P}(X = x+1)}{\mathbb{P}(X = x)} = \frac{\binom{r}{x+1} \binom{N-r}{n-x-1}}{\binom{r}{x} \binom{N-r}{n-x}} = \frac{(r-x)(n-x)}{(x+1)(N-r-n+x+1)},$$

because $\binom{r}{x+1}/\binom{r}{x} = (r-x)/(x+1)$ and $\binom{N-r}{n-x-1}/\binom{N-r}{n-x} = (n-x)/(N-r-n+x+1)$. In particular

$$\frac{\mathbb{P}(X = 1)}{\mathbb{P}(X = 0)} = \frac{rn}{N-r-n+1}, \quad \frac{\mathbb{P}(X = 2)}{\mathbb{P}(X = 1)} = \frac{(r-1)(n-1)}{2(N-r-n+2)}.$$

When a question states that one such probability is a given multiple of another, the $\binom{N}{n}$ denominators cancel and the ratio is a linear equation in the unknown r . Solve for r , then compute the probability actually asked for.

2.7 Worked examples

Example 2.1: Committee with a constraint

A group of 12 policyholders consists of 7 who hold auto coverage only and 5 who hold both auto and homeowners coverage. A committee of 5 is chosen at random from the group. Calculate the probability that the committee includes at least 2 policyholders who hold both coverages.

- (A) 0.44 (B) 0.61 (C) 0.69 (D) 0.75 (E) 0.81

Solution

Total committees: $\binom{12}{5} = 792$. Use the complement: fewer than 2 from the “both” group means 0 or 1.

$$\binom{5}{0}\binom{7}{5} + \binom{5}{1}\binom{7}{4} = 21 + 5 \cdot 35 = 196.$$

$$\mathbb{P}(\text{at least 2}) = 1 - \frac{196}{792} = \frac{596}{792} = 0.7525.$$

Answer: (D).

The direct sum $\binom{5}{2}\binom{7}{3} + \binom{5}{3}\binom{7}{2} + \binom{5}{4}\binom{7}{1} + \binom{5}{5}\binom{7}{0} = 350 + 210 + 35 + 1 = 596$ agrees. Choice (A) is the probability of exactly 2, choice (C) is the probability of at most 2.

Example 2.2: Sampling a portfolio with three policy types

An insurer’s portfolio consists of 20 policies: 8 auto, 7 homeowners, and 5 life. Four policies are selected at random without replacement for audit. Calculate the probability that the audit sample contains exactly 2 auto, 1 homeowners, and 1 life policy.

- (A) 0.006 (B) 0.058 (C) 0.202 (D) 0.263 (E) 0.350

Solution

$$\mathbb{P} = \frac{\binom{8}{2}\binom{7}{1}\binom{5}{1}}{\binom{20}{4}} = \frac{28 \cdot 7 \cdot 5}{4845} = \frac{980}{4845} = 0.2023.$$

Answer: (C).

Choice (B) results from writing $8 \cdot 7 \cdot 5$ in the numerator (forgetting that the two auto policies are an unordered pair, $\binom{8}{2} = 28$, not $8 \cdot 7$ or 8). Choice (A) results from an ordered numerator $8 \cdot 7 \cdot 5 \cdot 3$ over the ordered denominator ${}_{20}P_4$ without multiplying by the $4!/2! = 12$ arrangements of the types.

Example 2.3: At least one match: birthday pattern

A claims department has 8 adjusters. Each of 4 incoming claims is assigned at random to one of the 8 adjusters, independently of the other claims. Calculate the probability that at least two of the four claims are assigned to the same adjuster.

- (A) 0.41 (B) 0.50 (C) 0.55 (D) 0.59 (E) 0.66

Solution

Assignments are with replacement and ordered (claim 1 to adjuster x , claim 2 to adjuster y , ...), so the sample space has $8^4 = 4096$ equally likely outcomes. The complement of “at least two share” is “all four different”, with ${}_8P_4 = 8 \cdot 7 \cdot 6 \cdot 5 = 1680$ outcomes.

$$\mathbb{P}(\text{at least one shared adjuster}) = 1 - \frac{1680}{4096} = 1 - 0.4102 = 0.5898.$$

Answer: (D).

The same structure gives the birthday problem: with 23 people and 365 equally likely birthdays, $1 - {}_{365}P_{23}/365^{23} = 0.507$.

Example 2.4: Arrangement with a separation restriction

Seven claim files are placed in a queue in random order. Three of the files are auto claims and four are homeowners claims. Calculate the probability that no two auto claim files are adjacent in the queue.

- (A) 1/7 (B) 2/7 (C) 3/7 (D) 4/7 (E) 5/7

Solution

Treat the files as distinct. Total orderings: $7! = 5040$. Arrange the four homeowners files first: $4! = 24$ ways. They create 5 gaps (three interior, two ends). Choose 3 of the 5 gaps for the auto files, $\binom{5}{3} = 10$, and order the auto files within the chosen gaps, $3! = 6$. Favourable orderings: $24 \cdot 10 \cdot 6 = 1440$.

$$\mathbb{P} = \frac{1440}{5040} = \frac{2}{7}.$$

Answer: (B).

The same answer is reached treating the files as indistinguishable within type: the queue is a pattern of 3 A's and 4 H's, $\binom{7}{3} = 35$ patterns, of which $\binom{5}{3} = 10$ have no two A's adjacent; $10/35 = 2/7$. Either model works if used for both numerator and denominator. For contrast, the probability that the three auto files are all adjacent is $5! \cdot 3!/7! = 720/5040 = 1/7$, choice (A).

Example 2.5: Multinomial partition of claims among adjusters

Twelve claims of different sizes are to be divided at random among three adjusters, with each adjuster receiving exactly four claims. Calculate the probability that the three largest claims are all assigned to the same adjuster.

- (A) 0.018 (B) 0.027 (C) 0.036 (D) 0.045 (E) 0.055

Solution

Counting. Total partitions: $\frac{12!}{4!4!4!} = 34650$. Favourable: choose which adjuster receives the three largest claims (3 ways), choose that adjuster's fourth claim from the remaining 9 ($\binom{9}{1} = 9$), then split the remaining 8 claims between the other two adjusters ($\binom{8}{4} = 70$).

Favourable count $3 \cdot 9 \cdot 70 = 1890$.

$$\mathbb{P} = \frac{1890}{34650} = \frac{3}{55} = 0.0545.$$

Sequential. Wherever the largest claim goes, that adjuster has 3 open slots among the 11 remaining; the second largest lands there with probability $3/11$, and then the third largest with probability $2/10$: $\frac{3}{11} \cdot \frac{2}{10} = \frac{6}{110} = \frac{3}{55}$.

Answer: **(E)**.

Choice (A) is $\binom{4}{3}/\binom{12}{3}$, which is the probability that a specified adjuster gets all three; multiplying by 3 adjusters gives the correct 0.055.

Example 2.6: Solving for the group size

An insurer will audit 2 of its n agents, chosen at random. There are exactly 45 different pairs of agents that could be audited. The insurer instead decides to audit 3 agents chosen at random. Calculate the probability that a particular agent is among the three audited.

(A) 0.10 (B) 0.20 (C) 0.30 (D) 0.33 (E) 0.45

Solution

$\binom{n}{2} = \frac{n(n-1)}{2} = 45$, so $n(n-1) = 90 = 10 \cdot 9$ and $n = 10$ (the other root of $n^2 - n - 90 = 0$ is $n = -9$, rejected).

With $n = 10$, the number of 3-subsets containing the particular agent is $\binom{9}{2} = 36$ (the agent is in; choose 2 companions from the other 9), out of $\binom{10}{3} = 120$. Probability $36/120 = 0.30$. Equivalently, $k/n = 3/10$.

Answer: **(C)**.

Example 2.7: With replacement: every option chosen at least once

A survey asks 5 customers to select one of 3 deductible options. Each customer selects an option at random, independently, with each option equally likely. Calculate the probability that every option is selected by at least one customer.

(A) 0.383 (B) 0.617 (C) 0.667 (D) 0.753 (E) 0.802

Solution

Ordered outcomes with replacement: $3^5 = 243$. Let A_i be the event that option i is selected by nobody. By inclusion-exclusion,

$$\mathbb{P}(A_1 \cup A_2 \cup A_3) = 3 \left(\frac{2}{3}\right)^5 - 3 \left(\frac{1}{3}\right)^5 + 0 = \frac{3 \cdot 32 - 3 \cdot 1}{243} = \frac{93}{243}.$$

$$\mathbb{P}(\text{every option used}) = 1 - \frac{93}{243} = \frac{150}{243} = 0.6173.$$

Answer: **(B)**.

Direct count: the 5 customers split over the 3 options as 3+1+1 or 2+2+1. Pattern 3+1+1: choose the option used three times (3), choose those three customers ($\binom{5}{3} = 10$), assign the remaining two customers to the two remaining options ($2! = 2$): 60. Pattern 2+2+1: choose the option used once (3), choose that customer (5), split the remaining four customers into two labelled pairs ($\binom{4}{2} = 6$): 90. Total 150, as before. Choice (A) is the probability of the complement.

Example 2.8: Difference of two dice: lattice count

Two fair eight-sided dice, each numbered 1 to 8, are rolled. Calculate the probability that the two numbers rolled differ by at least 3.

(A) 15/32 (B) 17/32 (C) 9/16 (D) 21/32 (E) 3/4

Solution

The $8^2 = 64$ ordered pairs are equally likely. “Differ by at least 3” is $|x - y| > 2$, so $n = 8$ and $k = 2$ in the band formula:

$$\#\{|x - y| > 2\} = (8 - 2)(8 - 3) = 30, \quad \mathbb{P} = \frac{30}{64} = \frac{15}{32} = 0.469.$$

Answer: (A).

Check by diagonals: $|x - y| = 3, 4, 5, 6, 7$ has 5, 4, 3, 2, 1 points on each side of the main diagonal, $2(5 + 4 + 3 + 2 + 1) = 30$. The complement, $|x - y| \leq 2$, has $8 + 2(7 + 6) = 34$ points, giving choice (B). Choice (D), $42/64$, is $\mathbb{P}(|X - Y| \geq 2)$, the result of reading “at least 3” as “more than 1”.

Example 2.9: Solving for the width of the band

Two players each choose an integer from 1 to 30 inclusive, independently and at random. The probability that the two integers differ by more than k is 0.38. Calculate k .

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Solution

There are $30^2 = 900$ equally likely ordered pairs, and the pairs with $|x - y| > k$ number $(30 - k)(29 - k)$. Hence

$$(30 - k)(29 - k) = 0.38 \cdot 900 = 342 = 19 \cdot 18,$$

a product of consecutive integers, so $30 - k = 19$ and $k = 11$.

Answer: (E).

Check: the band $|x - y| \leq 11$ has $30 + 2(11)(30) - 11 \cdot 12 = 558$ points, and $558 + 342 = 900$. Without the closed form, the row-by-row count of the sample-exam style ($x = 1$ gives 12 values of y , $x = 2$ gives 13, ...) is available but slow for $n = 30$; the closed form turns the question into factoring 342.

Example 2.10: Items in an array: no two in the same row or column

A warehouse stores 48 boxes in a rack with 8 rows and 6 columns, one box per position. Three boxes are selected at random. Calculate the probability that no two of the selected boxes are in the same row or the same column.

(A) 0.389 (B) 0.454 (C) 0.583 (D) 0.648 (E) 0.778

Solution

Total selections: $\binom{48}{3} = 17296$. Favourable: choose the 3 columns, $\binom{6}{3} = 20$, then assign distinct rows to them in order, $8 \cdot 7 \cdot 6 = 336$; $20 \cdot 336 = 6720$. The same count arises from rows first, $\binom{8}{3} \cdot 6 \cdot 5 \cdot 4 = 56 \cdot 120 = 6720$.

$$\mathbb{P} = \frac{6720}{17296} = 0.3885.$$

Answer: **(A)**.

Sequential count: the first box can be any of 48. The second must avoid the first box's row (6 positions) and column (8 positions), which overlap in the first box itself, so $48 - 6 - 8 + 1 = 35$ positions remain. The third must avoid two rows and two columns, $48 - 12 - 16 + 4 = 24$. The product $48 \cdot 35 \cdot 24 = 40320$ counts ordered selections; dividing by $3!$ gives 6720. Placing 40320 over the unordered denominator 17296 gives 2.33, an impossible probability, which is the signal that the numerator and denominator use different conventions.

Example 2.11: Defectives pooled from two sources

A distributor holds 90 smoke detectors: 40 from manufacturer X, of which 15% are faulty, and 50 from manufacturer Y, of which 6% are faulty. An inspector selects 6 detectors at random without replacement from the 90. Calculate the probability that at least two of the six selected detectors are faulty.

(A) 0.079 (B) 0.089 (C) 0.096 (D) 0.104 (E) 0.108

Solution

The stock contains $0.15(40) + 0.06(50) = 6 + 3 = 9$ faulty detectors and 81 good ones. The sample is drawn from the pooled stock, so the hypergeometric pattern applies with $a = 9$, $b = 81$, $n = 6$. Use the complement.

$$\mathbb{P}(X = 0) = \frac{\binom{81}{6}}{\binom{90}{6}} = \frac{324540216}{622614630} = 0.5213, \quad \mathbb{P}(X = 1) = \frac{\binom{9}{1}\binom{81}{5}}{\binom{90}{6}} = \frac{230594364}{622614630} = 0.3704.$$

$$\mathbb{P}(X \geq 2) = 1 - 0.5213 - 0.3704 = 0.1084.$$

Answer: **(E)**.

Choice (C) is $\mathbb{P}(X = 2) = \binom{9}{2}\binom{81}{4}/\binom{90}{6} = 0.0962$, the answer to “exactly two”. Treating the draws as independent with $p = 9/90 = 0.1$ gives $1 - 0.9^6 - 6(0.1)(0.9)^5 = 0.114$, which is close but wrong: the sample is without replacement.

Example 2.12: Unknown count from a ratio of probabilities

An insurer holds 40 policies, an unknown number of which carry an earthquake rider. Five policies are selected at random without replacement. The probability that exactly two of the five carry the rider equals the probability that exactly one carries the rider. Calculate the probability that exactly three of the five selected policies carry the rider.

(A) 0.029 (B) 0.123 (C) 0.153 (D) 0.184 (E) 0.347

Solution

Let r be the number of policies with the rider. The denominators $\binom{40}{5}$ cancel in the ratio:

$$\frac{\mathbb{P}(X = 2)}{\mathbb{P}(X = 1)} = \frac{\binom{r}{2}\binom{40-r}{3}}{\binom{r}{1}\binom{40-r}{4}} = \frac{r-1}{2} \cdot \frac{4}{37-r} = \frac{2(r-1)}{37-r} = 1,$$

so $2r - 2 = 37 - r$ and $r = 13$. Then

$$\mathbb{P}(X = 3) = \frac{\binom{13}{3}\binom{27}{2}}{\binom{40}{5}} = \frac{286 \cdot 351}{658008} = \frac{100386}{658008} = 0.1526.$$

Answer: **(C)**.

Choice (E) is the common value $\mathbb{P}(X = 1) = \mathbb{P}(X = 2) = 0.347$; choice (B) is $\mathbb{P}(X = 0) = \binom{27}{5}/\binom{40}{5} = 0.123$; choice (D) is $\mathbb{P}(X \geq 3) = 0.184$; choice (A) is $\mathbb{P}(X = 4)$. The ratio $\binom{40-r}{3}/\binom{40-r}{4} = 4/(37-r)$ is the step that saves time; expanding both binomial coefficients in full is unnecessary.

2.8 Traps**Trap: Ordered numerator over unordered denominator, or the reverse**

The most common combinatorics error is counting the favourable outcomes as ordered sequences (products such as $8 \cdot 7 \cdot 5$) while the denominator is an unordered count $\binom{N}{n}$, or the reverse. Decide at the start whether the sample space is ordered or unordered and count both sets the same way. When in doubt, use unordered counts with $(\cdot)$ throughout.

Trap: Double counting in “at least k ” problems

To count committees with at least 2 from a group of 5, it is wrong to “choose 2 from the 5, then choose the other 3 from the remaining 10”: $\binom{5}{2}\binom{10}{3} = 1200$ exceeds the total $\binom{12}{5} = 792$, because a committee with three from the group is counted three times. Either sum the exact cases $(\binom{5}{2}\binom{7}{3} + \binom{5}{3}\binom{7}{2} + \cdots)$ or use the complement.

Trap: Forgetting to divide by $k!$

${}_nP_k$ counts ordered selections; $\binom{n}{k} = {}_nP_k/k!$ counts unordered ones. A question that asks for a committee, a sample, or a hand wants $\binom{n}{k}$. Writing $8 \cdot 7 = 56$ for the number of pairs of auto policies, when the number is $\binom{8}{2} = 28$, doubles the numerator.

Trap: Identical objects treated as distinct, or the reverse

The word “identical” or “indistinguishable” in a question signals a multinomial coefficient $n!/(n_1! \cdots n_r!)$, not $n!$. Conversely, when policies or claims are described by type but are physically distinct items, they may be treated as distinct, provided the denominator uses the same convention. The safe approach for probability questions is to treat all objects as distinct in both numerator and denominator.

Trap: “At least one” by direct summation

Summing the cases $1, 2, \dots, n$ for “at least one” wastes time and multiplies the chances of an arithmetic slip. Compute $1 - \mathbb{P}(\text{none})$. For “at least two”, compute $1 - \mathbb{P}(0) - \mathbb{P}(1)$.

2.9 Practice problems

Practice 2.1

A claims department has 8 senior adjusters and 6 junior adjusters. A review team of 4 is selected at random from the 14. Calculate the probability that the team consists of exactly 2 senior and 2 junior adjusters.

- (A) 0.21 (B) 0.32 (C) 0.42 (D) 0.51 (E) 0.60

Practice 2.2

Of 20 claims submitted in a month, 3 are fraudulent. An auditor selects 5 of the 20 claims at random without replacement. Calculate the probability that at least one fraudulent claim is selected.

- (A) 0.40 (B) 0.52 (C) 0.56 (D) 0.60 (E) 0.75

Practice 2.3

Five auto claim files and three homeowners claim files are placed in a processing queue in random order. Calculate the probability that the three homeowners files are processed consecutively.

- (A) $1/56$ (B) $3/28$ (C) $1/8$ (D) $3/14$ (E) $3/8$

Practice 2.4

The number of distinct pairs of policies that can be formed from a portfolio of n policies is 66. Four policies are selected at random from the portfolio. Calculate the probability that two specified policies are both among the four selected.

- (A) $1/66$ (B) $1/33$ (C) $1/22$ (D) $1/12$ (E) $1/11$

Practice 2.5

Nine claims are divided at random among three adjusters so that each adjuster receives exactly three claims. Calculate the probability that two particular claims are assigned to

the same adjuster.

- (A) $1/4$ (B) $2/7$ (C) $1/3$ (D) $4/9$ (E) $1/2$

Practice 2.6

Four customers each independently select one of 5 deductible options, with every option equally likely. Calculate the probability that the four customers select four different options.

- (A) 0.096 (B) 0.192 (C) 0.288 (D) 0.384 (E) 0.480

Practice 2.7

A pricing committee of 4 is selected at random from 5 actuaries and 4 underwriters. Calculate the probability that the committee includes at least one actuary and at least one underwriter.

- (A) 0.833 (B) 0.881 (C) 0.905 (D) 0.952 (E) 0.976

Practice 2.8

An integer X is chosen at random from 1 to 12 inclusive, and independently an integer Y is chosen at random from 1 to 20 inclusive. Calculate the probability that X and Y differ by at most 2.

- (A) 0.238 (B) 0.250 (C) 0.288 (D) 0.375 (E) 0.475

Practice 2.9

Three fair six-sided dice are rolled. Calculate the probability that the largest and smallest of the three numbers rolled differ by at most 1.

- (A) $1/12$ (B) $1/9$ (C) $1/8$ (D) $5/36$ (E) $1/6$

Practice 2.10

Thirty-six policy files are arranged in a 6-by-6 array, one file per position. Three files are selected at random. Calculate the probability that at least two of the selected files are in the same row or in the same column.

- (A) 0.664 (B) 0.700 (C) 0.750 (D) 0.806 (E) 0.860

Practice 2.11

An insurer's file of 75 policies was written by three brokers: 30 by broker A, of which 20% contain a rating error; 25 by broker B, of which 12% contain a rating error; and 20 by broker C, of which 5% contain a rating error. Five policies are selected at random without replacement from the 75. Calculate the probability that exactly two of the five contain a rating error.

- (A) 0.014 (B) 0.089 (C) 0.114 (D) 0.116 (E) 0.392

Practice 2.12

Of 30 claims received in a week, an unknown number have been flagged for review. Three of the 30 claims are selected at random without replacement. The probability that none of the three is flagged equals the probability that exactly one is flagged. Calculate the probability that at least two of the three selected claims are flagged.

- (A) 0.044 (B) 0.078 (C) 0.101 (D) 0.119 (E) 0.128

2.10 Answers to practice problems

1. (C). $\binom{8}{2}\binom{6}{2}/\binom{14}{4} = 28 \cdot 15/1001 = 420/1001 = 0.4196$.
2. (D). $1 - \binom{17}{5}/\binom{20}{5} = 1 - 6188/15504 = 1 - 0.3991 = 0.6009$. Choice (C), $1 - (17/20)^5 = 0.556$, wrongly assumes replacement.
3. (B). Glue the three homeowners files into one block: $6!$ arrangements of six blocks, times $3!$ internal orders, over $8!$: $6 \cdot 720/40320 = 4320/40320 = 3/28$.
4. (E). $\binom{n}{2} = 66 \Rightarrow n(n-1) = 132 = 12 \cdot 11$, $n = 12$. Both specified policies in: $\binom{10}{2}/\binom{12}{4} = 45/495 = 1/11$. Sequentially: $\frac{4}{12} \cdot \frac{3}{11} = 1/11$.
5. (A). Wherever the first claim goes, that adjuster has 2 open slots among the 8 remaining claims: $2/8 = 1/4$. Counting: $3 \cdot \binom{7}{1}\binom{6}{3}/\frac{9!}{3!3!3!} = 420/1680 = 1/4$.
6. (B). ${}_5P_4/5^4 = 120/625 = 0.192$.
7. (D). Complement: all actuaries or all underwriters, $[\binom{5}{4} + \binom{4}{4}]/\binom{9}{4} = 6/126$; probability $120/126 = 20/21 = 0.952$.
8. (A). The grid is 12-by-20, not square, so count row by row. $X = 1$ allows $Y \in \{1, 2, 3\}$ (3 values); $X = 2$ allows $\{1, \dots, 4\}$ (4); each of $X = 3, \dots, 12$ allows 5 values, all within 1 to 20: $3 + 4 + 10 \cdot 5 = 57$ pairs out of 240; $57/240 = 0.2375$. Choice (D) applies the square band formula with $n = 12$ ($54/144$). The $n = 20$ square value, $94/400 = 0.235$, is also wrong: X never exceeds 12, so the band is clipped on one side only.
9. (E). Outcomes with all three equal: 6. Outcomes with all three values in $\{a, a+1\}$ but not all equal: $2^3 - 2 = 6$ for each $a = 1, \dots, 5$, giving 30. Total 36 of 216; $36/216 = 1/6$.
10. (A). Complement: no two share a row or column, $\binom{6}{3} \cdot 6 \cdot 5 \cdot 4 = 2400$ out of $\binom{36}{3} = 7140$; $1 - 2400/7140 = 1 - 0.3361 = 0.6639$.
11. (C). Policies with errors: $0.20(30) + 0.12(25) + 0.05(20) = 6 + 3 + 1 = 10$. $\binom{10}{2}\binom{65}{3}/\binom{75}{5} = 45 \cdot 43680/17259390 = 0.1139$. Choice (D) is the binomial value $\binom{5}{2}(10/75)^2(65/75)^3 = 0.1157$, which assumes replacement; choice (A) is $\mathbb{P}(X = 3)$; choice (E) is $\mathbb{P}(X = 1)$.
12. (E). With r flagged, $\mathbb{P}(X = 1)/\mathbb{P}(X = 0) = 3r/(28 - r) = 1$ gives $r = 7$. Then $\mathbb{P}(X \geq 2) = [\binom{7}{2}\binom{23}{1} + \binom{7}{3}]/\binom{30}{3} = (483 + 35)/4060 = 0.1276$. Choice (D) is $\mathbb{P}(X = 2)$ alone.

2.11 Sample exam cross-reference

- Bank 1, Q97: items in an array, no two in the same row or column; choose the columns as a set, assign rows in order, or count sequentially and divide by $k!$.
- Bank 1, Q113: two dice, absolute difference less than 3; band around the diagonal, $n + 2kn - k(k+1)$ with $n = 6$, $k = 2$.
- Bank 1, Q134: two integers from 1 to 20 differing by at most 3; band count 128 of 400, or complement $(n-k)(n-k-1) = 17 \cdot 16$.
- Bank 2, Q90: defectives pooled from two sources; form the integer count $0.2(30) + 0.08(50) = 10$, then the hypergeometric pattern with $a = 10$, $b = 70$, $n = 5$.

- Bank 2, Q105: unknown count from a ratio of hypergeometric probabilities; $\mathbb{P}(X = 1)/\mathbb{P}(X = 0) = rn/(N - r - n + 1)$, solve for r , then compute $\mathbb{P}(X = 2)$.
- Bank 2, Q122: three categories sampled without replacement; product of three binomial coefficients over $\binom{12}{6}$.
- Bank 2, Q128: two types, sample of two, exactly one of each; $\binom{a}{1}\binom{b}{1}/\binom{a+b}{2}$, or the sequential product summed over the two orders.

2.12 One-page summary

- Classical probability: $\mathbb{P}(A) = |A|/|S|$ for equally likely outcomes. Count numerator and denominator by the same convention (both ordered or both unordered).
- Multiplication principle: $n_1 n_2 \cdots n_k$ for a k -stage procedure.
- Ordered, no replacement: ${}_n P_k = n!/(n - k)!$; all n : $n!$.
- Unordered, no replacement: $\binom{n}{k} = n!/[k!(n - k)!] = {}_n P_k/k!$.
- Ordered, with replacement: n^k . Unordered, with replacement: $\binom{n+k-1}{k}$.
- Identities: $\binom{n}{k} = \binom{n}{n-k}$; $\binom{n}{2} = n(n - 1)/2$; $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$; $\sum_k \binom{n}{k} = 2^n$; $(x + y)^n = \sum_k \binom{n}{k} x^k y^{n-k}$.
- A random k -subset of n contains a specified object with probability k/n .
- Multinomial coefficient $n!/(n_1! \cdots n_r!)$: arrangements with identical objects, and partitions of n distinct objects into labelled groups of sizes $n_1, \dots, n_r$.
- Hypergeometric pattern: $\binom{a}{x}\binom{b}{n-x}/\binom{a+b}{n}$; extend with one factor per type. Equals the sequential product of conditional probabilities times the number of orderings.
- Circular arrangements: $(n - 1)!$. Stars and bars: $x_1 + \cdots + x_r = n$, $x_i \geq 0$, has $\binom{n+r-1}{r-1}$ solutions; $x_i \geq 1$ has $\binom{n-1}{r-1}$.
- Must be adjacent: glue into a block, arrange blocks, multiply by internal orderings. Must be separated: arrange the others, then choose gaps.
- “At least one”: $1 - \mathbb{P}(\text{none})$. Birthday pattern: $1 - {}_n P_k/n^k$.
- Solve $\binom{n}{2} = m$ by $n(n - 1) = 2m$ and factoring into consecutive integers.
- Never “fix k then choose the rest” for “at least k ”; sum exact cases or use the complement.
- Lattice pairs $x, y \in \{1, \dots, n\}$: $\#\{|x - y| \leq k\} = n + 2kn - k(k + 1)$, $\#\{|x - y| > k\} = (n - k)(n - k - 1)$. Unequal ranges: count row by row.
- Array r -by- c , k items with no two in the same row or column: $\binom{c}{k} r!/(r - k)!$; sequentially $rc(r - 1)(c - 1) \cdots$ divided by $k!$.
- Pooled sources: number of defectives $D = \sum p_i N_i$ (an integer), then hypergeometric with $a = D$.
- Ratio of hypergeometric probabilities: $\mathbb{P}(X = 1)/\mathbb{P}(X = 0) = rn/(N - r - n + 1)$, $\mathbb{P}(X = 2)/\mathbb{P}(X = 1) = (r - 1)(n - 1)/[2(N - r - n + 2)]$; the $\binom{N}{n}$ cancels, solve the linear equation for r .

3 Outcome (c): Independence

3.1 What the outcome asks

The exam tests whether you can recognise independence from the wording of a word problem and then use the multiplication rule $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ without being told to. The most common question shapes are: the probability that all of several independent risks occur (multiply), the probability that at least one of them occurs (one minus the product of the complements), and the probability that exactly one occurs (sum of the disjoint “one yes, the rest no” products). A second family gives you $\mathbb{P}(A)$, $\mathbb{P}(A \cup B)$ and the word “independent” and asks you to solve for $\mathbb{P}(B)$, or gives both risks the same unknown probability and makes you solve a quadratic. A third family gives a table of joint probabilities and asks whether the events are independent, or asks for the missing cell that makes them independent. Two further shapes appear on the sample exam: a draw from each of two urns, where the urns are independent and one urn holds an unknown number of balls that you solve for from a “same colour” probability; and a “determine which of the following is true” question about three events built from two dice, which tests whether you can check pairwise and mutual independence by counting outcomes. Underlying all of these is the fact that if A and B are independent, so are A and B^c , which is what lets you convert “fails” probabilities into “works” probabilities freely.

3.2 Definitions

Key fact: Independence of two events

Events A and B are *independent* if and only if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

If $\mathbb{P}(B) > 0$ this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, and if $\mathbb{P}(A) > 0$ it is equivalent to $\mathbb{P}(B \mid A) = \mathbb{P}(B)$: knowing that one event happened does not change the probability of the other.

Use the product form when the question gives joint probabilities and the conditional form when it gives conditional probabilities.

Complements of independent events are independent. If A and B are independent, then each of the pairs (A, B^c) , (A^c, B) and (A^c, B^c) is also independent. Proof for (A, B^c) : since $A = (A \cap B) \cup (A \cap B^c)$ is a disjoint union,

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)(1 - \mathbb{P}(B)) = \mathbb{P}(A)\mathbb{P}(B^c).$$

Applying the same step again with the roles swapped gives (A^c, B^c) , and by symmetry (A^c, B) . The practical consequence: if two components fail independently, they also work independently, and one works independently of the other failing.

Key fact: Working with complements

For independent A and B :

$$\begin{aligned}\mathbb{P}(\text{both occur}) &= \mathbb{P}(A)\mathbb{P}(B), \\ \mathbb{P}(\text{neither occurs}) &= \mathbb{P}(A^c \cap B^c) = (1 - \mathbb{P}(A))(1 - \mathbb{P}(B)), \\ \mathbb{P}(\text{at least one occurs}) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A)\mathbb{P}(B) = 1 - (1 - \mathbb{P}(A))(1 - \mathbb{P}(B)), \\ \mathbb{P}(\text{exactly one occurs}) &= \mathbb{P}(A)(1 - \mathbb{P}(B)) + (1 - \mathbb{P}(A))\mathbb{P}(B).\end{aligned}$$

Mutual independence of n events. Events $A_1, A_2, \dots, A_n$ are *mutually independent* if for every subcollection $A_{i_1}, \dots, A_{i_k}$ ($2 \leq k \leq n$),

$$\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \dots \mathbb{P}(A_{i_k}).$$

The events are *pairwise independent* if the product rule holds for every pair only. Mutual independence implies pairwise independence; the converse fails.

Classic counterexample. Toss two fair coins. Let A = first coin is heads, B = second coin is heads, C = exactly one head. Then $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$, and

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \cap C) = \mathbb{P}(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2},$$

so the three events are pairwise independent. But $A \cap B \cap C = \emptyset$ (two heads is not exactly one head), so $\mathbb{P}(A \cap B \cap C) = 0 \neq \frac{1}{8}$. The three events are not mutually independent. When an exam question says “independent” about three or more events, it means mutually independent, and every product rule you need is available.

Key fact: Checking independence from an outcome table

When the experiment has N equally likely outcomes, $\mathbb{P}(A) = N(A)/N$, where $N(A)$ counts the outcomes in A , and the product rule for A and B becomes

$$N(A \cap B) \cdot N = N(A) \cdot N(B).$$

To settle a “determine which of the following is true” question about three events: (1) count $N(A)$, $N(B)$, $N(C)$; (2) count each of the three pairwise intersections and test the product rule for each pair; (3) count the triple intersection and test $N(A \cap B \cap C) \cdot N^2 = N(A)N(B)N(C)$. The events are mutually independent only if all four tests pass; they are pairwise independent if the three tests in step (2) pass. Two shortcuts: events defined on separate mechanisms (the red die and the blue die, the first toss and the second toss) are independent without any counting; an event that combines both mechanisms (the sum, the number of heads) must be checked against each of the others by counting, because it can be independent of each die separately and still be determined once both dice are known.

Key fact: At least one of n independent events

If $A_1, \dots, A_n$ are mutually independent with $\mathbb{P}(A_i) = p_i$, then

$$\mathbb{P}(\text{none occur}) = \prod_{i=1}^n (1 - p_i), \quad \mathbb{P}(\text{at least one occurs}) = 1 - \prod_{i=1}^n (1 - p_i).$$

When all $p_i = p$: $\mathbb{P}(\text{at least one}) = 1 - (1 - p)^n$. This is the single most used formula in this outcome.

Key fact: At least k of n independent events

Split “at least k ” into the disjoint cases “exactly k ”, “exactly $k + 1$ ”, $\dots$, “exactly n ” and add. Each “exactly j ” case is itself a sum over the $\binom{n}{j}$ patterns of which j events occur, and each pattern is a product of j probabilities and $n - j$ complements. For three events with probabilities p_1, p_2, p_3 ,

$$\mathbb{P}(\text{at least two}) = p_1 p_2 (1 - p_3) + p_1 (1 - p_2) p_3 + (1 - p_1) p_2 p_3 + p_1 p_2 p_3.$$

When every $p_i = p$, $\mathbb{P}(\text{exactly } j) = \binom{n}{j} p^j (1 - p)^{n-j}$, and for “at least two” the complement is shorter: $1 - (1 - p)^n - np(1 - p)^{n-1}$. With unequal probabilities, list the patterns explicitly; with four or more events and unequal probabilities, the complement route (none, exactly one) is still shorter than listing every “exactly two” pattern.

Independent versus mutually exclusive. Two events with positive probability cannot be both independent and mutually exclusive. Proof: if A and B are mutually exclusive then $\mathbb{P}(A \cap B) = 0$; if they are also independent then $0 = \mathbb{P}(A)\mathbb{P}(B)$, which forces $\mathbb{P}(A) = 0$ or $\mathbb{P}(B) = 0$, contradicting positive probability. Mutually exclusive events are in fact as dependent as possible: if one occurs, the other certainly does not.

Independent trials. An experiment is repeated so that the outcome of any trial does not affect any other trial (successive years of a policy, successive tosses, separate policyholders). Then any event defined by the outcomes of trial 1, any event defined by trial 2, and so on, are mutually independent. In particular, if each trial “succeeds” with probability p and $q = 1 - p$,

$$\mathbb{P}(\text{first success occurs on trial } k) = q^{k-1}p, \quad \mathbb{P}(\text{no success in the first } k \text{ trials}) = q^k.$$

The first formula is just the multiplication rule applied to “fail, fail, $\dots$, fail, succeed”. You do not need a named distribution to use it.

Key fact: Three unknown-solving patterns under independence

- *Union given, one marginal given (often as a complement).* Write $\mathbb{P}(A \cup B) = 1 - \mathbb{P}(A^c) \mathbb{P}(B^c)$, so

$$\mathbb{P}(B^c) = \frac{1 - \mathbb{P}(A \cup B)}{\mathbb{P}(A^c)}.$$

This is linear in the unknown; the quadratic appears only when both events share the same unknown probability. If the question gives a joint piece such as $\mathbb{P}(A \cap B^c)$ instead of a marginal, divide it by $\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A \cup B)$: the $\mathbb{P}(B^c)$ factors cancel and the ratio is $\mathbb{P}(A)/\mathbb{P}(A^c)$.

- *Ratio plus joint.* If $\mathbb{P}(A) = k \mathbb{P}(B)$ and $\mathbb{P}(A \cap B) = j$, independence gives $k \mathbb{P}(B)^2 = j$, so

$$\mathbb{P}(B) = \sqrt{j/k}, \quad \mathbb{P}(A) = k\sqrt{j/k}.$$

Keep the root exact until the final multiplication, then use complement independence

for “neither” or “exactly one”. The same pattern with the ratio stated between the complements gives $k \mathbb{P}(B^c)^2 = \mathbb{P}(A^c \cap B^c)$.

- *Two separate mechanisms with an unknown count.* One draw from each of two urns is independent across urns. If urn 2 holds x balls of one colour and known numbers of the others, every urn-2 probability has denominator $x + (\text{known count})$, and

$$\mathbb{P}(\text{same colour}) = \sum_{\text{colours } c} \mathbb{P}(\text{urn 1 gives } c) \mathbb{P}(\text{urn 2 gives } c).$$

Multiply through by the common denominator; the equation is linear in x . “Different colours” is $1 - \mathbb{P}(\text{same colour})$, and there is no “both draw a specific colour” double counting because the colour cases are disjoint.

3.3 How to recognise independence in a word problem

Wording that signals independence	Wording that signals dependence
“independent”, “independently”, “mutually independent”	“given that”, “of those who”, “among policyholders with” (conditional information)
“the events are independent of each other”	Any conditional probability $\mathbb{P}(A \mid B)$ quoted with $\mathbb{P}(A \mid B) \neq \mathbb{P}(A)$
Separate policies, separate policyholders, separate risks “each with probability”	“drawn without replacement”, “selected from the remaining”
Separate years of a policy where each year’s claim status is stated to be independent	One population with overlapping groups (e.g. 40% have auto, 30% have home, 15% have both)
“components fail independently”, “each component operates independently”	Components that share a power source, or a system where one failure causes another
Repeated tosses, rolls, draws with replacement, independent trials	Joint probabilities given in a table (independence must be checked, not assumed)

Two operating rules. First, if the question does not say “independent” and does not describe independent trials, do not multiply marginal probabilities; look for the joint or conditional information that the question gives instead. Second, if the question says “independent” about several events, treat that as mutual independence and use every product you need.

3.4 Worked examples

Example 3.1: Solve for an unknown claim probability

An insurer issues two policies to unrelated customers. The probability that a claim is filed on a given policy is the same for both policies, and claims on the two policies are independent. The probability that at least one of the two policies has a claim is 0.64. Calculate the probability that a claim is filed on a given policy.

(A) 0.20 (B) 0.32 (C) 0.36 (D) 0.40 (E) 0.60

Solution

Let p be the claim probability on each policy, and let A, B be the claim events. Method 1, complement: “at least one” is the complement of “neither”, and by independence

$$\mathbb{P}(\text{neither}) = (1 - p)^2 = 1 - 0.64 = 0.36 \implies 1 - p = 0.6 \implies p = 0.4.$$

Method 2, inclusion–exclusion with the independence product:

$$\mathbb{P}(A \cup B) = p + p - p^2 = 0.64 \implies p^2 - 2p + 0.64 = 0 \implies p = \frac{2 \pm \sqrt{4 - 2.56}}{2} = \frac{2 \pm 1.2}{2},$$

so $p = 0.4$ or $p = 1.6$; discard 1.6. Answer: **(D)**.

Example 3.2: Union given with a complement and a joint piece

For a randomly selected hospital claim, let X be the event that the claim includes imaging charges and Y the event that the claim includes pharmacy charges. The two events are independent. The probability that a claim includes imaging or pharmacy charges is 0.82, and the probability that a claim includes imaging charges but not pharmacy charges is 0.27. Calculate the probability that a claim includes pharmacy charges.

(A) 0.18 (B) 0.27 (C) 0.40 (D) 0.45 (E) 0.55

Solution

Let $x = \mathbb{P}(X)$ and $y = \mathbb{P}(Y)$. Under independence the union is one minus the product of the complements:

$$0.82 = \mathbb{P}(X \cup Y) = 1 - (1 - x)(1 - y) \implies (1 - x)(1 - y) = 0.18.$$

The joint piece is also a product, because X and Y^c are independent: $x(1 - y) = 0.27$. Dividing the two equations removes the unknown $1 - y$:

$$\frac{x(1 - y)}{(1 - x)(1 - y)} = \frac{0.27}{0.18} = 1.5 \implies x = 1.5(1 - x) \implies x = 0.6.$$

Then $1 - y = 0.27/0.6 = 0.45$ and $y = 0.55$. Answer: **(E)**. Check: $0.6 + 0.55 - (0.6)(0.55) = 0.82$. Choice (D) is $\mathbb{P}(Y^c)$, left unfinished. If the question had given a marginal directly, for instance “the probability that a claim does not include imaging charges is 0.40”, the division step is unnecessary: $0.18 = 0.40(1 - y)$ gives $y = 0.55$ in one line.

Example 3.3: Ratio between two independent purchases

An actuary studying renters makes the following conclusions:

- (i) A renter is three times as likely to purchase contents coverage as liability coverage.
- (ii) The event that a renter purchases contents coverage is independent of the event that the renter purchases liability coverage.
- (iii) The probability that a renter purchases both coverages is 0.15.

Calculate the probability that a renter purchases exactly one of the two coverages.

- (A) 0.594 (B) 0.671 (C) 0.744 (D) 0.776 (E) 0.850

Solution

Let L be the liability purchase and C the contents purchase, with $\mathbb{P}(L) = \ell$ and $\mathbb{P}(C) = 3\ell$. Independence turns the joint probability into a quadratic in one unknown:

$$\begin{aligned} 0.15 = \mathbb{P}(C \cap L) &= (3\ell)(\ell) = 3\ell^2 \implies \ell^2 = 0.05 \\ &\implies \ell = \sqrt{0.05} = 0.2236, \quad \mathbb{P}(C) = 3\ell = 0.6708. \end{aligned}$$

Both values lie in $[0, 1]$, so the ratio is consistent. Since C and L are independent, so are C and L^c and C^c and L :

$$\begin{aligned} \mathbb{P}(\text{exactly one}) &= \mathbb{P}(C)\mathbb{P}(L^c) + \mathbb{P}(C^c)\mathbb{P}(L) \\ &= (0.6708)(0.7764) + (0.3292)(0.2236) = 0.5208 + 0.0736 = 0.5944. \end{aligned}$$

Answer: **(A)**. Check: “exactly one” plus “both” is “at least one”, $0.5944 + 0.15 = 0.7444 = 1 - (0.3292)(0.7764)$, which is choice (C). Choice (B) is $\mathbb{P}(C)$ and choice (D) is $\mathbb{P}(L^c)$.

Example 3.4: Series–parallel system

A machine has three components, 1, 2 and 3, which operate independently. The machine works if component 1 works and at least one of components 2 and 3 works. The probabilities that components 1, 2 and 3 work are 0.9, 0.8 and 0.7 respectively. Calculate the probability that the machine works.

- (A) 0.504 (B) 0.720 (C) 0.846 (D) 0.940 (E) 0.994

Solution

Let W_i be the event that component i works. Components 2 and 3 are in parallel, so that block works unless both fail:

$$\mathbb{P}(W_2 \cup W_3) = 1 - \mathbb{P}(W_2^c \cap W_3^c) = 1 - (0.2)(0.3) = 0.94.$$

The complement step uses the fact that W_2^c and W_3^c are independent because W_2 and W_3 are. Component 1 is in series with that block, and W_1 is independent of any event built from W_2 and W_3 , so

$$\mathbb{P}(\text{machine works}) = \mathbb{P}(W_1) \mathbb{P}(W_2 \cup W_3) = (0.9)(0.94) = 0.846.$$

Answer: **(C)**. Choice (A) is the probability that all three work, which is the wrong system logic.

Example 3.5: At least two of three failures with unequal probabilities

A cooling plant has three pumps that fail independently during a year with probabilities 0.10, 0.20 and 0.25. The plant shuts down if and only if at least two of the pumps fail. Calculate the probability that the plant shuts down during the year.

- (A) 0.005 (B) 0.050 (C) 0.080 (D) 0.085 (E) 0.460

Solution

Write F for a failure and S for a survival, in pump order. “At least two fail” is the disjoint union of the three “exactly two” patterns and the “all three” pattern, and each pattern is a product by mutual independence (with complements inheriting independence):

$$\mathbb{P}(FFS) = (0.10)(0.20)(0.75) = 0.0150,$$

$$\mathbb{P}(FSF) = (0.10)(0.80)(0.25) = 0.0200,$$

$$\mathbb{P}(SFF) = (0.90)(0.20)(0.25) = 0.0450,$$

$$\mathbb{P}(FFF) = (0.10)(0.20)(0.25) = 0.0050.$$

The sum is $0.0150 + 0.0200 + 0.0450 + 0.0050 = 0.0850$. Answer: **(D)**. Choice (C) omits the all-fail pattern. Choice (E) is “at least one fails”, $1 - (0.90)(0.80)(0.75) = 0.46$. The complement route agrees: $\mathbb{P}(\text{none}) = 0.54$, $\mathbb{P}(\text{exactly one}) = (0.10)(0.80)(0.75) + (0.90)(0.20)(0.75) + (0.90)(0.80)(0.25) = 0.375$, and $1 - 0.54 - 0.375 = 0.085$.

Example 3.6: At least one claim among many policyholders

An insurer has 8 policyholders. In a given year each policyholder files a claim with probability 0.05, independently of the others.

Calculate the probability that at least one of the 8 policyholders files a claim during the year.

- (A) 0.264 (B) 0.337 (C) 0.400 (D) 0.663 (E) 0.736

Solution

The complement of “at least one claim” is “no claims”. By independence,

$$\mathbb{P}(\text{no claims}) = (1 - 0.05)^8 = 0.95^8 = 0.6634, \quad \mathbb{P}(\text{at least one}) = 1 - 0.6634 = 0.3366.$$

Answer: **(B)**. Choice (C) is 8×0.05 , the result of adding the probabilities, which is only an upper bound. Choice (D) is the complement, left unfinished.

Example 3.7: Independence from a joint-probability table

For a randomly selected policyholder, let S be the event that the policyholder is a smoker and C the event that the policyholder filed a claim last year. An insurer’s records give $\mathbb{P}(S \cap C) = 0.12$ and $\mathbb{P}(S \cap C^c) = 0.28$. Assume S and C are independent.

Calculate the probability that a randomly selected policyholder is a nonsmoker who did not

file a claim.

- (A) 0.18 (B) 0.30 (C) 0.42 (D) 0.48 (E) 0.60

Solution

The two given cells make up all of S : $\mathbb{P}(S) = 0.12 + 0.28 = 0.40$. Independence then gives $\mathbb{P}(C)$ from the joint cell:

$$\mathbb{P}(S \cap C) = \mathbb{P}(S)\mathbb{P}(C) \implies \mathbb{P}(C) = \frac{0.12}{0.40} = 0.30.$$

Since S and C are independent, so are S^c and C^c :

$$\mathbb{P}(S^c \cap C^c) = \mathbb{P}(S^c)\mathbb{P}(C^c) = (0.60)(0.70) = 0.42.$$

Answer: **(C)**. The completed table is

	C	C^c	Total
S	0.12	0.28	0.40
S^c	0.18	0.42	0.60
Total	0.30	0.70	1.00

and every cell equals its row total times its column total, which is the test for independence when a full table is given. If the exam instead gives you the full table and asks whether the events are independent, check one cell: $0.12 = 0.40 \times 0.30$.

Example 3.8: Two boxes with an unknown count

Box 1 contains 12 tickets: 3 green and 9 yellow. Box 2 contains 18 yellow tickets and an unknown number of green tickets. One ticket is drawn at random from each box. The probability that the two tickets are of different colours is 0.55.

Calculate the number of green tickets in Box 2.

- (A) 9 (B) 18 (C) 21 (D) 27 (E) 36

Solution

Let n be the number of green tickets in Box 2, so Box 2 holds $n + 18$ tickets. The draws from the two boxes are independent, and “same colour” splits into the disjoint cases both green and both yellow:

$$\mathbb{P}(\text{same colour}) = \mathbb{P}(G_1)\mathbb{P}(G_2) + \mathbb{P}(Y_1)\mathbb{P}(Y_2) = \frac{3}{12} \cdot \frac{n}{n+18} + \frac{9}{12} \cdot \frac{18}{n+18} = \frac{0.25n + 13.5}{n+18}.$$

“Different colours” is the complement, so $\mathbb{P}(\text{same colour}) = 1 - 0.55 = 0.45$, and

$$0.25n + 13.5 = 0.45(n + 18) = 0.45n + 8.1 \implies 5.4 = 0.2n \implies n = 27.$$

Answer: **(D)**. Check: $(6.75 + 13.5)/45 = 20.25/45 = 0.45$. Choice (B) is the count that

would give Box 2 the same green-to-yellow ratio as Box 1 (18 : 18 is not 3 : 9; the value $n = 18$ makes $\mathbb{P}(\text{same colour})$ exactly 0.5), and choice (E) results from solving with 0.55 on the wrong side of the complement.

Example 3.9: Exactly one of two independent claims

A policyholder has an auto policy and a homeowners policy. In a given year the probability of an auto claim is 0.25 and the probability of a homeowners claim is 0.10. The two claim events are independent.

Calculate the probability that the policyholder files a claim on exactly one of the two policies during the year.

- (A) 0.025 (B) 0.300 (C) 0.325 (D) 0.350 (E) 0.675

Solution

Let A = auto claim, H = homeowners claim. “Exactly one” is the disjoint union $(A \cap H^c) \cup (A^c \cap H)$, and each piece is a product by independence (including complement independence):

$$\mathbb{P}(\text{exactly one}) = (0.25)(0.90) + (0.75)(0.10) = 0.225 + 0.075 = 0.300.$$

Answer: **(B)**. Check via the other route: $\mathbb{P}(A \cup B) = 0.25 + 0.10 - 0.025 = 0.325$ is “at least one” (choice (C)); subtracting “both” = 0.025 gives 0.300.

Example 3.10: First claim in a given year

A policyholder files at most one claim per year. In each year the probability of filing a claim is 0.2, independently of all other years.

Calculate the probability that the policyholder’s first claim occurs in the fourth year.

- (A) 0.0819 (B) 0.1024 (C) 0.4096 (D) 0.5120 (E) 0.5904

Solution

“First claim in year 4” means no claim in years 1, 2, 3 and a claim in year 4. The four yearly events are independent, so multiply:

$$\mathbb{P}(\text{first claim in year 4}) = (0.8)(0.8)(0.8)(0.2) = 0.8^3 \times 0.2 = 0.512 \times 0.2 = 0.1024.$$

Answer: **(B)**. Choice (A) is $0.8^4 \times 0.2$, an off-by-one error (that is the probability the first claim is in year 5). Choice (D) is the probability of no claim in the first three years, which ignores the claim in year 4. Choice (E) is the probability of at least one claim in the first four years.

Example 3.11: First claim in an even-numbered year

As in the previous example, a policyholder files a claim in each year with probability 0.2, independently across years.

Calculate the probability that the first claim occurs in an even-numbered year.

- (A) 0.160 (B) 0.200 (C) 0.444 (D) 0.500 (E) 0.556

Solution

Let $q = 0.8$ and $p = 0.2$. The events “first claim in year 2”, “first claim in year 4”, ... are mutually exclusive, and each is a product by independence:

$$\mathbb{P}(\text{even}) = qp + q^3p + q^5p + \cdots = \frac{qp}{1 - q^2} = \frac{qp}{(1 - q)(1 + q)} = \frac{q}{1 + q} = \frac{0.8}{1.8} = 0.4444.$$

Answer: **(C)**. Sanity check: an odd year is more likely than an even year because year 1 comes first, and indeed $\mathbb{P}(\text{odd}) = 1/(1 + q) = 0.5556 > 0.4444$.

Example 3.12: Pairwise versus mutual independence from an outcome table

Two fair four-sided dice, one white and one black, are rolled. Each die shows 1, 2, 3 or 4 with equal probability.

Let A be the event that the white die shows an odd number.

Let B be the event that the black die shows an odd number.

Let C be the event that the sum of the two numbers is 4.

Determine which of the following is true.

- (A) A, B, C mutually independent (B) each pair independent, not mutually
(C) exactly two pairs independent (D) no pair independent (E) exactly one pair independent

Solution

There are $N = 4 \times 4 = 16$ equally likely outcomes (w, b) . Count each event:

$$N(A) = 8, \quad N(B) = 8, \quad C = \{(1, 3), (2, 2), (3, 1)\}, \quad N(C) = 3.$$

Test each pair with $N(\text{pair}) \cdot N = N(\cdot) N(\cdot)$:

$$\begin{array}{lll} A \cap B = \{(1, 1), (1, 3), (3, 1), (3, 3)\}, & 4 \times 16 = 64 = 8 \times 8 & \text{independent,} \\ A \cap C = \{(1, 3), (3, 1)\}, & 2 \times 16 = 32 \neq 8 \times 3 = 24 & \text{not independent,} \\ B \cap C = \{(1, 3), (3, 1)\}, & 2 \times 16 = 32 \neq 24 & \text{not independent.} \end{array}$$

Exactly one pair is independent. Answer: **(E)**. The pair (A, B) needed no counting: the two dice are separate mechanisms. The event C combines both dice, so it had to be checked against each of A and B ; here $\mathbb{P}(C | A) = 2/8 = 0.25 \neq 3/16$, which is the conditional form of the same test. Once a pair fails, the events cannot be mutually independent, so the triple count is not needed. A parity event such as “the sum is odd” behaves differently from “the

sum is 4”: it is independent of each die separately but is fixed once both dice are known, which is the two-coin phenomenon described under mutual independence; test it the same way, and when all three pairs pass, go on to the triple count.

3.5 Traps

Trap: Adding when you should multiply

“Each of 8 policyholders files a claim with probability 0.05, independently; find the probability at least one files” is not $8 \times 0.05 = 0.40$. Adding gives $\mathbb{P}(A_1) + \cdots + \mathbb{P}(A_n)$, which counts overlaps multiple times and can exceed 1. The correct route is $1 - 0.95^8 = 0.3366$. Adding is correct only for mutually exclusive events, and independent events with positive probability are never mutually exclusive.

Trap: Treating independent as mutually exclusive

Independent means $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, which is positive whenever both probabilities are. Mutually exclusive means $\mathbb{P}(A \cap B) = 0$. They are different, and for events with positive probability they are incompatible. If a question says two risks are independent and you write $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$, you have silently assumed they are mutually exclusive; the correct formula is $\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A)\mathbb{P}(B)$.

Trap: Assuming independence when it is not stated

If a question says 40% of policyholders have auto coverage and 30% have homeowners coverage and asks for the probability a policyholder has both, 0.12 is wrong unless the question says the coverages are independent. Without that word, the question gives $\mathbb{P}(A \cap B)$, $\mathbb{P}(A \cup B)$ or a conditional probability somewhere, and that is what to use. Overlapping populations, “given that”, and sampling without replacement all signal dependence.

Trap: Pairwise is not mutual

If you are told only that $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, $\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C)$ and $\mathbb{P}(B \cap C) = \mathbb{P}(B)\mathbb{P}(C)$, you may not conclude $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. The two-coin example (first head, second head, exactly one head) has all three pairwise products correct and the triple product wrong. Use the triple product only when the question says the events are (mutually) independent, or when the question describes independent trials.

Trap: Forgetting that complements inherit independence

Being told only that the “works” events W_2, W_3 are independent is enough: W_2^c and W_3^c are then independent too, so $\mathbb{P}(W_2^c \cap W_3^c) = (1 - 0.8)(1 - 0.7) = 0.06$ directly. The same fact is what makes “exactly one” equal to $p_1(1 - p_2) + (1 - p_1)p_2$ and “none” equal to $\prod(1 - p_i)$.

3.6 Practice problems

Practice 3.1

Events A and B are independent, $\mathbb{P}(A) = 0.4$ and $\mathbb{P}(A \cup B) = 0.7$. Calculate $\mathbb{P}(B)$.

- (A) 0.30 (B) 0.42 (C) 0.50 (D) 0.58 (E) 0.75

Practice 3.2

A system has three components that operate independently, each of which works with probability 0.9. The system works if at least two of the three components work. Calculate the probability that the system works.

- (A) 0.729 (B) 0.810 (C) 0.900 (D) 0.972 (E) 0.999

Practice 3.3

An insurer has 10 policies in force. Each policy has a claim during the year with probability 0.03, and the policies are independent. Calculate the probability that the insurer receives at least one claim during the year.

- (A) 0.030 (B) 0.263 (C) 0.300 (D) 0.737 (E) 0.970

Practice 3.4

A policyholder has three coverages. In a given year the probabilities of a claim on the three coverages are 0.1, 0.2 and 0.3, and the three claim events are mutually independent. Calculate the probability that the policyholder has a claim on exactly one of the three coverages.

- (A) 0.006 (B) 0.398 (C) 0.496 (D) 0.504 (E) 0.600

Practice 3.5

Two independent risks each have the same probability p of producing a loss in a year. The probability that at least one of the two risks produces a loss is 0.51. Calculate p .

- (A) 0.255 (B) 0.300 (C) 0.490 (D) 0.510 (E) 0.700

Practice 3.6

A driver has an accident in any given year with probability 0.1, independently of other years. Calculate the probability that the driver's first accident occurs in the third year.

- (A) 0.081 (B) 0.090 (C) 0.100 (D) 0.271 (E) 0.810

Practice 3.7

Events A , B and C each have probability 0.5 and are pairwise independent, but $\mathbb{P}(A \cap B \cap C) = 0$.

Calculate $\mathbb{P}(A \cup B \cup C)$.

- (A) 0.500 (B) 0.625 (C) 0.750 (D) 0.875 (E) 1.000

Practice 3.8

Urn 1 contains 4 red, 4 white and 2 blue balls. Urn 2 contains 5 red balls, 3 blue balls and an unknown number of white balls. One ball is drawn at random from each urn. The probability that the two balls are of different colours is 0.65.

Calculate the number of white balls in Urn 2.

- (A) 4 (B) 6 (C) 8 (D) 12 (E) 16

Practice 3.9

Two risks produce losses independently of each other. The probability that risk 1 produces no loss during the year is twice the probability that risk 2 produces no loss during the year. The probability that at least one of the two risks produces a loss during the year is 0.68.

Calculate the probability that exactly one of the two risks produces a loss during the year.

- (A) 0.12 (B) 0.32 (C) 0.44 (D) 0.48 (E) 0.56

Practice 3.10

Events A , B and C are mutually independent, $\mathbb{P}(A \cup B \cup C) = 0.874$, $\mathbb{P}(A^c) = 0.6$ and $\mathbb{P}(B) = 0.3$.

Calculate $\mathbb{P}(C)$.

- (A) 0.30 (B) 0.42 (C) 0.58 (D) 0.63 (E) 0.70

Practice 3.11

A fair coin is tossed three times. Let A be the event that the first toss is heads, B the event that the second toss is heads, and C the event that the three tosses show an odd number of heads.

Determine which of the following is true.

- (A) A , B , C mutually independent (B) each pair independent, not mutually
(C) exactly two pairs independent (D) exactly one pair independent (E) no pair
independent

Practice 3.12

A server has four cooling fans that fail independently, each with probability 0.1 during a year. The server overheats if and only if at least two of the fans fail. Calculate the probability that the server overheats during the year.

- (A) 0.0037 (B) 0.0100 (C) 0.0486 (D) 0.0523 (E) 0.3439

3.7 Answers to practice problems

- (C). $0.7 = 0.4 + b - 0.4b = 0.4 + 0.6b$, so $b = 0.3/0.6 = 0.5$. Equivalently $\mathbb{P}(A^c \cap B^c) = 0.3 = 0.6(1 - b)$.
- (D). $\mathbb{P}(\text{exactly two}) + \mathbb{P}(\text{all three}) = 3(0.9)^2(0.1) + 0.9^3 = 0.243 + 0.729 = 0.972$. The three “exactly two” patterns are disjoint and each is a product by independence.
- (B). $1 - 0.97^{10} = 1 - 0.7374 = 0.2626$. Choice (C) is the sum 10×0.03 .
- (B). $(0.1)(0.8)(0.7) + (0.9)(0.2)(0.7) + (0.9)(0.8)(0.3) = 0.056 + 0.126 + 0.216 = 0.398$. Choice (C) is “at least one”, $1 - (0.9)(0.8)(0.7) = 0.496$.
- (B). $\mathbb{P}(\text{neither}) = (1 - p)^2 = 1 - 0.51 = 0.49$, so $1 - p = 0.7$ and $p = 0.3$. Via the quadratic, $2p - p^2 = 0.51$ gives the same root.
- (A). No accident in years 1 and 2, accident in year 3: $(0.9)^2(0.1) = 0.081$. Choice (D) is “first accident within three years”, $1 - 0.9^3$.
- (C). Inclusion–exclusion: $3(0.5) - 3(0.25) + 0 = 0.75$. The pairwise products supply the three pair terms; the triple term is given as 0, not as 0.125. This is the two-coin example (first head, second head, exactly one head), where $A \cup B \cup C$ is “at least one head” = $3/4$.
- (A). Let x be the number of white balls, so Urn 2 holds $8 + x$ balls. $\mathbb{P}(\text{same colour}) = 1 - 0.65 = 0.35$, and colour by colour, $\mathbb{P}(\text{same}) = [(0.4)(5) + (0.4)(x) + (0.2)(3)]/(8 + x) = (2.6 + 0.4x)/(8 + x)$. Then $2.6 + 0.4x = 2.8 + 0.35x$, so $0.05x = 0.2$ and $x = 4$. Check: $(2.6 + 1.6)/12 = 0.35$.
- (E). Let q_i be the no-loss probability of risk i , with $q_1 = 2q_2$. “At least one loss” = 0.68 gives $\mathbb{P}(\text{no loss at all}) = q_1q_2 = 0.32$, so $2q_2^2 = 0.32$, $q_2 = 0.4$, $q_1 = 0.8$. The loss probabilities are $p_1 = 0.2$ and $p_2 = 0.6$, and $\mathbb{P}(\text{exactly one}) = (0.2)(0.4) + (0.8)(0.6) = 0.08 + 0.48 = 0.56$. Choice (A) is “both”, $(0.2)(0.6) = 0.12$; choice (B) is “neither”.
- (E). $\mathbb{P}(\text{none}) = 1 - 0.874 = 0.126 = \mathbb{P}(A^c)\mathbb{P}(B^c)\mathbb{P}(C^c) = (0.6)(0.7)(1 - c)$, so $1 - c = 0.126/0.42 = 0.3$ and $c = 0.7$. Choice (A) is $\mathbb{P}(C^c)$.
- (A). $N = 8$. $C = \{HHH, HTT, THT, TTH\}$, so $N(A) = N(B) = N(C) = 4$. Pairs: $A \cap B = \{HHH, HHT\}$, $A \cap C = \{HHH, HTT\}$, $B \cap C = \{HHH, THT\}$, each of size 2, and $2 \times 8 = 16 = 4 \times 4$, so all three pairs are independent. Triple: $A \cap B \cap C = \{HHH\}$, and $1 \times 8^2 = 64 = 4 \times 4 \times 4$, so the triple product also holds and the events are mutually independent. This differs from the two-coin case because a third toss decides the parity whatever the first two tosses show, so C carries no information about $A \cap B$. Do not assume a parity event always breaks mutual independence; count.
- (D). Complement of “none or exactly one”: $1 - 0.9^4 - 4(0.1)(0.9)^3 = 1 - 0.6561 - 0.2916 = 0.0523$. Choice (C) is “exactly two” alone, $\binom{4}{2}(0.1)^2(0.9)^2 = 0.0486$; choice (A) is “at least three”; choice (E) is “at least one”.

3.8 Sample exam cross-reference

- Bank 1, Q4: one draw from each of two urns is independent across urns; write $\mathbb{P}(\text{same colour})$ as the sum of colour-by-colour products with the unknown count in every urn-2 denominator, clear the denominator, and solve the linear equation (Example “Two boxes with an unknown count”, Practice 8).
- Bank 1, Q10: a ratio $\mathbb{P}(A) = k\mathbb{P}(B)$ plus the joint probability under independence gives $k\mathbb{P}(B)^2 = \mathbb{P}(A \cap B)$, a square root; then “neither” is $\mathbb{P}(A^c)\mathbb{P}(B^c)$ by complement independence (Example “Ratio between two independent purchases”, Practice 9).
- Bank 1, Q16: union and one marginal given, the marginal as a complement; write $\mathbb{P}(A \cup B) = 1 - \mathbb{P}(A^c)\mathbb{P}(B^c)$ and solve the linear equation for the missing complement (Example “Union given with a complement and a joint piece”, Practice 1 and Practice 10).
- Bank 1, Q120: three events on two dice, “determine which is true”; count $N(\cdot)$ for each event, each pair and the triple, test $N(A \cap B)N = N(A)N(B)$ for every pair and $N(A \cap B \cap C)N^2 = N(A)N(B)N(C)$ for the triple; events on separate dice are independent without counting, an event on the sum must be counted (Example “Pairwise versus mutual independence from an outcome table”, Practice 7 and Practice 11).
- Bank 1, Q125: “at least two of three fail” is the disjoint union of the three “exactly two” patterns and the “all three” pattern, each a product of failure and survival probabilities (Example “At least two of three failures with unequal probabilities”, Practice 2 and Practice 12).

3.9 One-page summary

- Definition: A and B independent $\iff \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Equivalent when $\mathbb{P}(B) > 0$: $\mathbb{P}(A | B) = \mathbb{P}(A)$.
- If A, B independent then so are (A, B^c) , (A^c, B) , (A^c, B^c) . Proof: $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B^c)$.
- Two events with positive probability cannot be both independent and mutually exclusive ($\mathbb{P}(A)\mathbb{P}(B) > 0 = \mathbb{P}(A \cap B)$ is impossible).
- For independent A, B : $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A)\mathbb{P}(B) = 1 - (1 - \mathbb{P}(A))(1 - \mathbb{P}(B))$.
- Exactly one of two: $p_1(1 - p_2) + (1 - p_1)p_2$. Neither: $(1 - p_1)(1 - p_2)$.
- n mutually independent events: $\mathbb{P}(\text{none}) = \prod(1 - p_i)$; $\mathbb{P}(\text{at least one}) = 1 - \prod(1 - p_i)$; equal p : $1 - (1 - p)^n$.
- Exactly k of n equal-probability independent events: $\binom{n}{k}p^k(1 - p)^{n-k}$ (each pattern is a product; the patterns are disjoint).
- Mutual independence requires the product rule for every subcollection; pairwise independence does not imply it. Counterexample: two coins, A = first H, B = second H, C = exactly one H; $\mathbb{P}(A \cap B \cap C) = 0 \neq 1/8$.
- “Independent” about three or more events on the exam means mutually independent.
- Independent trials, success probability p , $q = 1 - p$: $\mathbb{P}(\text{first success on trial } k) = q^{k-1}p$; $\mathbb{P}(\text{no success in } k \text{ trials}) = q^k$; $\mathbb{P}(\text{first success in an even trial}) = q/(1 + q)$.
- Series block: multiply the “works” probabilities. Parallel block: $1 - \prod(\text{fails})$. Reduce a mixed system block by block.
- Solving for an unknown: write $\mathbb{P}(A \cup B) = a + b - ab$ (linear in one unknown) or $2p - p^2 / (1 - p)^2$ (quadratic; take the root in $[0, 1]$).
- Union with one marginal known (possibly as a complement): $\mathbb{P}(B^c) = (1 - \mathbb{P}(A \cup B))/\mathbb{P}(A^c)$. A joint piece $\mathbb{P}(A \cap B^c)$ divided by $\mathbb{P}(A^c \cap B^c)$ gives $\mathbb{P}(A)/\mathbb{P}(A^c)$.

- Ratio plus joint: $\mathbb{P}(A) = k\mathbb{P}(B)$ and $\mathbb{P}(A \cap B) = j$ give $\mathbb{P}(B) = \sqrt{j/k}$; the same with complements if the ratio is stated between the “no” events.
- Two urns, one draw each: $\mathbb{P}(\text{same colour}) = \sum_c \mathbb{P}_1(c)\mathbb{P}_2(c)$; an unknown count x in urn 2 sits in every urn-2 denominator, and clearing it leaves a linear equation. “Different colours” $= 1 - \mathbb{P}(\text{same})$.
- At least k of n : add the disjoint “exactly j ” cases for $j \geq k$; each pattern is a product of j probabilities and $n-j$ complements. At least two of n with equal p : $1 - (1-p)^n - np(1-p)^{n-1}$.
- Outcome-table test with N equally likely outcomes: pair independent iff $N(A \cap B)N = N(A)N(B)$; triple product iff $N(A \cap B \cap C)N^2 = N(A)N(B)N(C)$. Events on separate mechanisms (two dice, two tosses) are independent without counting; an event built from both (sum, parity) must be counted against each other event, and the answer can go either way.
- Table check: events are independent if and only if each cell equals its row total times its column total; checking one cell is enough for a 2×2 table.
- Do not multiply marginals unless the problem says “independent” or describes independent trials; do not add unless the events are mutually exclusive.

4 Outcome (d): Mutually exclusive events

4.1 What the outcome asks

The exam tests whether you can recognise when two or more events cannot occur together (mutually exclusive, or disjoint, events) and whether you then add their probabilities correctly. It also tests the reverse direction: given $\mathbb{P}(A)$, $\mathbb{P}(B)$ and $\mathbb{P}(A \cup B)$, decide whether the events are disjoint, or solve for whatever intersection probability makes the numbers consistent. Underneath both directions sits one technique that drives almost every Topic 1 problem: split the event you want into non-overlapping pieces, find the probability of each piece, and add. Partitions of the sample space (a family of disjoint events that together cover everything) are the standard tool for that split. The phrases “exactly one”, “at least one”, “at most one” and “none” are all evaluated by writing the event as a union of disjoint pieces and adding.

4.2 Definitions and rules

Two events A and B are *mutually exclusive* (equivalently *disjoint*) if $A \cap B = \emptyset$, so they cannot both occur on the same trial. A family $A_1, A_2, \dots$ is *pairwise disjoint* if $A_i \cap A_j = \emptyset$ for every $i \neq j$. A family is *exhaustive* if its union is the whole sample space S . A family that is both pairwise disjoint and exhaustive is a *partition* of S : every outcome lies in exactly one of the events.

Key fact: Addition rule for disjoint events

If $A \cap B = \emptyset$ then

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

More generally, if $A_1, \dots, A_n$ are pairwise disjoint then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mathbb{P}(A_i),$$

and the same holds for a countably infinite family $A_1, A_2, \dots$ (countable additivity, one of the probability axioms). Additivity is the *only* rule that lets you add probabilities directly; it applies exactly when the events are disjoint.

Key fact: Complement as a partition

For any event A , the pair $\{A, A^c\}$ is a partition of S , so

$$\mathbb{P}(A) + \mathbb{P}(A^c) = 1, \quad \mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

More generally, if $B_1, \dots, B_n$ partition S then $\mathbb{P}(B_1) + \dots + \mathbb{P}(B_n) = 1$. This is how the exam hides one probability: it gives $n - 1$ of them and expects you to recover the last one by subtraction.

Key fact: Disjoint decomposition identities

For any events A and B (disjoint or not), the following unions are unions of disjoint pieces, so their probabilities add:

$$\begin{aligned} A &= (A \cap B) \cup (A \cap B^c), & \mathbb{P}(A) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c), \\ A \cup B &= A \cup (B \cap A^c), & \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B \cap A^c), \\ A \cup B &= (A \cap B^c) \cup (A \cap B) \cup (A^c \cap B), & \mathbb{P}(A \cup B) &= \mathbb{P}(A \cap B^c) + \mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B). \end{aligned}$$

The first line is the law of total probability in its simplest form: condition on whether B happens or not. With a partition $B_1, \dots, B_n$ it reads $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i)$.

Key fact: Inclusion–exclusion collapses when events are disjoint

For any two events, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. For three,

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

When the events are pairwise disjoint every intersection has probability 0 and the formula reduces to plain addition. Conversely, if the given numbers satisfy $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ then $\mathbb{P}(A \cap B) = 0$; if $\mathbb{P}(A \cup B) < \mathbb{P}(A) + \mathbb{P}(B)$ the events overlap with positive probability.

Key fact: Disjoint is not independent

If A and B are mutually exclusive and $\mathbb{P}(A) > 0$, $\mathbb{P}(B) > 0$, then A and B are *dependent*. *Proof.* Disjointness gives $\mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0$, while $\mathbb{P}(A)\mathbb{P}(B) > 0$. Hence $\mathbb{P}(A \cap B) \neq \mathbb{P}(A)\mathbb{P}(B)$, which is the definition of dependence. Equivalently, $\mathbb{P}(A \mid B) = 0 \neq \mathbb{P}(A)$: knowing B occurred rules A out completely. $\square$

The three adjectives the exam uses for a family of events answer three different questions.

Term	Definition	Consequence for probabilities
Mutually exclusive	$A \cap B = \emptyset$	$\mathbb{P}(A \cap B) = 0$; $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$
Independent	$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$	$\mathbb{P}(A \mid B) = \mathbb{P}(A)$; can both occur
Exhaustive	$A \cup B = S$	$\mathbb{P}(A \cup B) = 1$; at least one always occurs

Mutually exclusive is a statement about the *sets* (they do not overlap). Independent is a statement about the *numbers* (the product rule holds). Exhaustive is a statement about *coverage* (nothing is left out). Any combination is possible except mutually exclusive together with independent when both probabilities are positive.

4.3 Disjoint decomposition as a technique

The standard move on a Topic 1 word problem is to rewrite the event asked for as a union of pieces that cannot overlap, then add the probabilities of the pieces. Two things make the move reliable. First, the pieces are usually intersections of the given events and their complements, so each piece is a single “cell” of the Venn diagram. Second, every cell probability can be recovered from the quantities the exam typically gives: the marginals $\mathbb{P}(A)$, $\mathbb{P}(B)$, $\mathbb{P}(C)$, the pairwise intersections and the triple intersection.

Two events. The four cells $A \cap B$, $A \cap B^c$, $A^c \cap B$ and $A^c \cap B^c$ have probabilities $\mathbb{P}(AB)$, $\mathbb{P}(A) - \mathbb{P}(AB)$, $\mathbb{P}(B) - \mathbb{P}(AB)$ and $1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(AB)$, where $\mathbb{P}(AB)$ abbreviates $\mathbb{P}(A \cap B)$. “Exactly one” is $(A \cap B^c) \cup (A^c \cap B)$, two disjoint cells, so its probability is $\mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(AB)$. “At least one” adds the cell $A \cap B$ back in; “at most one” removes it from the whole space; “neither” is the single cell $A^c \cap B^c$.

Three events. Let $S_1 = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C)$, $S_2 = \mathbb{P}(AB) + \mathbb{P}(AC) + \mathbb{P}(BC)$ and $S_3 = \mathbb{P}(ABC)$. The eight cells of the three-set Venn diagram are disjoint. “Exactly two” is the union of the three cells $A \cap B \cap C^c$, $A \cap B^c \cap C$ and $A^c \cap B \cap C$; each pairwise intersection contains the triple intersection, so each cell has probability (pairwise) $-S_3$, and adding gives $S_2 - 3S_3$. Cell-by-cell reasoning of the same kind gives the rest of the list below.

Key fact: “Exactly”, “at least”, “at most” formulas

Two events, with $\mathbb{P}(A \cap B)$ written $\mathbb{P}(AB)$:

$$\begin{aligned}\mathbb{P}(\text{exactly one}) &= \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(AB), \\ \mathbb{P}(\text{at least one}) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(AB), \\ \mathbb{P}(\text{at most one}) &= 1 - \mathbb{P}(AB), \\ \mathbb{P}(\text{none}) &= 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(AB).\end{aligned}$$

Three events, with $S_1 = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C)$, $S_2 = \mathbb{P}(AB) + \mathbb{P}(AC) + \mathbb{P}(BC)$ and $S_3 = \mathbb{P}(ABC)$:

$$\begin{aligned}\mathbb{P}(\text{exactly three}) &= S_3, \\ \mathbb{P}(\text{exactly two}) &= S_2 - 3S_3, \\ \mathbb{P}(\text{exactly one}) &= S_1 - 2S_2 + 3S_3, \\ \mathbb{P}(\text{at least one}) &= S_1 - S_2 + S_3, \\ \mathbb{P}(\text{at least two}) &= S_2 - 2S_3, \\ \mathbb{P}(\text{at most one}) &= 1 - S_2 + 2S_3, \\ \mathbb{P}(\text{none}) &= 1 - S_1 + S_2 - S_3.\end{aligned}$$

Check on any problem: exactly one + exactly two + exactly three = at least one.

When the given events are themselves mutually exclusive, $S_2 = S_3 = 0$ and every formula above collapses: “at least one” and “exactly one” coincide and equal S_1 , “at most one” has probability 1, and “none” has probability $1 - S_1$.

4.4 Partitions pinned down by linear constraints

A second family of exam questions never gives a single cell probability directly. Instead the sample space is partitioned into n cells $p_1, \dots, p_n$ and the problem states relationships among them: ratios (“three times as many”), comparisons with a combination (“one-fourth of the other two combined”), cumulative totals over an ordered list (“stage 2 or lower”), or marginal proportions that overlap because every member is counted more than once. Each statement is one linear equation in the cell probabilities, and the partition supplies one more: $p_1 + \dots + p_n = 1$. The question is then solved as a small linear system. Three patterns cover the syllabus.

Key fact: Ratio-specified partition

Choose one cell as the base t and express every other cell as a multiple of t using the ratio statements. A statement of the form “ X is one-fourth of Y and Z combined” is the equation $p_X = \frac{1}{4}(p_Y + p_Z)$; substitute the multiples already found for p_Y and p_Z . The partition equation then reads (sum of multiples) $t = 1$, which gives t , and the event asked for is a sum of the relevant multiples times t . If a statement is absolute rather than relative (“the proportion of X is 0.05 less than the proportion of Y ”, or “ X accounts for 10%”), the equation has a constant term; the system is still linear and is solved the same way, with the constant carried into the partition equation.

Key fact: Cumulative statements over an ordered partition

When the cells are ordered (stages 0, 1, ..., severity levels, claim counts), each statement is a sum of consecutive cells. Write every statement as such a sum, together with $p_0 + \dots + p_n = 1$, and combine:

$$\begin{aligned}\mathbb{P}(k \text{ or higher}) &= 1 - (p_0 + \dots + p_{k-1}), \\ \mathbb{P}(\text{other than } k) &= 1 - p_k, \\ \mathbb{P}(k \text{ or lower}) + \mathbb{P}(k \text{ or higher}) &= 1 + p_k, \\ \mathbb{P}(j \text{ or lower}) - \mathbb{P}(i \text{ or lower}) &= p_{i+1} + \dots + p_j \quad (i < j).\end{aligned}$$

The third line is the workhorse: two cumulative statements that overlap in exactly one cell sum to 1 plus that cell. In general subtracting one sum of cells from another leaves the cells that are in one but not the other. There are usually fewer statements than cells; the target cell is still determined, and the cells that are never separated (for example the top two levels) are simply carried as a block.

Key fact: Counting each member k times

Suppose a group G of the population is described by proportions $\mathbb{P}(A_1), \dots, \mathbb{P}(A_m)$, and every member of G belongs to *exactly* k of the events A_i while nobody outside G belongs to any. Then

$$\mathbb{P}(A_1) + \dots + \mathbb{P}(A_m) = k \mathbb{P}(G), \quad \mathbb{P}(\text{none}) = 1 - \mathbb{P}(G) = 1 - \frac{\mathbb{P}(A_1) + \dots + \mathbb{P}(A_m)}{k}.$$

Proof. Each member of G contributes to exactly k of the m proportions, so the sum of the proportions counts G exactly k times. $\square$

The same argument with mixed membership gives the general identity

$$\sum_{i=1}^m \mathbb{P}(A_i) = \sum_{j=1}^m j \mathbb{P}(\text{exactly } j \text{ of the } A_i),$$

because a member in exactly j of the events is counted j times on the left. For three events this is $S_1 = \mathbb{P}(\text{exactly one}) + 2\mathbb{P}(\text{exactly two}) + 3\mathbb{P}(\text{exactly three})$, which agrees with the formulas of the previous key fact. When “exactly two or none” is stated, the identity collapses to $S_1 = 2\mathbb{P}(\text{exactly two})$ and no individual pairwise cell is needed. When the individual cells

are wanted, write each proportion as the sum of the cells it contains and solve the resulting system; with three events and “exactly two” the three equations $x + y = \mathbb{P}(A)$, $x + z = \mathbb{P}(B)$, $y + z = \mathbb{P}(C)$ give $x = \frac{1}{2}[\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(C)]$ and similarly for y and z .

4.5 Worked examples

Example 4.1: Claim size bands

An insurer classifies each claim into exactly one of four size bands. The probability that a claim is small (under 1,000) is 0.42, medium (from 1,000 to under 5,000) is 0.28, and large (from 5,000 to under 25,000) is 0.18. All other claims are catastrophic. Calculate the probability that a randomly selected claim is either medium or catastrophic.

- (A) 0.40 (B) 0.46 (C) 0.58 (D) 0.34 (E) 0.30

Solution

The four bands partition the sample space, so their probabilities sum to 1:

$$\mathbb{P}(\text{catastrophic}) = 1 - 0.42 - 0.28 - 0.18 = 0.12.$$

Medium and catastrophic are disjoint bands, so the addition rule applies directly:

$$\mathbb{P}(\text{medium} \cup \text{catastrophic}) = 0.28 + 0.12 = 0.40.$$

Answer: **(A)**.

Example 4.2: Are the events disjoint?

For a randomly chosen customer of an insurer, the probability of holding an auto policy is 0.35, the probability of holding a homeowners policy is 0.45, and the probability of holding at least one of the two is 0.70. Calculate the probability that the customer holds both policies.

- (A) 0.00 (B) 0.20 (C) 0.15 (D) 0.10 (E) 0.05

Solution

Let A be auto and B be homeowners. If A and B were disjoint, $\mathbb{P}(A \cup B)$ would equal $0.35 + 0.45 = 0.80$. The stated value 0.70 is smaller, so the events overlap and the general addition rule is required:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) = 0.35 + 0.45 - 0.70 = 0.10.$$

Answer: **(D)**. The gap between the disjoint-case value 0.80 and the actual value 0.70 is exactly the intersection probability.

Example 4.3: Exactly one of two

A policyholder has both an auto policy and a homeowners policy with the same company. In a given year the probability of a claim on the auto policy is 0.40, the probability of a claim on the homeowners policy is 0.25, and the probability of claims on both is 0.10. Calculate the probability that the policyholder files a claim on exactly one of the two policies.

- (A) 0.65 (B) 0.45 (C) 0.60 (D) 0.35 (E) 0.55

Solution

Let A be an auto claim and H a homeowners claim. “Exactly one” is the disjoint union $(A \cap H^c) \cup (A^c \cap H)$. Each piece is a marginal minus the intersection:

$$\mathbb{P}(A \cap H^c) = 0.40 - 0.10 = 0.30, \quad \mathbb{P}(A^c \cap H) = 0.25 - 0.10 = 0.15.$$

Adding the disjoint pieces gives $0.30 + 0.15 = 0.45$, which matches the formula $\mathbb{P}(A) + \mathbb{P}(H) - 2\mathbb{P}(A \cap H) = 0.40 + 0.25 - 0.20 = 0.45$. Answer: **(B)**. Choice (E) is $\mathbb{P}(A \cup H) = 0.55$, the “at least one” answer.

Example 4.4: Exactly two of three

A group health plan offers hospitalisation (A), dental (B) and vision (C) riders. For a randomly selected member, $\mathbb{P}(A \cap B) = 0.12$, $\mathbb{P}(A \cap C) = 0.09$, $\mathbb{P}(B \cap C) = 0.07$ and $\mathbb{P}(A \cap B \cap C) = 0.03$. Calculate the probability that the member holds exactly two of the three riders.

- (A) 0.25 (B) 0.28 (C) 0.22 (D) 0.16 (E) 0.19

Solution

“Exactly two” is the disjoint union of three cells, each a pairwise intersection with the triple intersection removed:

$$\mathbb{P}(A \cap B \cap C^c) = 0.12 - 0.03 = 0.09,$$

$$\mathbb{P}(A \cap B^c \cap C) = 0.09 - 0.03 = 0.06,$$

$$\mathbb{P}(A^c \cap B \cap C) = 0.07 - 0.03 = 0.04.$$

Sum: $0.09 + 0.06 + 0.04 = 0.19$. Equivalently $S_2 - 3S_3 = 0.28 - 0.09 = 0.19$. Answer: **(E)**. Choice (B) is S_2 itself, which triple-counts the members holding all three riders.

Example 4.5: Mutually exclusive with an unknown

An applicant for life insurance is classified as either preferred or standard, never both, and some applicants are declined. Let A be the event that an applicant is classified preferred and B the event that an applicant is classified standard. The probability that an applicant is accepted (preferred or standard) is 0.60, and an applicant is twice as likely to be classified preferred as standard. Calculate $\mathbb{P}(B)$.

- (A) 0.40 (B) 0.30 (C) 0.20 (D) 0.15 (E) 0.60

Solution

The problem states that A and B cannot both occur, so they are mutually exclusive and $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$. With $\mathbb{P}(A) = 2\mathbb{P}(B)$,

$$0.60 = 2\mathbb{P}(B) + \mathbb{P}(B) = 3\mathbb{P}(B) \implies \mathbb{P}(B) = 0.20,$$

and $\mathbb{P}(A) = 0.40$. Answer: **(C)**. The probability of being declined is $1 - 0.60 = 0.40$, which is choice (A); read the question stem carefully.

Example 4.6: Partition specified by ratios and a difference

A dental plan classifies every claim as exactly one of preventive, basic, major or orthodontic. Claim records show that the number of basic claims is one-half the number of preventive and major claims combined, that three times as many preventive claims are filed as major claims, and that the proportion of orthodontic claims is 0.05 less than the proportion of major claims. Calculate the probability that the next claim filed is either basic or orthodontic.

- (A) 0.25 (B) 0.30 (C) 0.45 (D) 0.55 (E) 0.40

Solution

Let m be the probability that a claim is major. The four categories partition the sample space, so the statements translate into

$$\mathbb{P}(\text{preventive}) = 3m, \quad \mathbb{P}(\text{basic}) = \frac{1}{2}(3m + m) = 2m, \quad \mathbb{P}(\text{orthodontic}) = m - 0.05.$$

The partition equation $3m + 2m + m + (m - 0.05) = 1$ gives $7m = 1.05$, so $m = 0.15$. The cells are 0.45, 0.30, 0.15 and 0.10, which sum to 1. Basic and orthodontic are disjoint categories, so

$$\mathbb{P}(\text{basic} \cup \text{orthodontic}) = 2m + (m - 0.05) = 0.30 + 0.10 = 0.40.$$

Answer: **(E)**. Choice (C) is $\mathbb{P}(\text{preventive})$, and choice (A) is $\mathbb{P}(\text{major} \cup \text{orthodontic})$. The constant term -0.05 does not change the method; it is carried into the partition equation and the system stays linear.

Example 4.7: Every enrolled member counted three times

An employer offers four supplementary benefits, W , X , Y and Z . Each employee elects exactly three of the four benefits or elects none. The proportions of employees who elect W , X , Y and Z are 0.33, 0.42, 0.51 and 0.54 respectively. Calculate the probability that a randomly chosen employee elects exactly the benefits W , X and Y .

- (A) 0.06 (B) 0.09 (C) 0.40 (D) 0.46 (E) 0.60

Solution

Let G be the event that the employee elects benefits at all. Every member of G appears in exactly three of the four stated proportions, and nobody outside G appears in any, so

$$0.33 + 0.42 + 0.51 + 0.54 = 1.80 = 3\mathbb{P}(G) \implies \mathbb{P}(G) = 0.60,$$

and $\mathbb{P}(\text{none}) = 0.40$. An employee who elects exactly W , X and Y is an enrolled employee who omits Z . The enrolled employees split into four disjoint cells by which single benefit they omit, and everyone in G who does not omit Z holds Z , so

$$\mathbb{P}(\text{elects } W, X, Y \text{ only}) = \mathbb{P}(G) - \mathbb{P}(Z) = 0.60 - 0.54 = 0.06.$$

Answer: **(A)**. As a check, the other three cells are $0.60 - 0.33 = 0.27$, $0.60 - 0.42 = 0.18$ and $0.60 - 0.51 = 0.09$, and $0.27 + 0.18 + 0.09 + 0.06 = 0.60$. Choice (C) is the probability of electing nothing, and (D) is $1 - \mathbb{P}(Z)$, which wrongly treats “omits Z ” as “does not hold Z ” for the whole population.

Example 4.8: Six ordered levels, four overlapping statements

Each patient discharged from a hospital is assigned exactly one severity level: 0, 1, 2, 3, 4 or 5. Hospital records show that 60% of patients have level 2 or lower, 75% have level 1 or higher, 85% have a level other than 2, and 45% have level 1, 4 or 5. Calculate the probability that a randomly selected patient has severity level 4 or higher.

- (A) 0.15 (B) 0.20 (C) 0.25 (D) 0.30 (E) 0.40

Solution

Let p_i be the probability of level i . The six cells partition the sample space and the statements are

$$\begin{aligned} p_0 + p_1 + p_2 &= 0.60, & p_1 + p_2 + p_3 + p_4 + p_5 &= 0.75, \\ p_2 &= 1 - 0.85 = 0.15, & p_1 + p_4 + p_5 &= 0.45. \end{aligned}$$

The second statement is the complement of level 0, so $p_0 = 1 - 0.75 = 0.25$. The first statement then gives $p_1 = 0.60 - 0.25 - 0.15 = 0.20$, and the fourth gives

$$p_4 + p_5 = 0.45 - p_1 = 0.45 - 0.20 = 0.25.$$

Answer: **(C)**. Levels 4 and 5 are never separated by the data and do not need to be; the block $p_4 + p_5$ is exactly the event asked for. The remaining cell is $p_3 = 1 - 0.25 - 0.20 - 0.15 - 0.25 = 0.15$, which is non-negative, so the data are consistent. Choice (E) is $1 - 0.60$, the probability of level 3 or higher.

4.6 Traps

Trap: Adding without disjointness

$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ is valid only when $A \cap B = \emptyset$. If the problem does not state that the events cannot occur together (“exactly one of”, “never both”, “classified into one of the following categories”), use the general rule $\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. A quick sanity check: if the plain sum exceeds 1, the events cannot be disjoint. If the sum is at most 1, the events might still overlap; only the problem statement or a given intersection settles it.

Trap: Disjoint is not independent

Students frequently treat “mutually exclusive” as a synonym for “independent”. It is close to the opposite. Disjoint events with positive probability are always dependent, because $\mathbb{P}(A \cap B) = 0$ while $\mathbb{P}(A)\mathbb{P}(B) > 0$. When a question says “mutually exclusive”, set every intersection to 0; when it says “independent”, set every intersection to the product of the marginals. Never do both.

Trap: Conditioning on a disjoint event

If A and B are mutually exclusive and $\mathbb{P}(B) > 0$ then $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B) = 0$. Likewise $\mathbb{P}(A \cup B^c) = \mathbb{P}(B^c)$ because $A \subseteq B^c$. Answer choices in such questions are built from the numbers $\mathbb{P}(A)$ and $\mathbb{P}(B)$ to catch anyone who computes $\mathbb{P}(A)/\mathbb{P}(B)$ or $\mathbb{P}(A) + \mathbb{P}(B^c)$ mechanically.

Trap: “Exactly” versus “at least”

“Exactly one” excludes the intersection; “at least one” includes it. For two events the difference is exactly $\mathbb{P}(A \cap B)$, and the exam usually lists both values among the choices. For three events, “exactly two” is $S_2 - 3S_3$, not S_2 : every pairwise intersection already contains the triple intersection, so S_2 counts the triple cell three times. “At least two” is $S_2 - 2S_3$, which keeps the triple cell once.

Trap: Overlapping proportions are not a partition

When a question lists proportions holding coverages A , B and C that sum to more than 1, the proportions overlap and cannot be added to get “at least one”. If the question also says each person holds exactly two (or exactly k) of the coverages or none, do not reach for inclusion–exclusion with unknown intersections; the sum of the proportions is exactly k times the probability of holding any coverage, and “none” follows at once. The distractor $1 - (\text{sum of proportions})$ is negative or meaningless in such questions, and the distractor $1 - \mathbb{P}(A)$ for “does not hold A ” is not the same as “holds the others but not A ”; the latter is $\mathbb{P}(G) - \mathbb{P}(A)$ where G is the enrolled group.

Trap: Reading cumulative statements

“Stage 2 or lower” includes stage 2; “stage 1 or higher” includes stage 1; “other than stage 2” is the complement of a single cell. Write every statement as an explicit sum of cell probabilities before doing any arithmetic. A ratio statement inside an ordered partition (“level 3 is twice as likely as level 4”) is one more linear equation, not a conditional probability.

4.7 Practice problems

Practice 4.1

Claims on a commercial property policy fall into exactly one of four bands. The probability a claim is under 500 is 0.35, from 500 to under 2,000 is 0.40, and 10,000 or more is 0.05. Calculate the probability that a claim is at least 2,000.

- (A) 0.45 (B) 0.20 (C) 0.05 (D) 0.25 (E) 0.30

Practice 4.2

For a randomly selected policyholder, the probability of filing a medical claim in a year is 0.30, the probability of filing a prescription-drug claim is 0.50, and the probability of filing at least one of the two is 0.65. Calculate the probability that the policyholder files exactly one of the two types of claim.

- (A) 0.50 (B) 0.35 (C) 0.65 (D) 0.45 (E) 0.55

Practice 4.3

Each policy in a portfolio is in exactly one of three risk classes: A , B or C , or is unclassified. Let A , B and C also denote the events that a randomly chosen policy is in the respective class. You are given $\mathbb{P}(A) = 3\mathbb{P}(C)$, $\mathbb{P}(B) = 2\mathbb{P}(C)$ and $\mathbb{P}(A \cup B \cup C) = 0.90$. Calculate $\mathbb{P}(A \cup C)$.

- (A) 0.55 (B) 0.45 (C) 0.70 (D) 0.75 (E) 0.60

Practice 4.4

A survey of policyholders finds that 50% carry collision coverage (A), 40% carry comprehensive coverage (B) and 30% carry rental reimbursement (C). Also, 20% carry both A and B , 15% carry both A and C , 10% carry both B and C , and 5% carry all three. Calculate the probability that a randomly selected policyholder carries exactly two of the three coverages.

- (A) 0.45 (B) 0.30 (C) 0.35 (D) 0.25 (E) 0.40

Practice 4.5

For a randomly selected insured, let A be the event of a claim on a first-party coverage and B the event of a claim on a third-party coverage. You are given $\mathbb{P}(A \cap B) = 0.12$, $\mathbb{P}(A \cap B^c) = 0.28$ and $\mathbb{P}(A^c \cap B) = 0.18$. Calculate the probability that the insured has no claim on either coverage.

- (A) 0.60 (B) 0.30 (C) 0.40 (D) 0.42 (E) 0.58

Practice 4.6

Events A and B are mutually exclusive with $\mathbb{P}(A) = 0.30$ and $\mathbb{P}(B) = 0.50$. Calculate $\mathbb{P}(A \cup B^c)$.

- (A) 0.50 (B) 0.80 (C) 0.30 (D) 0.20 (E) 0.65

Practice 4.7

A pet insurer writes policies on dogs, cats, birds and exotic pets only, and each policy covers exactly one kind of animal. The number of cat policies is one-half the number of dog and bird policies combined; four times as many dog policies are written as bird policies; and the proportion of exotic-pet policies is 0.02 less than the proportion of bird policies. Calculate the probability that the next policy written covers a cat or a bird.

- (A) 0.30 (B) 0.40 (C) 0.42 (D) 0.58 (E) 0.60

Practice 4.8

Members of a group plan may enrol in exactly two of the optional riders A , B and C , in all three, or in none. The proportions of members enrolled in A , B and C are 0.35, 0.40 and 0.45 respectively, and 0.12 of the members are enrolled in all three. Calculate the probability that a randomly chosen member is enrolled in no rider.

- (A) 0.40 (B) 0.42 (C) 0.54 (D) 0.58 (E) 0.46

Practice 4.9

A workers' compensation insurer assigns each claim exactly one severity code from 1 to 6. You are given that 55% of claims have severity 3 or lower, 88% have severity 2 or higher, 40% have severity 3 or 4, and a claim is equally likely to have severity 3 as severity 4. Calculate the probability that a randomly selected claim has severity 2.

- (A) 0.20 (B) 0.23 (C) 0.25 (D) 0.35 (E) 0.43

4.8 Answers to practice problems

1. **(D)** The bands partition the sample space, so the missing band (2,000 to under 10,000) has probability $1 - 0.35 - 0.40 - 0.05 = 0.20$. At least 2,000 is the union of the two disjoint top bands: $0.20 + 0.05 = 0.25$.
2. **(A)** $\mathbb{P}(A \cap B) = 0.30 + 0.50 - 0.65 = 0.15$. Exactly one is $\mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = 0.65 - 0.15 = 0.50$, or $0.30 + 0.50 - 2(0.15) = 0.50$.
3. **(E)** The classes are disjoint, so $3c + 2c + c = 6c = 0.90$ and $c = 0.15$. Then $\mathbb{P}(A \cup C) = 3c + c = 4c = 0.60$.
4. **(B)** $S_2 = 0.20 + 0.15 + 0.10 = 0.45$, $S_3 = 0.05$, and exactly two is $S_2 - 3S_3 = 0.45 - 0.15 = 0.30$. Check: exactly one is $1.20 - 0.90 + 0.15 = 0.45$, and $0.45 + 0.30 + 0.05 = 0.80 = S_1 - S_2 + S_3$.
5. **(D)** The three given cells are disjoint and make up $A \cup B$, so $\mathbb{P}(A \cup B) = 0.12 + 0.28 + 0.18 = 0.58$ and the answer is $1 - 0.58 = 0.42$. (Here $\mathbb{P}(A) = 0.40$, $\mathbb{P}(B) = 0.30$ and $\mathbb{P}(A)\mathbb{P}(B) = 0.12 = \mathbb{P}(A \cap B)$, so these events happen to be independent, and they are not disjoint.)
6. **(A)** Since $A \cap B = \emptyset$, $A \subseteq B^c$, so $A \cup B^c = B^c$ and the probability is $1 - 0.50 = 0.50$. Choice (B) is the wrong result $\mathbb{P}(A) + \mathbb{P}(B^c) = 0.80$, which double counts A .
7. **(C)** Let b be the bird probability. Then dog = $4b$, cat = $\frac{1}{2}(4b + b) = 2.5b$ and exotic = $b - 0.02$. The partition equation $4b + 2.5b + b + b - 0.02 = 8.5b - 0.02 = 1$ gives $b = 0.12$, so the cells are 0.48, 0.30, 0.12 and 0.10. Cat or bird is $0.30 + 0.12 = 0.42$.
8. **(E)** Each member enrolled in exactly two riders is counted twice in the three proportions and each member in all three is counted three times, so $0.35 + 0.40 + 0.45 = 1.20 =$

$2\mathbb{P}(\text{exactly two}) + 3(0.12)$, giving $\mathbb{P}(\text{exactly two}) = 0.42$. Enrolled at all is $0.42 + 0.12 = 0.54$ and none is $1 - 0.54 = 0.46$. Check by cells: with x, y, z the exactly-two cells, $x + y = 0.23$, $x + z = 0.28$, $y + z = 0.33$, so $x = 0.09$, $y = 0.14$, $z = 0.19$, which sum to 0.42.

9. **(B)** From “88% severity 2 or higher”, $p_1 = 0.12$. From $p_3 + p_4 = 0.40$ and $p_3 = p_4$, $p_3 = 0.20$. Then $p_2 = 0.55 - p_1 - p_3 = 0.55 - 0.12 - 0.20 = 0.23$. Codes 5 and 6 together carry $1 - 0.55 - 0.20 = 0.25$ and are never separated. Choice (E) is $0.55 - 0.12$, which forgets to remove p_3 ; choice (D) is $0.55 - 0.20$, which forgets p_1 .

4.9 Sample exam cross-reference

- Bank 1, Q14: every covered employee counted in exactly two of three proportions, so the sum of the proportions is twice the probability of holding coverage; none is one minus that (key fact “Counting each member k times”, Example on four benefits).
- Bank 1, Q92: three disjoint categories given by ratios, one of them relative to the other two combined; write all cells as multiples of a base cell and use the partition equation (key fact “Ratio-specified partition”, Example on dental claims).
- Bank 1, Q147: five ordered cells with overlapping cumulative statements; write each statement as a sum of cells, use the complement for “or higher” and “other than”, and subtract (key fact “Cumulative statements over an ordered partition”, Example on severity levels).

4.10 One-page summary

- Mutually exclusive (disjoint): $A \cap B = \emptyset$, so $\mathbb{P}(A \cap B) = 0$ and $\mathbb{P}(A | B) = 0$. Pairwise disjoint: no two of the family overlap. Partition: pairwise disjoint and exhaustive; probabilities sum to 1.
- Addition rule: for disjoint events, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$, and $\mathbb{P}(\bigcup A_i) = \sum \mathbb{P}(A_i)$ for any finite or countable pairwise disjoint family. This is the only situation in which probabilities add directly.
- General rule: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. If the given union equals the sum of the marginals, the events are disjoint; if it is smaller, the shortfall is $\mathbb{P}(A \cap B)$.
- Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. A partition with one probability missing: subtract the others from 1.
- Decomposition identities: $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$; $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B)$; $\mathbb{P}(A \cup B) = \mathbb{P}(A \cap B^c) + \mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B)$.
- Disjoint with positive probabilities implies dependent: $\mathbb{P}(A \cap B) = 0 \neq \mathbb{P}(A)\mathbb{P}(B)$. Mutually exclusive, independent and exhaustive are three unrelated properties.
- Two events: exactly one = $\mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(AB)$; at least one = $\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(AB)$; at most one = $1 - \mathbb{P}(AB)$; none = $1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(AB)$.
- Three events with S_1, S_2, S_3 : exactly three = S_3 ; exactly two = $S_2 - 3S_3$; exactly one = $S_1 - 2S_2 + 3S_3$; at least one = $S_1 - S_2 + S_3$; at least two = $S_2 - 2S_3$; none = $1 - S_1 + S_2 - S_3$.
- Technique: rewrite the target event as a union of Venn-diagram cells, compute each cell from marginals and intersections, then add. Check that exactly one + exactly two + exactly three = at least one.
- If A and B are disjoint then $A \subseteq B^c$, so $\mathbb{P}(A \cup B^c) = \mathbb{P}(B^c)$ and $\mathbb{P}(A \cap B^c) = \mathbb{P}(A)$.
- Ratio-specified partition: write every cell as a multiple of one base cell t (“ X is one-fourth of Y and Z combined” means $p_X = \frac{1}{4}(p_Y + p_Z)$), set the sum of the multiples times t equal to 1, solve for t . Absolute statements (“0.05 less than”, “10% of claims”) add a constant; the system stays linear.

- Ordered partition with cumulative statements: “ k or higher” = $1 - (p_0 + \cdots + p_{k-1})$; “other than k ” = $1 - p_k$; “ k or lower” + “ k or higher” = $1 + p_k$; the difference of two cumulative sums is the block of cells between them. Cells never separated by the data are carried as a block.
- Counting k times: if every member of G lies in exactly k of $A_1, \dots, A_m$ and nobody else lies in any, $\sum \mathbb{P}(A_i) = k\mathbb{P}(G)$. In general $\sum_i \mathbb{P}(A_i) = \sum_j j \mathbb{P}(\text{exactly } j)$; for three events $S_1 = \mathbb{P}(\text{exactly one}) + 2\mathbb{P}(\text{exactly two}) + 3S_3$. “Exactly two or none” with three coverages: $\mathbb{P}(\text{none}) = 1 - S_1/2$.

5 Outcome (e): Addition and multiplication rules

5.1 What the outcome asks

The exam tests two rules and the habit of choosing the right one. The addition rule (inclusion-exclusion) gives the probability that at least one of several events occurs; the exam states it for two or three events, usually with one quantity missing, and asks for that quantity or for a derived one such as “neither” or “exactly one”. The multiplication rule (chain rule) gives the probability that several events occur in sequence, each conditioned on the ones before it; the exam uses it for draws without replacement and for claim histories over successive years. Many questions combine the two: split the outcome into disjoint cases with the addition rule, then compute each case with the multiplication rule. A tree diagram is the bookkeeping device for the multiplication rule, and its branches are the cases that the addition rule adds up.

5.2 Addition rule

Key fact: General addition rule, two events

For any events A and B ,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

The subtraction removes the double count of outcomes that lie in both events. The rule holds whether or not A and B are independent or disjoint.

Key fact: General addition rule, three events

For any events A, B, C ,

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

Sign pattern: singles added, pairs subtracted, triple added. Seven terms on the right; the exam typically gives six of them (or the union) and asks for the seventh.

Special cases reduce the number of terms.

- **Disjoint (mutually exclusive).** If $A \cap B = \emptyset$ then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$. For three pairwise disjoint events the union is the sum of the three probabilities.
- **Independent.** If A and B are independent then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, so $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A)\mathbb{P}(B)$.
- **Complement shortcut.** By De Morgan, $(A \cup B)^c = A^c \cap B^c$, so $\mathbb{P}(A \cup B) = 1 - \mathbb{P}(A^c \cap B^c)$ and $\mathbb{P}(\text{neither}) = 1 - \mathbb{P}(A \cup B)$. For n events, $\mathbb{P}(\text{at least one}) = 1 - \mathbb{P}(\text{none})$; when the events are independent this is $1 - \prod_{i=1}^n (1 - \mathbb{P}(A_i))$.
- **Bound.** Without any information about intersections, $\mathbb{P}(A_1 \cup \dots \cup A_n) \leq \sum_{i=1}^n \mathbb{P}(A_i)$ (Boole's inequality), with equality only when the events are pairwise disjoint.

Key fact: Exactly one of two events

$$\mathbb{P}(\text{exactly one of } A, B) = \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B) = \mathbb{P}(A \cup B) - \mathbb{P}(A \cap B).$$

Given any two of $\mathbb{P}(A) + \mathbb{P}(B)$, $\mathbb{P}(A \cap B)$, $\mathbb{P}(A \cup B)$, $\mathbb{P}(\text{exactly one})$, the remaining ones follow.

Key fact: Counting identity for three events

Write e_1, e_2, e_3 for the probabilities that exactly one, exactly two, exactly three of A, B, C occur. Then

$$\mathbb{P}(A \cup B \cup C) = e_1 + e_2 + e_3, \quad \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) = e_1 + 2e_2 + 3e_3.$$

The second identity holds because an outcome in exactly k of the events is counted k times in the sum of the marginals. The exam uses these two linear equations to recover e_2 or e_3 from “at least one” and “exactly one” data.

Solving-for-unknown patterns. The exam rarely asks for the union directly. The standard forms are:

1. Given $\mathbb{P}(A \cup B)$, $\mathbb{P}(A)$, $\mathbb{P}(B)$: the intersection is $\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)$.
2. Given $\mathbb{P}(A^c \cap B^c)$ (“neither”): first $\mathbb{P}(A \cup B) = 1 - \mathbb{P}(A^c \cap B^c)$, then proceed as in pattern 1.
3. Given $\mathbb{P}(\text{exactly one})$ and the two marginals: $\mathbb{P}(A \cap B) = \frac{1}{2}[\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(\text{exactly one})]$.
4. Three events with six of the seven inclusion-exclusion terms and either the union or “none” given: solve the seven-term equation for the missing term.
5. Three events with “at least one” and “exactly one” given together with the sum of the marginals: solve the two linear equations in e_2 and e_3 .

Key fact: Difference rule and the four Venn regions

Two events cut the sample space into four disjoint regions: $A \cap B$ (both), $A \cap B^c$ (A only), $A^c \cap B$ (B only), and $A^c \cap B^c$ (neither). Their probabilities sum to 1. Each marginal is the sum of the two regions inside it, which gives the difference rule

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B), \quad \mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Any two-event question is a set of linear equations in these four region probabilities. Given any three independent facts about them (marginals, “both”, “neither”, “ A only”, a union, or “exactly one”), fill in the four regions and read off the answer. “Neither” is often the fastest route: $\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A \text{ only}) - \mathbb{P}(B \text{ only}) - \mathbb{P}(\text{both})$.

Key fact: A relation between the marginals replaces a marginal

When the question gives “both” and “neither” but neither marginal, it supplies instead a relation between them: a difference, $\mathbb{P}(A) = \mathbb{P}(B) + d$, or a ratio, $\mathbb{P}(A) = k\mathbb{P}(B)$. Write

$x = \mathbb{P}(B)$, substitute into inclusion-exclusion, and solve the single linear equation

$$1 - \mathbb{P}(\text{neither}) = \mathbb{P}(A \cup B) = \mathbb{P}(A) + x - \mathbb{P}(A \cap B),$$

with $\mathbb{P}(A) = x + d$ or $\mathbb{P}(A) = kx$. A difference gives $2x + d - \mathbb{P}(A \cap B)$ on the right; a ratio gives $(k+1)x - \mathbb{P}(A \cap B)$. Once x is known, the other marginal and the “only” regions follow from the difference rule.

Key fact: Data stated in complements

The exam often states every probability as “no A ”, “no B ”, and “neither”. Convert first: $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$, $\mathbb{P}(B) = 1 - \mathbb{P}(B^c)$, and by De Morgan $\mathbb{P}(A \cup B) = 1 - \mathbb{P}(A^c \cap B^c)$. Then inclusion-exclusion gives

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) = \mathbb{P}(A^c \cap B^c) - \mathbb{P}(A^c) - \mathbb{P}(B^c) + 1.$$

“At least one A and at least one B ” is $\mathbb{P}(A \cap B)$ when A is “at least one A ”; the phrase “at least one” inside each event does not change the algebra.

5.3 Multiplication rule

Key fact: General multiplication rule, two events

For any events A and B with positive probability,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B \mid A) = \mathbb{P}(B) \mathbb{P}(A \mid B).$$

This is the definition of conditional probability rearranged. Use whichever ordering matches the information given.

Key fact: Chain rule, three or more events

$$\mathbb{P}(A_1 \cap A_2 \cap A_3) = \mathbb{P}(A_1) \mathbb{P}(A_2 \mid A_1) \mathbb{P}(A_3 \mid A_1 \cap A_2),$$

and in general $\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1) \prod_{k=2}^n \mathbb{P}(A_k \mid A_1 \cap \cdots \cap A_{k-1})$. Each factor is conditioned on everything that happened before it.

Key fact: Independence collapses the chain

If $A_1, \dots, A_n$ are mutually independent, every conditional factor equals the marginal and $\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1) \mathbb{P}(A_2) \cdots \mathbb{P}(A_n)$. The product of marginals is a special case of the chain rule, not a separate rule. Multiply unconditional probabilities only when the problem states independence or the setup implies it (sampling with replacement, separate policyholders described as independent).

Sequential sampling without replacement. When items are drawn one at a time without replacement, the composition of the pool changes after each draw, so the draws are dependent. The chain rule handles this directly: the k th factor is the number of favourable items remaining

divided by the number of items remaining. From a pool of N items of which m are of type X ,

$$\mathbb{P}(\text{first two both type } X) = \frac{m}{N} \cdot \frac{m-1}{N-1}, \quad \mathbb{P}(\text{first is } X, \text{ second is not}) = \frac{m}{N} \cdot \frac{N-m}{N-1}.$$

Key fact: Symmetry of draws without replacement

The unconditional probability that the k th draw is of type X equals the probability that the first draw is of type X , namely m/N , for every $k \leq N$. Every ordering of the N items is equally likely, so the item in position k is a uniformly random item from the pool. Once the results of earlier draws are known, the conditional probability changes.

Tree diagrams. A tree diagram lists the first event's outcomes as branches, then each second event's outcomes as sub-branches, and so on. Each branch is labelled with a conditional probability given the path that leads to it; the probability of a complete path is the product of its branch labels, which is exactly the chain rule. Because the paths are disjoint, the probability of any event is the sum of the probabilities of the paths in it, which is the addition rule for disjoint cases. As a table, the tree for a two-year claim history with $\mathbb{P}(C_1) = 0.20$, $\mathbb{P}(C_2 | C_1) = 0.35$, $\mathbb{P}(C_2 | C_1^c) = 0.15$ has one row per path:

Year 1 (branch)	Year 2 (branch)	Product	Path probability
Claim (0.20)	Claim (0.35)	0.20×0.35	0.070
Claim (0.20)	No claim (0.65)	0.20×0.65	0.130
No claim (0.80)	Claim (0.15)	0.80×0.15	0.120
No claim (0.80)	No claim (0.85)	0.80×0.85	0.680
Total			1.000

The path probabilities sum to 1.000, which is the standard check. Any event is a set of rows and its probability is the sum of those rows.

5.4 Combining the rules

The most common exam structure is a two-stage experiment: a classification or a first draw, then a second outcome whose probability depends on the first. The method is fixed: list the disjoint cases for the first stage (the branches of the tree), which must cover every possibility without overlapping; for each case compute $\mathbb{P}(\text{case}) \times \mathbb{P}(\text{target} | \text{case})$ with the multiplication rule; add the results with the addition rule for disjoint events. Written out for cases $B_1, \dots, B_n$ that partition the sample space and a target event A ,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(B_i) \mathbb{P}(A | B_i).$$

This is the law of total probability. Outcome (g) develops it fully and adds Bayes' theorem for reversing the conditioning; for this outcome it is enough to recognise that "split into cases, multiply along each path, add the paths" is the same computation. Example: 70% of policyholders are low risk with claim probability 0.10 and 30% are high risk with claim probability 0.30; the probability that a randomly selected policyholder has a claim is $0.70 \times 0.10 + 0.30 \times 0.30 = 0.07 + 0.09 = 0.16$.

A question that gives conditional probabilities along a chain and asks for "at least one" over the chain is usually fastest through the "none" path: multiply along it and subtract from 1.

Key fact: Group rates over a two-event Venn

When each Venn region has its own rate for a second event R (a renewal, a claim, a lapse), the overall rate is a weighted sum over the regions. The region sizes come from the difference rule, and the multiplication rule applies region by region:

$$\begin{aligned}\mathbb{P}(R) = & \mathbb{P}(A \text{ only}) \mathbb{P}(R \mid A \text{ only}) + \mathbb{P}(B \text{ only}) \mathbb{P}(R \mid B \text{ only}) \\ & + \mathbb{P}(\text{both}) \mathbb{P}(R \mid \text{both}) + \mathbb{P}(\text{neither}) \mathbb{P}(R \mid \text{neither}).\end{aligned}$$

Regions with no stated rate contribute zero only when the question makes that explicit (a policyholder with no policy cannot renew). If the overall rate is given and one region's rate is missing, the same equation is solved for the missing rate.

Key fact: Conditional on one group, marginal overall

A question may give $\mathbb{P}(M)$, the marginal $\mathbb{P}(S)$, and a conditional $\mathbb{P}(S \mid M)$ on one group only. The multiplication rule gives the joint on that group, $\mathbb{P}(S \cap M) = \mathbb{P}(M)\mathbb{P}(S \mid M)$, and the difference rule gives the joint on the complementary group, $\mathbb{P}(S \cap M^c) = \mathbb{P}(S) - \mathbb{P}(M)\mathbb{P}(S \mid M)$. The four-region table then completes:

$$\mathbb{P}(M^c \cap S^c) = 1 - \mathbb{P}(M) - \mathbb{P}(S) + \mathbb{P}(M)\mathbb{P}(S \mid M).$$

If the conditional is stated for the complement of S (“of those in M , 70% have no claim”), take $\mathbb{P}(S \mid M) = 1 - 0.70$ before multiplying.

5.5 Worked examples

Example 5.1: Three coverages, solve for the triple intersection

An insurer's policyholders may hold auto, homeowners, and life coverage. Among them, 60% hold auto, 45% hold homeowners, 30% hold life, 25% hold both auto and homeowners, 15% hold both auto and life, 10% hold both homeowners and life, and 5% hold none of the three. Calculate the percentage of policyholders who hold all three coverages.

(A) 10% (B) 5% (C) 8% (D) 12% (E) 15%

Solution

Let A , H , L be the events that a policyholder holds auto, homeowners, life. The complement of the union is “none”, so $\mathbb{P}(A \cup H \cup L) = 1 - 0.05 = 0.95$. Three-event inclusion-exclusion with $x = \mathbb{P}(A \cap H \cap L)$:

$$0.95 = 0.60 + 0.45 + 0.30 - 0.25 - 0.15 - 0.10 + x = 0.85 + x,$$

so $x = 0.10$. Answer **(A)**.

Example 5.2: Claims in two successive years

For a policyholder, the probability of at least one claim in year 1 is 0.20. If the policyholder has a claim in year 1, the probability of a claim in year 2 is 0.35; if there is no claim in year 1, the probability of a claim in year 2 is 0.15. Calculate the probability that the policyholder has claims in both years or in neither year.

- (A) 0.68 (B) 0.70 (C) 0.80 (D) 0.75 (E) 0.93

Solution

Let C_1, C_2 be the claim events. By the multiplication rule,

$$\begin{aligned}\mathbb{P}(C_1 \cap C_2) &= \mathbb{P}(C_1)\mathbb{P}(C_2 \mid C_1) = 0.20 \times 0.35 = 0.07, \\ \mathbb{P}(C_1^c \cap C_2^c) &= \mathbb{P}(C_1^c)\mathbb{P}(C_2^c \mid C_1^c) = 0.80 \times 0.85 = 0.68.\end{aligned}$$

“Both” and “neither” are disjoint, so the addition rule gives $0.07 + 0.68 = 0.75$. Answer **(D)**. As a check, “exactly one year” is $0.20 \times 0.65 + 0.80 \times 0.15 = 0.25$ and $0.75 + 0.25 = 1$.

Example 5.3: Two policies of the same type

A portfolio contains 20 policies: 10 auto, 6 homeowners, and 4 life. An auditor selects two policies at random without replacement. Calculate the probability that the two selected policies are of the same type.

- (A) 0.30 (B) 0.35 (C) 0.33 (D) 0.38 (E) 0.40

Solution

“Same type” splits into three disjoint cases: both auto, both homeowners, both life. Each case is a chain-rule product, and the cases are added:

$$\frac{10}{20} \cdot \frac{9}{19} + \frac{6}{20} \cdot \frac{5}{19} + \frac{4}{20} \cdot \frac{3}{19} = \frac{90 + 30 + 12}{380} = \frac{132}{380} = \frac{33}{95} \approx 0.3474.$$

Answer **(B)**. Counting unordered pairs gives the same value, $(45 + 15 + 6)/190$.

Example 5.4: Type of the second policy drawn

From the portfolio of the previous example (10 auto, 6 homeowners, 4 life), the auditor selects two policies at random without replacement. Calculate the probability that the second policy selected is a life policy.

- (A) 0.15 (B) 0.16 (C) 0.19 (D) 0.21 (E) 0.20

Solution

Condition on the first draw. Either the first policy is life (probability $4/20$, leaving 3 life among 19) or it is not (probability $16/20$, leaving 4 life among 19):

$$\mathbb{P}(\text{second is life}) = \frac{4}{20} \cdot \frac{3}{19} + \frac{16}{20} \cdot \frac{4}{19} = \frac{12 + 64}{380} = \frac{76}{380} = \frac{4}{20} = 0.20.$$

Answer **(E)**. The result equals the first-draw probability $4/20$. This is not a coincidence: if nothing is known about the first draw, the second policy is a uniformly random policy from the 20, so its type distribution is the same as the pool's. Algebraically, $\frac{m}{N} \cdot \frac{m-1}{N-1} + \frac{N-m}{N} \cdot \frac{m}{N-1} = \frac{m(N-1)}{N(N-1)} = \frac{m}{N}$. Recognising the symmetry saves the computation; the computation serves as a check.

Example 5.5: Three coverages from “at least one” and “exactly one”

A survey of an insurer's policyholders finds that 60% hold auto coverage, 35% hold homeowners coverage, and 20% hold life coverage. Also, 80% hold at least one of the three coverages and 50% hold exactly one. Calculate the percentage of policyholders who hold all three coverages.

- (A) 0% (B) 10% (C) 5% (D) 15% (E) 25%

Solution

Let e_1, e_2, e_3 be the proportions holding exactly one, two, three coverages. “At least one” gives $e_1 + e_2 + e_3 = 0.80$ and $e_1 = 0.50$, so $e_2 + e_3 = 0.30$. The sum of the marginals counts each policyholder once per coverage held:

$$e_1 + 2e_2 + 3e_3 = 0.60 + 0.35 + 0.20 = 1.15 \implies 2e_2 + 3e_3 = 0.65.$$

Subtract twice the first equation from the second: $e_3 = 0.65 - 2(0.30) = 0.05$, and then $e_2 = 0.25$. Answer **(C)**. The individual pairwise intersections are neither needed nor determined by the data; only their sum $e_2 + 3e_3 = 0.40$ is.

Example 5.6: Complement of a union from count data

Of 200 liability claims closed last year, 90 involved property damage, 60 involved bodily injury, and 120 involved property damage or bodily injury or both. A claim is selected at random. Calculate the probability that the selected claim involved neither property damage nor bodily injury.

- (A) 0.40 (B) 0.15 (C) 0.25 (D) 0.45 (E) 0.60

Solution

Work in counts, then convert. The union has 120 claims, so its complement has $200 - 120 = 80$ claims and the probability is $80/200 = 0.40$. Answer **(A)**. The intersection is $90 + 60 - 120 = 30$ claims, and “exactly one” is $120 - 30 = 90$ claims. The 90 and 60 are counts, not percentages; treating them as proportions produces a nonsense answer.

Example 5.7: Chain rule over three years

A policyholder's claim history is modelled year by year. The probability of no claim in year 1 is 0.80. Given no claim in year 1, the probability of no claim in year 2 is 0.85. Given no claim in years 1 and 2, the probability of no claim in year 3 is 0.90. Calculate the probability that the policyholder has at least one claim during the three years.

- (A) 0.29 (B) 0.35 (C) 0.45 (D) 0.39 (E) 0.55

Solution

Let N_k be the event of no claim in year k . By the chain rule,

$$\mathbb{P}(N_1 \cap N_2 \cap N_3) = \mathbb{P}(N_1)\mathbb{P}(N_2 \mid N_1)\mathbb{P}(N_3 \mid N_1 \cap N_2) = 0.80 \times 0.85 \times 0.90 = 0.612.$$

At least one claim is the complement: $1 - 0.612 = 0.388$. Answer **(D)**. The problem gives no conditional probabilities after a claim, so the paths containing a claim cannot be computed and the complement route is the only one available.

Example 5.8: Exactly one claim type from complement data

For a dental plan member in a given year, the probability of no preventive claim is 0.55, the probability of no restorative claim is 0.72, and the probability of neither a preventive claim nor a restorative claim is 0.42. Calculate the probability that the member has exactly one of the two claim types during the year.

- (A) 0.15 (B) 0.43 (C) 0.45 (D) 0.58 (E) 0.73

Solution

Let P and R be the events of at least one preventive claim and at least one restorative claim. Convert every given quantity out of its complement: $\mathbb{P}(P) = 1 - 0.55 = 0.45$, $\mathbb{P}(R) = 1 - 0.72 = 0.28$, and by De Morgan $\mathbb{P}(P \cup R) = 1 - 0.42 = 0.58$. Inclusion-exclusion gives the intersection,

$$\mathbb{P}(P \cap R) = 0.45 + 0.28 - 0.58 = 0.15,$$

and exactly one is the union less the intersection, $0.58 - 0.15 = 0.43$. Answer **(B)**. The regions are P only 0.30, R only 0.13, both 0.15, neither 0.42; they sum to 1. Choice (A) is the intersection, which answers “both” rather than “exactly one”.

Example 5.9: Ratio between the marginals

Among an auto insurer’s policyholders, 20% carry both roadside assistance and rental reimbursement, and 30% carry neither. A policyholder is 1.5 times as likely to carry roadside assistance as to carry rental reimbursement. Calculate the probability that a randomly selected policyholder carries roadside assistance but not rental reimbursement.

- (A) 0.16 (B) 0.30 (C) 0.34 (D) 0.36 (E) 0.54

Solution

Let R be roadside assistance and E be rental reimbursement, and let $x = \mathbb{P}(E)$, so $\mathbb{P}(R) = 1.5x$. “Neither” gives the union, $\mathbb{P}(R \cup E) = 1 - 0.30 = 0.70$, and inclusion-exclusion becomes one equation in x :

$$0.70 = 1.5x + x - 0.20 = 2.5x - 0.20 \implies x = 0.36, \quad \mathbb{P}(R) = 0.54.$$

The difference rule gives roadside only: $\mathbb{P}(R \cap E^c) = 0.54 - 0.20 = 0.34$. Answer **(C)**. Choice (D) is $\mathbb{P}(E)$ and choice (E) is $\mathbb{P}(R)$; both are intermediate values, not the region asked for. Check: $0.34 + 0.16 + 0.20 + 0.30 = 1$.

Example 5.10: Neither claim type from an “only” region

For a class of auto policyholders, the probability of filing a collision claim in a year is 0.09 and the probability of filing a comprehensive claim is 0.05. The probability of filing a comprehensive claim but not a collision claim is 0.03. Calculate the probability that a randomly selected policyholder in this class files no claim of either type during the year.

- (A) 0.86 (B) 0.87 (C) 0.89 (D) 0.91 (E) 0.88

Solution

Let C be a collision claim and M a comprehensive claim. The “only” region given is $\mathbb{P}(M \cap C^c) = 0.03$, so the difference rule on M gives the intersection, $\mathbb{P}(C \cap M) = 0.05 - 0.03 = 0.02$, and the difference rule on C gives the other “only” region, $\mathbb{P}(C \cap M^c) = 0.09 - 0.02 = 0.07$. The four regions sum to 1:

$$\mathbb{P}(\text{neither}) = 1 - 0.07 - 0.02 - 0.03 = 0.88.$$

Answer **(E)**. Equivalently, $\mathbb{P}(C \cup M) = 0.09 + 0.05 - 0.02 = 0.12$ and neither is $1 - 0.12 = 0.88$. Choice (A) results from subtracting both marginals without adding back the intersection, $1 - 0.09 - 0.05$.

Example 5.11: Weighted rates over four Venn regions

An insurer’s customers may hold an auto policy, a homeowners policy, both, or neither (customers with neither hold only a life policy). Records show that 60% of customers hold an auto policy, 45% hold a homeowners policy, and 20% hold both. The insurer estimates that next year 30% of auto-only customers, 40% of homeowners-only customers, 70% of customers with both, and 10% of customers with neither will contact the service centre. Calculate the percentage of customers expected to contact the service centre next year.

- (A) 34.0% (B) 36.0% (C) 37.5% (D) 41.5% (E) 45.0%

Solution

Let A and H be the auto and homeowners events. The difference rule sizes the regions: auto only $0.60 - 0.20 = 0.40$, homeowners only $0.45 - 0.20 = 0.25$, both 0.20, and neither $1 - 0.40 - 0.25 - 0.20 = 0.15$. The regions are disjoint and exhaust the customers, so the multiplication rule on each region followed by the addition rule gives

$$0.40(0.30) + 0.25(0.40) + 0.20(0.70) + 0.15(0.10) = 0.120 + 0.100 + 0.140 + 0.015 = 0.375.$$

Answer **(C)**. Weighting the marginals instead of the “only” regions, $0.60(0.30) + 0.45(0.40) + \dots$, counts the “both” customers in three regions and overstates the total.

Example 5.12: Joint on the complementary group from a conditional on one group

An insurer examines its book of personal lines policyholders and finds: (i) every policyholder holds at least one policy; (ii) 60% of policyholders hold more than one policy; (iii) 25% of policyholders filed a claim last year; (iv) of the policyholders who hold more than one policy, 70% filed no claim last year. Calculate the probability that a randomly selected policyholder holds exactly one policy and filed no claim last year.

- (A) 0.07 (B) 0.15 (C) 0.18 (D) 0.28 (E) 0.33

Solution

Let M be “more than one policy” and S be “filed a claim”. By (i), M^c is “exactly one policy”. Statement (iv) is a conditional on M for the complement of S , so $\mathbb{P}(S^c | M) = 1 - 0.70 = 0.30$ and the multiplication rule gives $\mathbb{P}(S^c \cap M) = 0.60(0.30) = 0.18$. The difference rule moves to the other group: $\mathbb{P}(S^c \cap M^c) = 0.25 - 0.18 = 0.07$. The target is the remaining region of M^c ,

$$\mathbb{P}(M^c \cap S^c) = \mathbb{P}(M^c) - \mathbb{P}(S^c \cap M) = 0.40 - 0.07 = 0.33,$$

or, in one line through De Morgan, $1 - \mathbb{P}(M \cup S) = 1 - 0.60 - 0.25 + 0.18 = 0.33$. Answer **(E)**. Choice (D) is 0.40×0.70 , which treats the 70% as if it applied to single-policy holders; the conditional was given for the other group.

5.6 Traps

Trap: Adding without subtracting the intersection

$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ only when A and B are disjoint. Policyholders can hold two coverages and a claim can involve two causes. Unless the question says “mutually exclusive” or the events cannot occur together by construction, subtract $\mathbb{P}(A \cap B)$. A quick check: if $\mathbb{P}(A) + \mathbb{P}(B) > 1$, the events cannot be disjoint.

Trap: Multiplying marginals without independence

$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ requires independence. Draws without replacement, successive years of the same policyholder’s claims, and risk class followed by claim status are all dependent. The correct factor for the second event is the conditional probability given the first. If the question supplies $\mathbb{P}(B | A)$, it is telling you the events are not independent.

Trap: Confusing the joint with the conditional

$\mathbb{P}(A \cap B)$ is the probability that both happen; $\mathbb{P}(B | A)$ is the probability of B in the world where A has already happened. “The probability that a policyholder with a claim in year 1 has a claim in year 2” is $\mathbb{P}(C_2 | C_1)$. “The probability that a policyholder has claims in both years” is $\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2 | C_1)$. The conditional is never smaller than the joint.

Trap: Sign pattern in three-event inclusion-exclusion

Singles are added, pairwise intersections are subtracted, and the triple intersection is added back. The common errors are subtracting the triple intersection or omitting it. Write the full seven-term equation before substituting. Check the result: each pairwise intersection must be at least the triple intersection, and each marginal at least any pairwise intersection containing it.

Trap: Counts, percentages, and the base

Some questions give raw counts (90 of 200 claims), some give percentages of the whole group, and some give percentages of a subgroup (“of those with auto coverage, 40% also hold homeowners”). The third kind is a conditional probability, not a joint probability, and must be multiplied by the subgroup’s proportion before it enters the addition rule. With counts, work in counts and divide by the total once at the end.

5.7 Practice problems**Practice 5.1**

An insurer’s records show that 55% of policyholders are male, 40% are over age 50, and 20% are males over age 50. A policyholder is selected at random. Calculate the probability that the selected policyholder is a female who is age 50 or younger.

- (A) 0.05 (B) 0.25 (C) 0.20 (D) 0.30 (E) 0.45

Practice 5.2

Among an insurer’s policyholders, 50% hold auto coverage, 40% hold homeowners coverage, 30% hold life coverage, 20% hold auto and homeowners, 15% hold auto and life, 10% hold homeowners and life, and 5% hold all three. Calculate the percentage of policyholders who hold none of the three coverages.

- (A) 10% (B) 15% (C) 25% (D) 30% (E) 20%

Practice 5.3

A claims adjuster has 12 open files: 5 auto, 4 homeowners, and 3 life. The adjuster selects three files at random without replacement to review first. Calculate the probability that none of the three selected files is a life file.

- (A) 0.30 (B) 0.35 (C) 0.38 (D) 0.42 (E) 0.45

Practice 5.4

A batch of 25 newly issued policies contains 7 with a data-entry error. A quality reviewer examines the policies one at a time in random order without replacement. Calculate the probability that the third policy examined contains an error.

- (A) 0.28 (B) 0.24 (C) 0.26 (D) 0.30 (E) 0.32

Practice 5.5

For a policyholder, the probability of a claim in year 1 is 0.15. In any year, if the policyholder had a claim in the preceding year, the probability of a claim is 0.40; if the policyholder had no claim in the preceding year, the probability of a claim is 0.10. Calculate the probability that the policyholder has a claim in exactly one of years 1 and 2.

- (A) 0.135 (B) 0.150 (C) 0.200 (D) 0.175 (E) 0.235

Practice 5.6

Among an auto insurer's policyholders, 45% carry collision coverage, 35% carry comprehensive coverage, and 50% carry exactly one of the two. Calculate the percentage of policyholders who carry both collision and comprehensive coverage.

- (A) 5% (B) 15% (C) 10% (D) 20% (E) 30%

Practice 5.7

A life insurer classifies applicants as standard, preferred, or substandard; 60% of applicants are standard, 30% are preferred, and 10% are substandard. The probability of a claim within the first policy year is 0.08 for a standard applicant, 0.03 for a preferred applicant, and 0.20 for a substandard applicant. Calculate the probability that a randomly selected applicant has a claim within the first policy year.

- (A) 0.058 (B) 0.065 (C) 0.085 (D) 0.103 (E) 0.077

Practice 5.8

For a health plan member in a given year, the probability of no outpatient visit is 0.64, the probability of no prescription claim is 0.55, and the probability of neither an outpatient visit nor a prescription claim is 0.40. Calculate the probability that the member has at least one outpatient visit but no prescription claim during the year.

- (A) 0.09 (B) 0.21 (C) 0.24 (D) 0.15 (E) 0.39

Practice 5.9

Among the employees of a company, 12% are enrolled in both the dental plan and the vision plan, and 28% are enrolled in neither. An employee is twice as likely to be enrolled in the dental plan as in the vision plan. Calculate the percentage of employees enrolled in the vision plan but not the dental plan.

- (A) 16% (B) 28% (C) 44% (D) 56% (E) 60%

Practice 5.10

For a class of homeowners, the probability of filing a liability claim in a year is 0.05, the probability of filing a liability claim but no property claim is 0.02, and the probability of filing no claim of either type is 0.85. Calculate the probability that a randomly selected homeowner in this class files a property claim during the year.

- (A) 0.10 (B) 0.12 (C) 0.13 (D) 0.15 (E) 0.18

Practice 5.11

An insurer sells auto and homeowners policies, and every policyholder holds at least one of the two. Records show that 70% of policyholders hold an auto policy and 55% hold a homeowners policy. Next year, 20% of auto-only policyholders and 30% of homeowners-only policyholders are expected to qualify for a loyalty discount. Overall, 38% of policyholders are expected to qualify. Calculate the percentage of policyholders holding both policies who are expected to qualify for the discount.

- (A) 60% (B) 70% (C) 75% (D) 80% (E) 85%

Practice 5.12

An insurer's auto customers each insure at least one vehicle. Of the customers, 55% insure more than one vehicle and 30% insure at least one vehicle that is subject to a lien. Of the customers who insure exactly one vehicle, 40% insure a vehicle subject to a lien. Calculate the probability that a randomly selected customer insures more than one vehicle and none of the customer's vehicles is subject to a lien.

- (A) 0.43 (B) 0.12 (C) 0.18 (D) 0.37 (E) 0.45

5.8 Answers to practice problems

1. **(B)** 0.25. $\mathbb{P}(M \cup O) = 0.55 + 0.40 - 0.20 = 0.75$; female and 50 or younger is the complement of the union, $1 - 0.75 = 0.25$.
2. **(E)** 20%. Union = $0.50 + 0.40 + 0.30 - 0.20 - 0.15 - 0.10 + 0.05 = 0.80$; none = $1 - 0.80 = 0.20$.
3. **(C)** 0.38. Nine non-life files among 12: $\frac{9}{12} \cdot \frac{8}{11} \cdot \frac{7}{10} = \frac{504}{1320} = \frac{21}{55} \approx 0.3818$.
4. **(A)** 0.28. By symmetry of draws without replacement, the third policy is a uniformly random policy from the batch, so the probability is $7/25 = 0.28$.
5. **(D)** 0.175. Claim then no claim: $0.15 \times 0.60 = 0.090$; no claim then claim: $0.85 \times 0.10 = 0.085$; disjoint, so add: 0.175.
6. **(B)** 15%. Exactly one = $\mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B)$, so $0.50 = 0.80 - 2x$ and $x = 0.15$.
7. **(E)** 0.077. Split by class and multiply along each path: $0.60(0.08) + 0.30(0.03) + 0.10(0.20) = 0.048 + 0.009 + 0.020 = 0.077$.
8. **(D)** 0.15. Convert: $\mathbb{P}(O) = 0.36$, $\mathbb{P}(R) = 0.45$, $\mathbb{P}(O \cup R) = 0.60$. Intersection = $0.36 + 0.45 - 0.60 = 0.21$; outpatient but no prescription = $0.36 - 0.21 = 0.15$.
9. **(A)** 16%. Let $x = \mathbb{P}(V)$, so $\mathbb{P}(D) = 2x$. Union = $1 - 0.28 = 0.72 = 3x - 0.12$, so $x = 0.28$; vision only = $0.28 - 0.12 = 0.16$.
10. **(C)** 0.13. Both = $0.05 - 0.02 = 0.03$; union = $1 - 0.85 = 0.15$; the union is liability-only plus property, so $\mathbb{P}(P) = 0.15 - 0.02 = 0.13$ (property only 0.10).
11. **(D)** 80%. Union is 1, so both = $0.70 + 0.55 - 1 = 0.25$, auto only 0.45, homeowners only 0.30. Then $0.45(0.20) + 0.30(0.30) + 0.25x = 0.38$ gives $0.25x = 0.20$, $x = 0.80$.
12. **(A)** 0.43. Single-vehicle customers with a lien: $0.45(0.40) = 0.18$. Multi-vehicle with a lien: $0.30 - 0.18 = 0.12$. Multi-vehicle without: $0.55 - 0.12 = 0.43$.

5.9 Sample exam cross-reference

- Bank 1, Q2: “neither” plus two marginals; De Morgan to the union, then inclusion-exclusion for “both” (pattern 2; Example on exactly one claim type from complement data).
- Bank 1, Q7: size the regions auto only, homeowners only, both by the difference rule, then weight each region’s renewal rate and add (Example on weighted rates over four Venn regions).
- Bank 1, Q8: “both”, “neither”, and a difference between the marginals; one linear equation in the unknown marginal (Example on ratio between the marginals, with d in place of k).
- Bank 1, Q99: liability, property, and “liability but not property”; difference rule for the intersection, then the four regions sum to 1 for “neither” (Example on neither claim type from an “only” region).
- Bank 1, Q130: “no ER”, “no hospital”, “neither”; convert complements, De Morgan for the union, inclusion-exclusion for “at least one of each” (Key fact on data stated in complements).
- Bank 2, Q9: $\mathbb{P}(M)$, $\mathbb{P}(S)$, and $\mathbb{P}(S \mid M)$ on one group; multiplication rule for the joint on M , De Morgan for “exactly one car and not sports” (Example on the joint on the complementary group).

5.10 One-page summary

- Two events: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Always subtract the intersection unless the events are disjoint.
- Three events: add the three singles, subtract the three pairs, add the triple. Seven terms; the exam hides one.
- Complement of the union is “neither” or “none”: $\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A \cup B)$. “At least one” $= 1 - \mathbb{P}(\text{none})$.
- Exactly one of two events: $\mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B) = \mathbb{P}(A \cup B) - \mathbb{P}(A \cap B)$.
- Four Venn regions (A only, B only, both, neither) sum to 1. Difference rule: $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$. Any three independent facts fix all four.
- Marginals related by a difference or ratio: set $x = \mathbb{P}(B)$, write $\mathbb{P}(A) = x + d$ or kx , and solve $1 - \mathbb{P}(\text{neither}) = \mathbb{P}(A) + x - \mathbb{P}(A \cap B)$ for x .
- Data in complements: $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$, union $= 1 - \mathbb{P}(\text{neither})$, then $\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)$.
- Three events with e_1, e_2, e_3 (exactly one, two, three): union $= e_1 + e_2 + e_3$ and sum of marginals $= e_1 + 2e_2 + 3e_3$.
- Multiplication rule: $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A) = \mathbb{P}(B)\mathbb{P}(A \mid B)$. Chain rule: $\mathbb{P}(A_1 \cap A_2 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1)\mathbb{P}(A_3 \mid A_1 \cap A_2)$; each factor conditions on all earlier events.
- Under independence every conditional factor becomes the marginal and the chain is a plain product of marginals. Do not use the plain product otherwise.

- Without replacement: k th factor is (favourable items remaining)/(items remaining). Unconditionally, the k th draw has the same type distribution as the first: m/N .
- Tree diagram: branch labels are conditional probabilities; a path's probability is the product of its labels; an event's probability is the sum of its paths. Paths sum to 1.
- Two-stage problems: split into disjoint cases, multiply along each case, add. This is the law of total probability, $\mathbb{P}(A) = \sum_i \mathbb{P}(B_i)\mathbb{P}(A | B_i)$ (outcome (g)).
- Rates by Venn region: size the regions with the difference rule, multiply each region by its rate, add. Do not weight the marginals; “both” would be counted twice.
- Conditional on one group only: $\mathbb{P}(S \cap M) = \mathbb{P}(M)\mathbb{P}(S | M)$, then $\mathbb{P}(S \cap M^c) = \mathbb{P}(S) - \mathbb{P}(S \cap M)$, and $\mathbb{P}(M^c \cap S^c) = 1 - \mathbb{P}(M) - \mathbb{P}(S) + \mathbb{P}(M)\mathbb{P}(S | M)$.
- Joint versus conditional: $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B | A) \leq \mathbb{P}(B | A)$. A percentage “of those with A ” is a conditional probability. With counts, divide by the total once at the end.

6 Outcome (f): Conditional probability

6.1 What the outcome asks

The exam tests whether you can read the phrase “given that” (or one of its disguises) and respond by restricting the sample space to the conditioning event before you count or divide. Questions supply the information in one of a few forms: joint and marginal probabilities stated in words, a two-way table of counts or percentages, a Venn setup with two or three coverages, or a small population from which items are drawn. You compute $\mathbb{P}(A \mid B)$ from that information, and you apply the complement and addition rules *inside* the conditional world when the question asks for “not” or “or” after “given”. Questions that give $\mathbb{P}(A \mid B)$ and ask for $\mathbb{P}(B \mid A)$ across several classes are outcome (g) (Bayes and total probability); the definition that makes those work is the subject of this section, and the two-class version is solved here directly from the definition.

6.2 Definition and properties

Key fact: Definition of conditional probability

For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Read: the probability of A given that B has occurred. The denominator is the probability of the *conditioning* event, the one named after “given”.

Interpretation. Conditioning on B throws away every outcome outside B and treats B as the new sample space. Probabilities of outcomes inside B are then rescaled by dividing by $\mathbb{P}(B)$ so that they again sum to 1. Only the part of A that lies inside B , namely $A \cap B$, survives the restriction.

Because $\mathbb{P}(\cdot \mid B)$ is a genuine probability measure on the reduced sample space B , it satisfies all three axioms: $\mathbb{P}(A \mid B) \geq 0$, $\mathbb{P}(B \mid B) = 1$, and it is additive over disjoint events. Every rule derived from the axioms therefore has a conditional version, obtained by writing “ $\mid B$ ” after every event.

Key fact: Conditional complement and addition rules

Holding the conditioning event B fixed:

$$\begin{aligned}\mathbb{P}(A^c \mid B) &= 1 - \mathbb{P}(A \mid B), \\ \mathbb{P}(A \cup C \mid B) &= \mathbb{P}(A \mid B) + \mathbb{P}(C \mid B) - \mathbb{P}(A \cap C \mid B), \\ \mathbb{P}(A \cap C^c \mid B) &= \mathbb{P}(A \mid B) - \mathbb{P}(A \cap C \mid B).\end{aligned}$$

These hold because the event *after* the bar is the same throughout.

Key fact: What is NOT true

Changing the event *after* the bar is not a complement operation:

$$\mathbb{P}(A \mid B^c) \neq 1 - \mathbb{P}(A \mid B) \quad \text{in general.}$$

$\mathbb{P}(A | B)$ and $\mathbb{P}(A | B^c)$ live on two different reduced sample spaces and need not be related at all. To get $\mathbb{P}(A | B^c)$ you need $\mathbb{P}(A \cap B^c)$ and $\mathbb{P}(B^c)$, typically via $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$.

Key fact: Multiplication rule and chain rule

Rearranging the definition gives

$$\mathbb{P}(A \cap B) = \mathbb{P}(B) \mathbb{P}(A | B) = \mathbb{P}(A) \mathbb{P}(B | A),$$

and for three events

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A) \mathbb{P}(B | A) \mathbb{P}(C | A \cap B).$$

Conversely, conditioning on an intersection is the same definition with a compound conditioning event:

$$\mathbb{P}(C | A \cap B) = \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(A \cap B)}.$$

Key fact: Link to independence

If $\mathbb{P}(B) > 0$, then A and B are independent exactly when $\mathbb{P}(A | B) = \mathbb{P}(A)$, equivalently $\mathbb{P}(A | B) = \mathbb{P}(A | B^c)$. Knowing B happened does not change the probability of A . If $\mathbb{P}(A | B) \neq \mathbb{P}(A)$ the events are dependent, and the multiplication rule must use the conditional probability, not the marginal.

Two consequences. $\mathbb{P}(A | B)$ and $\mathbb{P}(B | A)$ share the numerator $\mathbb{P}(A \cap B)$ but have different denominators, so $\mathbb{P}(B | A) = \mathbb{P}(A | B) \mathbb{P}(B) / \mathbb{P}(A)$, which reverses a single condition whenever $\mathbb{P}(A)$ is given. If $A \subseteq B$ then $\mathbb{P}(A | B) = \mathbb{P}(A) / \mathbb{P}(B)$; if $A \cap B = \emptyset$ then $\mathbb{P}(A | B) = 0$.

6.3 Three computational settings

(i) Two-way (contingency) table

The conditioning event names a row or a column. Restrict to that row or column, then divide the cell you want by the row or column total. Nothing outside that row or column is used.

A table of counts. An insurer classifies 200 life policyholders by sex and smoking status.

	Smoker	Nonsmoker	Total
Male	30	90	120
Female	24	56	80
Total	54	146	200

- $\mathbb{P}(\text{Smoker} | \text{Male}) = 30/120 = 0.25$: restrict to the Male row.
- $\mathbb{P}(\text{Smoker} | \text{Female}) = 24/80 = 0.30$: restrict to the Female row.
- $\mathbb{P}(\text{Male} | \text{Smoker}) = 30/54 = 0.5556$: restrict to the Smoker column.
- $\mathbb{P}(\text{Male} \cap \text{Smoker}) = 30/200 = 0.15$: no restriction; divide by the grand total.

Since $\mathbb{P}(\text{Smoker} | \text{Male}) \neq \mathbb{P}(\text{Smoker} | \text{Female})$, sex and smoking status are dependent in this population.

A table of proportions. The same procedure works when the cells are joint probabilities that sum to 1; the row and column totals are then the marginal probabilities.

	Claim	No claim	Total
Young	0.12	0.28	0.40
Old	0.09	0.51	0.60
Total	0.21	0.79	1.00

- $\mathbb{P}(\text{Claim} \mid \text{Young}) = 0.12/0.40 = 0.30$.
- $\mathbb{P}(\text{Claim} \mid \text{Old}) = 0.09/0.60 = 0.15$.
- $\mathbb{P}(\text{Young} \mid \text{Claim}) = 0.12/0.21 = 0.5714$.
- $\mathbb{P}(\text{No claim} \mid \text{Young}) = 1 - 0.30 = 0.70 = 0.28/0.40$: conditional complement, same row.
- If the table had several age rows, conditioning on “age at least 30” would mean adding the relevant rows first: the denominator is the sum of those row totals and the numerator is the sum of their Claim cells.

Exam problems often give the table indirectly: a marginal (“40% of policyholders are young”), a conditional (“30% of young policyholders filed a claim”), and an overall rate (“21% filed a claim”). Rebuild the table: Young–Claim = $0.40 \times 0.30 = 0.12$, Old–Claim = $0.21 - 0.12 = 0.09$, the rest by subtraction. Every conditional is then a ratio of two entries.

Key fact: Filling a table from mixed information

A problem may describe a 2×2 or 3×2 table by a mixture of marginals, conditionals, and raw counts. Convert each sentence to a table entry:

- “ $x\%$ are B ” or “there are n in B ”: a row or column total.
- “Of those in B , a fraction p are A ”: the cell $A \cap B$ equals $p \times (\text{total of } B)$. The conditional supplies a cell only after it is multiplied by its own marginal.
- “Of those in A , a fraction q are B ”: the same cell $A \cap B$, now as $q \times (\text{total of } A)$. Conditionals may point in either direction.
- A total not stated is 1 (or the population size) minus the totals that are stated.

Fill the cells that are products first, then every remaining cell by subtraction from its row or column total. Only then read off the requested cell or the requested ratio. If the population size is a count and some information is in percentages, convert everything to counts (or everything to proportions) before filling the table.

(ii) Venn diagram / region bookkeeping

With two or three overlapping coverages, label each disjoint region with its probability (or count), then

$$\mathbb{P}(A \mid B) = \frac{\text{total of the regions inside } A \cap B}{\text{total of the regions inside } B}.$$

For three sets, the “only” regions come from inclusion–exclusion: $\mathbb{P}(A \text{ only}) = \mathbb{P}(A) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) + \mathbb{P}(A \cap B \cap C)$. “Of those with auto, what fraction also have home” means the denominator is every region inside the auto circle and the numerator is the lens $A \cap H$ (both of its regions, with and without life).

(iii) Sequential / tree

When the experiment happens in stages (select a class, then observe a claim; draw one file, then another), the branch labels on a probability tree are conditional probabilities given the path so far. Multiplying along a path gives the joint probability of the path. “Given the first was X , what is the probability the second is Y ” is read directly off a branch label; “given the second was Y ” reverses the tree, which is outcome (g).

(iv) An unknown fixed by a stated conditional

Some problems state the value of a conditional probability and ask for a quantity inside it. Write the conditional from the definition with the unknown in place, equate to the stated value, and solve. Two versions appear.

Unknown region. With three symmetric risk factors, the probabilities of the “only” and “exactly two” regions are given, the triple region x is not, and the problem states $\mathbb{P}(A \cap B \cap C \mid A \cap B) = r$. Since $\mathbb{P}(A \cap B) = x + (\text{exactly } A \text{ and } B)$, the equation is $x/(x + \text{exactly two}) = r$, linear in x . Once x is known, every region is known, and any later conditional is a ratio of region totals. Conditioning on A^c means the denominator is $1 - \mathbb{P}(A)$ with $\mathbb{P}(A) = (A \text{ only}) + 2(\text{exactly two}) + x$ by symmetry.

Unknown count. A student knows the answers to N of n questions and guesses the rest with success probability g . Then

$$\mathbb{P}(K \mid C) = \frac{\mathbb{P}(K \cap C)}{\mathbb{P}(C)} = \frac{N/n}{N/n + g(n - N)/n} = \frac{N}{N + g(n - N)},$$

where K = knows and C = correct. The unknown N sits in both numerator and denominator; cross-multiplying gives a linear equation. If a known answer is correct only with probability $k < 1$, the numerator becomes kN and the denominator $kN + g(n - N)$. The answer must be an integer; if it is not, the equation has been set up wrongly.

Phrases that signal conditioning

Phrase in the problem	Meaning
“given that B ” / “if it is known that B ”	Denominator is $\mathbb{P}(B)$.
“of those who B ” / “among B policyholders”	The B group is the whole population; take the fraction inside it.
“restricted to B ” / “within the B class”	Same.
“ $x\%$ of B policyholders are A ”	$\mathbb{P}(A \mid B) = x/100$, not $\mathbb{P}(A \cap B)$.
“ $x\%$ of policyholders are both A and B ”	Joint $\mathbb{P}(A \cap B)$; no conditioning.
“the probability that a B policyholder is A ”	$\mathbb{P}(A \mid B)$.

6.4 Worked examples

Example 6.1: Two-way table of policyholders

An auto insurer has 500 policyholders, classified by age and sex as follows.

	Male	Female	Total
Young	140	85	225
Old	160	115	275
Total	300	200	500

A policyholder is selected at random. Calculate the probability that the policyholder is female, given that the policyholder is young.

- (A) 0.17 (B) 0.38 (C) 0.40 (D) 0.43 (E) 0.45

Solution

The conditioning event is “young”, so restrict to the Young row, which has 225 policyholders. Of these, 85 are female.

$$\mathbb{P}(\text{Female} \mid \text{Young}) = \frac{85}{225} = 0.3778.$$

The distractors are the other ratios one could form from the same cell: $85/500 = 0.17$ is $\mathbb{P}(\text{Female} \cap \text{Young})$; $200/500 = 0.40$ is the marginal $\mathbb{P}(\text{Female})$; $85/200 = 0.425$ is the reversed conditional $\mathbb{P}(\text{Young} \mid \text{Female})$. **Answer: (B).**

Example 6.2: Two-way count table from mixed information

An insurer has 1,200 auto policyholders. Of these, 35% are under age 30. Of the policyholders aged 30 or over, 30% have had a prior claim. In total, 480 policyholders have had a prior claim. A policyholder with no prior claim is selected at random. Calculate the probability that the policyholder is under age 30.

- (A) 0.242 (B) 0.350 (C) 0.414 (D) 0.513 (E) 0.600

Solution

Work in counts. Under 30: $0.35 \times 1,200 = 420$; aged 30 or over: $1,200 - 420 = 780$. The conditional “of those 30 or over, 30% have a prior claim” gives the cell (Over, Claim) $= 0.30 \times 780 = 234$. The column total Claim is 480, so (Under, Claim) $= 480 - 234 = 246$ and (Under, No claim) $= 420 - 246 = 174$. The No-claim column has $1,200 - 480 = 720$.

	Claim	No claim	Total
Under 30	246	174	420
30 or over	234	546	780
Total	480	720	1200

The conditioning event is No claim, so

$$\mathbb{P}(\text{Under 30} \mid \text{No claim}) = \frac{174}{720} = 0.2417.$$

Distractors: $174/420 = 0.414$ is the reversed conditional; 0.35 is the marginal; $246/480 = 0.513$ conditions on Claim instead of No claim; $720/1,200 = 0.60$ is $\mathbb{P}(\text{No claim})$. **Answer: (A).**

Example 6.3: Three-by-two table from mixed information

An insurer classifies its policyholders by territory (urban, suburban, or rural) and by whether they filed a claim last year. It finds that 30% are urban, 45% are suburban, and 18% filed a claim. Of those who filed a claim, 40% are urban. Of the rural policyholders, one-fifth filed a claim. Calculate the probability that a policyholder who did not file a claim is suburban.

- (A) 0.129 (B) 0.392 (C) 0.478 (D) 0.820 (E) 0.871

Solution

Rural is the unstated territory: $1 - 0.30 - 0.45 = 0.25$. The Claim row has total 0.18 and the No-claim row 0.82. The two conditionals point in different directions and each supplies one cell:

$$\mathbb{P}(\text{Urban} \cap \text{Claim}) = 0.40 \times 0.18 = 0.072, \quad \mathbb{P}(\text{Rural} \cap \text{Claim}) = \frac{1}{5} \times 0.25 = 0.050.$$

The Claim row then gives Suburban–Claim = $0.18 - 0.072 - 0.050 = 0.058$, and the Suburban column gives Suburban–No claim = $0.45 - 0.058 = 0.392$.

	Urban	Suburban	Rural	Total
Claim	0.072	0.058	0.050	0.18
No claim	0.228	0.392	0.200	0.82
Total	0.30	0.45	0.25	1.00

The conditioning event is No claim:

$$\mathbb{P}(\text{Suburban} \mid \text{No claim}) = \frac{0.392}{0.82} = 0.4780.$$

Choice (B) is the joint probability 0.392; choice (E), $0.392/0.45 = 0.871$, is $\mathbb{P}(\text{No claim} \mid \text{Suburban})$, the reversed conditional; choice (A) is $\mathbb{P}(\text{Claim} \mid \text{Suburban}) = 0.058/0.45$. **Answer: (C).**

Example 6.4: Both, given at least one

For a randomly selected policyholder holding both auto and homeowners coverage with an insurer, the probability of an auto claim in a given year is 0.30, the probability of a homeowners claim is 0.20, and the probability of both is 0.08. Given that the policyholder files at least one claim in the year, calculate the probability that the policyholder files both.

- (A) 0.08 (B) 0.16 (C) 0.19 (D) 0.27 (E) 0.42

Solution

Let A = auto claim and H = homeowners claim. “At least one” is $A \cup H$, and “both” is $A \cap H$, which is a subset of $A \cup H$, so the intersection of the two events is just $A \cap H$.

$$\mathbb{P}(A \cup H) = 0.30 + 0.20 - 0.08 = 0.42, \quad \mathbb{P}(A \cap H \mid A \cup H) = \frac{0.08}{0.42} = 0.1905.$$

Answer: (C).

Example 6.5: Venn with three coverages

A survey of an insurer’s customers finds that 55% have auto coverage, 35% have homeowners coverage, and 25% have life coverage. Also, 15% have both auto and homeowners, 12% have both auto and life, 8% have both homeowners and life, and 5% have all three. Of those customers who have auto coverage, calculate the proportion who have neither homeowners nor life coverage.

- (A) 0.33 (B) 0.40 (C) 0.45 (D) 0.55 (E) 0.60

Solution

Let A , H , L be the three coverages. The denominator is $\mathbb{P}(A) = 0.55$. The numerator is the “auto only” region:

$$\mathbb{P}(A \cap H^c \cap L^c) = \mathbb{P}(A) - \mathbb{P}(A \cap H) - \mathbb{P}(A \cap L) + \mathbb{P}(A \cap H \cap L) = 0.55 - 0.15 - 0.12 + 0.05 = 0.33.$$

Then

$$\mathbb{P}(H^c \cap L^c \mid A) = \frac{0.33}{0.55} = 0.60.$$

Choice (A) is the joint probability 0.33, not the conditional. **Answer: (E).**

Example 6.6: Symmetric Venn with an unknown triple cell

An actuary studies three optional coverages, A , B , and C , among a block of policyholders. For each coverage, the probability that a policyholder has only that coverage (and neither of the other two) is 0.08. For each pair of coverages, the probability that a policyholder has exactly those two (and not the third) is 0.05. The probability that a policyholder has all three coverages, given that the policyholder has A and C , is $2/7$. Calculate the probability that a policyholder has at least one of coverages A and C , given that the policyholder does not have coverage B .

- (A) 0.210 (B) 0.263 (C) 0.350 (D) 0.738 (E) 0.800

Solution

Let $x = \mathbb{P}(A \cap B \cap C)$. The event $A \cap C$ consists of the “exactly A and C ” region and the triple region, so $\mathbb{P}(A \cap C) = 0.05 + x$ and

$$\frac{2}{7} = \mathbb{P}(A \cap B \cap C \mid A \cap C) = \frac{x}{x + 0.05} \implies 7x = 2x + 0.10 \implies x = 0.02.$$

Every region is now known. By symmetry, $\mathbb{P}(B) = 0.08 + 2(0.05) + 0.02 = 0.20$, so $\mathbb{P}(B^c) = 0.80$. The region outside all three circles is

$$1 - 3(0.08) - 3(0.05) - 0.02 = 0.59.$$

Inside B^c , the regions are A only (0.08), C only (0.08), exactly A and C (0.05), and none (0.59); these sum to 0.80 as they must. “At least one of A and C ” within B^c is the first three: $0.08 + 0.08 + 0.05 = 0.21$. Then

$$\mathbb{P}(A \cup C \mid B^c) = \frac{0.21}{0.80} = 0.2625.$$

Choice (A) is the joint probability 0.21; choice (D), $0.59/0.80 = 0.7375$, is the complementary conditional $\mathbb{P}(\text{none} \mid B^c)$. **Answer: (B).**

Example 6.7: Reversing one condition from the definition

In a block of life policies, 30% of policyholders are smokers. Among smokers, 20% filed a claim last year. Overall, 12% of policyholders filed a claim last year. Calculate the probability that a policyholder who filed a claim last year is a smoker.

(A) 0.06 (B) 0.20 (C) 0.30 (D) 0.50 (E) 0.60

Solution

Let S = smoker and C = filed a claim. Given: $\mathbb{P}(S) = 0.30$, $\mathbb{P}(C \mid S) = 0.20$, $\mathbb{P}(C) = 0.12$. The joint probability is

$$\mathbb{P}(S \cap C) = \mathbb{P}(S) \mathbb{P}(C \mid S) = 0.30 \times 0.20 = 0.06,$$

and the requested conditional is

$$\mathbb{P}(S \mid C) = \frac{\mathbb{P}(S \cap C)}{\mathbb{P}(C)} = \frac{0.06}{0.12} = 0.50.$$

No Bayes machinery is needed because $\mathbb{P}(C)$ is supplied directly; the definition does the whole job. **Answer: (D).**

Example 6.8: Conditioning on the complement

A life insurer classifies 60% of its policyholders as standard and the rest as preferred. The probability that a standard policyholder lapses within one year is 0.15. The probability that a randomly selected policyholder lapses within one year is 0.11. Calculate the probability that a preferred policyholder lapses within one year.

(A) 0.02 (B) 0.05 (C) 0.085 (D) 0.15 (E) 0.85

Solution

Let S = standard and L = lapse. Given: $\mathbb{P}(S) = 0.60$, so $\mathbb{P}(S^c) = 0.40$; $\mathbb{P}(L \mid S) = 0.15$; $\mathbb{P}(L) = 0.11$. Build the pieces of L :

$$\mathbb{P}(L \cap S) = 0.60 \times 0.15 = 0.09, \quad \mathbb{P}(L \cap S^c) = \mathbb{P}(L) - \mathbb{P}(L \cap S) = 0.11 - 0.09 = 0.02.$$

Then

$$\mathbb{P}(L \mid S^c) = \frac{0.02}{0.40} = 0.05.$$

Choice (E), $0.85 = 1 - 0.15$, is $\mathbb{P}(L^c \mid S)$, the probability that a *standard* policyholder does *not* lapse; it says nothing about preferred policyholders. Choice (A) is the joint probability, not the conditional. **Answer: (B).**

Example 6.9: Counts: conditioning on a derived complement

An insurer reviews the records of 1,200 drivers and finds that 300 of them had an accident last year. Of the drivers, 420 are under age 25 and 180 are over age 60; of the under-25 drivers, 126 had an accident, and of the over-60 drivers, 54 had an accident. Calculate the probability that a randomly selected driver had an accident last year, given that the driver is between ages 25 and 60 inclusive.

- (A) 0.200 (B) 0.223 (C) 0.250 (D) 0.300 (E) 0.400

Solution

Let M = driver is between 25 and 60, the complement of the union of the two described age groups, and A = accident. Neither the count of M nor the count of $A \cap M$ is stated; both are obtained by subtracting the described groups from the totals:

$$N(M) = 1,200 - 420 - 180 = 600, \quad N(A \cap M) = 300 - 126 - 54 = 120.$$

Then

$$\mathbb{P}(A \mid M) = \frac{N(A \cap M)}{N(M)} = \frac{120}{600} = 0.20.$$

Distractors: $300/1,200 = 0.25$ is the unconditional accident rate; $126/420 = 0.30$ subtracts only the under-25 group; $126/420 = 0.30$ is the rate for the under-25 group; $120/300 = 0.40$ is the reversed conditional $\mathbb{P}(M \mid A)$. **Answer: (A).**

Example 6.10: Conditioning on an intersection

An insurer's records show that 50% of its customers have auto coverage, 40% have homeowners coverage, and 20% have both. Also, 12% of all customers filed a claim last year, 8% have auto coverage and filed a claim last year, and 5% have both coverages and filed a claim last year. Calculate the probability that a customer who has both auto and homeowners coverage filed a claim last year.

- (A) 0.05 (B) 0.10 (C) 0.16 (D) 0.25 (E) 0.40

Solution

Let A = auto, H = homeowners, C = claim. The conditioning event is $A \cap H$ with $\mathbb{P}(A \cap H) = 0.20$; the numerator is $\mathbb{P}(A \cap H \cap C) = 0.05$.

$$\mathbb{P}(C \mid A \cap H) = \frac{0.05}{0.20} = 0.25.$$

The other figures are distractors: $\mathbb{P}(C \mid A) = 0.08/0.50 = 0.16$ conditions on auto alone; $\mathbb{P}(C) = 0.12$ and $\mathbb{P}(H) = 0.40$ are not needed. **Answer: (D).**

Example 6.11: Solving for an integer unknown

A candidate takes an examination of 30 multiple-choice questions, each with four choices. The candidate knows the answers to N of the questions and answers those correctly; on each of the remaining questions the candidate selects one of the four choices at random. The conditional probability that the candidate knew the answer to a question, given that the question was answered correctly, is 0.857. Calculate N .

(A) 10 (B) 12 (C) 14 (D) 16 (E) 18

Solution

Let K = the candidate knows the answer and C = the question is answered correctly. A question is selected at random, so $\mathbb{P}(K) = N/30$ and $\mathbb{P}(K^c) = (30 - N)/30$; also $\mathbb{P}(C \mid K) = 1$ and $\mathbb{P}(C \mid K^c) = 1/4$. By the multiplication rule,

$$\mathbb{P}(K \cap C) = \frac{N}{30}, \quad \mathbb{P}(C) = \mathbb{P}(K \cap C) + \mathbb{P}(K^c \cap C) = \frac{N}{30} + \frac{30 - N}{30} \cdot \frac{1}{4}.$$

The stated conditional is

$$0.857 = \mathbb{P}(K \mid C) = \frac{N/30}{N/30 + (30 - N)/120} = \frac{4N}{3N + 30}.$$

Cross-multiply: $0.857(3N + 30) = 4N$, so $2.571N + 25.71 = 4N$, giving $1.429N = 25.71$ and $N = 18.0$. Check: $4(18)/(54 + 30) = 72/84 = 0.857$. **Answer: (E).**

6.5 Traps

Trap: Dividing by the wrong marginal

The denominator is always the probability of the event named after “given” (or “of those who”, “among”). In a table, that means the total of the row or column you restricted to, never the grand total and never the other margin. Before dividing, say out loud which group is the new population.

Trap: $\mathbb{P}(A | B)$ versus $\mathbb{P}(B | A)$

These share the numerator $\mathbb{P}(A \cap B)$ but have different denominators and are generally different numbers. “20% of smokers filed a claim” is $\mathbb{P}(C | S)$; “50% of claimants are smokers” is $\mathbb{P}(S | C)$. Exam answer choices routinely include the reversed conditional as a distractor. Write the conditional with the bar before computing anything.

Trap: $\mathbb{P}(A | B^c)$ is not $1 - \mathbb{P}(A | B)$

The complement rule flips the event *before* the bar: $\mathbb{P}(A^c | B) = 1 - \mathbb{P}(A | B)$. Flipping the event *after* the bar changes the population, and the two conditionals need not add to 1. To find $\mathbb{P}(A | B^c)$, compute $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$ and divide by $\mathbb{P}(B^c) = 1 - \mathbb{P}(B)$.

Trap: Joint versus conditional in the wording

“ $x\%$ of policyholders are young and filed a claim” is a joint probability $\mathbb{P}(Y \cap C)$. “ $x\%$ of young policyholders filed a claim” is a conditional $\mathbb{P}(C | Y)$. “ $x\%$ of policyholders are young” is a marginal $\mathbb{P}(Y)$. Misreading one of these as another produces an answer that is present among the choices. When the problem mixes the three, rebuild the two-way table from them before answering.

Trap: Mixing counts and percentages

A conditional probability is a ratio, so it can be computed from counts or from proportions, but not from a mixture. If a row total is a count and a cell is a percentage of the whole population, convert both to the same units first. In a proportion table, the grand total is 1, and the marginals are row and column sums; do not divide a proportion by a count.

6.6 Practice problems

Practice 6.1

A life insurer classifies 400 policyholders as preferred or standard, and as smokers or non-smokers. There are 20 preferred smokers, 140 preferred nonsmokers, 90 standard smokers, and 150 standard nonsmokers. A policyholder is selected at random. Calculate the probability that the policyholder is preferred, given that the policyholder is a nonsmoker.

- (A) 0.35 (B) 0.40 (C) 0.48 (D) 0.72 (E) 0.88

Practice 6.2

For a randomly selected employee with a group benefits plan, the probability of a dental claim in a year is 0.45, the probability of a vision claim is 0.35, and the probability of both is 0.15. Given that the employee has at least one of the two types of claim in the year, calculate the probability that the employee has both.

- (A) 0.15 (B) 0.23 (C) 0.33 (D) 0.43 (E) 0.65

Practice 6.3

In a survey of an insurer's customers, 40% have auto coverage, 30% have homeowners coverage, and 25% have life coverage; 12% have auto and homeowners, 10% have auto and life, 8% have homeowners and life, and 4% have all three. Of those customers with homeowners coverage, calculate the proportion who have neither auto nor life coverage.

- (A) 0.14 (B) 0.40 (C) 0.47 (D) 0.53 (E) 0.60

Practice 6.4

An auto insurer classifies 20% of its policies as high risk. The probability that a high-risk policy has a claim in a year is 0.35. The probability that a randomly selected policy has a claim in a year is 0.16. Calculate the probability that a policy with a claim in the year is a high-risk policy.

- (A) 0.07 (B) 0.20 (C) 0.35 (D) 0.44 (E) 0.56

Practice 6.5

Of an insurer's policyholders, 55% are female. The probability that a female policyholder files a claim in a year is 0.10. The probability that a randomly selected policyholder files a claim in a year is 0.145. Calculate the probability that a male policyholder files a claim in the year.

- (A) 0.09 (B) 0.145 (C) 0.20 (D) 0.30 (E) 0.90

Practice 6.6

An insurer's records show that 60% of customers have auto coverage. Among customers with auto coverage, 50% carry collision coverage. Also, 12% of all customers have auto coverage with collision and filed a claim last year. Calculate the probability that a customer with auto coverage and collision coverage filed a claim last year.

- (A) 0.12 (B) 0.20 (C) 0.30 (D) 0.40 (E) 0.72

Practice 6.7

A file cabinet contains 12 claim files, of which 5 are fraudulent. An auditor selects 2 files at random without replacement. Given that at least one of the selected files is fraudulent, calculate the probability that both are fraudulent.

- (A) 0.15 (B) 0.22 (C) 0.36 (D) 0.41 (E) 0.68

Practice 6.8

An insurer has 1,600 insured drivers, of whom 45% are female. Among the male drivers, 30% are under age 30. The probability that a randomly selected driver under age 30 is female is 0.56. A female driver is selected at random. Calculate the probability that she is under age 30.

(A) 0.210 (B) 0.300 (C) 0.375 (D) 0.467 (E) 0.560

Practice 6.9

Employees of a company are enrolled in exactly one of three health plans: Basic, Standard, or Premium. Half of the employees are in the Basic plan, and the number in the Premium plan is half the number in the Standard plan. Last year 24% of employees filed a claim. Of the Premium enrollees, three-eighths filed a claim; of the employees who filed a claim, one-quarter are in the Basic plan. Calculate the probability that an employee who did not file a claim is in the Standard plan.

(A) 0.118 (B) 0.167 (C) 0.216 (D) 0.250 (E) 0.284

Practice 6.10

An actuary studies three risk factors, A , B , and C , in a population. The probabilities of having only A , only B , and only C are 0.09, 0.07, and 0.11 respectively. The probabilities of having exactly A and B , exactly A and C , and exactly B and C (and not the third factor) are 0.04, 0.06, and 0.03 respectively. The probability of having all three factors, given B and C , is 0.25. Calculate the probability that a member of the population has exactly one of the three risk factors, given that the member does not have factor C .

(A) 0.203 (B) 0.253 (C) 0.270 (D) 0.342 (E) 0.747

Practice 6.11

A health insurer reviews 1,850 hospital claims. Of these, 740 involved surgery and 610 came from network hospitals; of the surgical claims, 290 came from network hospitals. Also, 520 of the claims exceeded \$10,000, of which 200 came from network hospitals, and 310 of the claims both involved surgery and exceeded \$10,000, of which 150 came from network hospitals. A claim from a hospital that is not in the network is selected at random. Calculate the probability that it involved surgery or exceeded \$10,000.

(A) 0.492 (B) 0.508 (C) 0.514 (D) 0.621 (E) 0.766

Practice 6.12

A candidate takes an examination of 24 questions, each with three choices. The candidate has studied N of the questions and answers each of those correctly with probability 0.9; on each of the remaining questions the candidate guesses, selecting one of the three choices at random. The conditional probability that the candidate had studied a question, given that it was answered correctly, is 0.844. Calculate N .

(A) 8 (B) 10 (C) 12 (D) 14 (E) 16

6.7 Answers to practice problems

1. (C). Nonsmokers total $140 + 150 = 290$; of these 140 are preferred. $140/290 = 0.4828$. ($140/400 = 0.35$ is the joint; $160/400 = 0.40$ is the marginal; $140/160 = 0.875$ is the reversed conditional.)

2. **(B).** $\mathbb{P}(D \cup V) = 0.45 + 0.35 - 0.15 = 0.65$; $\mathbb{P}(D \cap V \mid D \cup V) = 0.15/0.65 = 0.2308$.
3. **(C).** Homeowners only $= 0.30 - 0.12 - 0.08 + 0.04 = 0.14$; divide by $\mathbb{P}(H) = 0.30$: $0.14/0.30 = 0.4667$. ($0.12/0.30 = 0.40$ is the fraction of homeowners customers with auto.)
4. **(D).** $\mathbb{P}(R \cap C) = 0.20 \times 0.35 = 0.07$; $\mathbb{P}(R \mid C) = 0.07/0.16 = 0.4375$.
5. **(C).** $\mathbb{P}(F \cap C) = 0.55 \times 0.10 = 0.055$; $\mathbb{P}(M \cap C) = 0.145 - 0.055 = 0.09$; $\mathbb{P}(C \mid M) = 0.09/0.45 = 0.20$. (Choice (E), $0.90 = 1 - 0.10$, is $\mathbb{P}(C^c \mid F)$, not a statement about males.)
6. **(D).** $\mathbb{P}(A \cap K) = 0.60 \times 0.50 = 0.30$; $\mathbb{P}(C \mid A \cap K) = 0.12/0.30 = 0.40$.
7. **(B).** $\mathbb{P}(\text{both}) = \binom{5}{2}/\binom{12}{2} = 10/66$; $\mathbb{P}(\text{none}) = \binom{7}{2}/\binom{12}{2} = 21/66$; $\mathbb{P}(\text{at least one}) = 45/66$; ratio $= 10/45 = 2/9 = 0.2222$. ($4/11 = 0.36$ is $\mathbb{P}(\text{second fraudulent} \mid \text{first fraudulent})$, a different condition.)
8. **(D).** Males $= 0.55 \times 1,600 = 880$, females $= 720$. Males under 30 $= 0.30 \times 880 = 264$. The reversed conditional fixes the under-30 total u : $\mathbb{P}(\text{Male} \mid \text{Under 30}) = 1 - 0.56 = 0.44 = 264/u$, so $u = 600$. Females under 30 $= 600 - 264 = 336$; $\mathbb{P}(\text{Under 30} \mid \text{Female}) = 336/720 = 0.4667$. (0.56 is the given reversed conditional; $600/1,600 = 0.375$ is the marginal; $336/1,600 = 0.21$ is the joint.)
9. **(E).** Standard s and Premium $s/2$ sum to 0.50 , so $s = 1/3$ and Premium $= 1/6$. Basic-Claim $= \frac{1}{4} \times 0.24 = 0.06$; Premium-Claim $= \frac{1}{6} \times \frac{3}{8} = 0.0625$; Standard-Claim $= 0.24 - 0.06 - 0.0625 = 0.1175$; Standard-No claim $= 1/3 - 0.1175 = 0.2158$; $\mathbb{P}(\text{No claim}) = 0.76$. $\mathbb{P}(\text{Standard} \mid \text{No claim}) = 0.2158/0.76 = 0.2840$. (0.216 is the joint; 0.118 is Standard-Claim; 0.167 is the Premium marginal.)
10. **(A).** $x/(x + 0.03) = 0.25$ gives $x = 0.01$. $\mathbb{P}(C) = 0.11 + 0.06 + 0.03 + 0.01 = 0.21$, so $\mathbb{P}(C^c) = 0.79$. Exactly one factor and not C means A only or B only: $0.09 + 0.07 = 0.16$. $0.16/0.79 = 0.2025$. ($0.27/0.79 = 0.342$ wrongly includes “ C only”, which is not in C^c ; $0.20/0.79 = 0.253$ wrongly includes the “exactly A and B ” region; $0.59/0.79 = 0.747$ is $\mathbb{P}(\text{none} \mid C^c)$.)
11. **(A).** Out-of-network total $= 1,850 - 610 = 1,240$. Within it: surgery $= 740 - 290 = 450$; over \$10,000 $= 520 - 200 = 320$; both $= 310 - 150 = 160$. Union $= 450 + 320 - 160 = 610$; $610/1,240 = 0.4919$. ($770/1,240 = 0.621$ omits the inclusion-exclusion subtraction; $950/1,850 = 0.514$ is the unconditional probability; $950/1,240 = 0.766$ mixes the whole-population numerator with the out-of-network denominator.)
12. **(E).** $\mathbb{P}(K \cap C) = 0.9N/24$ and $\mathbb{P}(C) = 0.9N/24 + \frac{1}{3}(24 - N)/24$, so $0.844 = 0.9N/(0.9N + (24 - N)/3) = 2.7N/(1.7N + 24)$. Then $0.844(1.7N + 24) = 2.7N$, $1.435N + 20.26 = 2.7N$, $N = 20.26/1.265 = 16.0$. Check: $14.4/(14.4 + 2.667) = 0.8438$.

6.8 Sample exam cross-reference

- Bank 2, Q6: counts; condition on the complement of a described group, deriving both the complement’s count and the intersection’s count by subtraction from the totals.
- Bank 2, Q11: 3×2 proportion table filled from two marginals, one row total, and two conditionals pointing in different directions; read off one cell.
- Bank 2, Q12: three symmetric risk factors; solve $x/(x + \text{exactly two}) = r$ for the triple cell, then $\mathbb{P}(\text{none} \mid A^c) = (\text{none})/(1 - \mathbb{P}(A))$.
- Bank 2, Q127: reverse one condition from the definition, $\mathbb{P}(B \mid C) = \mathbb{P}(C \mid B)\mathbb{P}(B)/\mathbb{P}(C)$, with both marginals given.
- Bank 2, Q132: 2×2 count table from a percentage marginal, a conditional, and a count marginal; read off a conditional on the unstated column.
- Bank 2, Q135: know-or-guess; $\mathbb{P}(K \mid C) = N/(N + g(n - N))$ with $g = 1/2$; cross-multiply and solve for the integer N .

6.9 One-page summary

- Definition: $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$, $\mathbb{P}(B) > 0$. The event after the bar is the new sample space; divide by its probability.
- Signal words for conditioning: given, of those who, among, if it is known that, restricted to, within the class, “ $x\%$ of B policyholders are A ”.
- “ $x\%$ of policyholders are A and B ” is joint; “ $x\%$ of B policyholders are A ” is conditional; “ $x\%$ of policyholders are B ” is marginal.
- Two-way table: restrict to the row or column named after “given”, divide the cell by that row or column total. Works for counts or proportions, not a mix.
- Rebuild the table from words: cell = marginal $\times$ conditional; remaining cells by subtraction from marginals.
- Mixed information (2×2 or 3×2): each conditional supplies one cell as conditional $\times$ its own marginal, in whichever direction it points; unstated totals by subtraction; convert to one unit (counts or proportions) first.
- Complement of a described group: $N(B^c) = N - N(B)$ and $N(A \cap B^c) = N(A) - N(A \cap B)$; with two disjoint described groups subtract both.
- Symmetric three-factor Venn with unknown triple x : $\mathbb{P}(\text{all} \mid A \cap B) = x/(x + \text{exactly two})$ fixes x ; $\mathbb{P}(A) = \text{only} + 2(\text{exactly two}) + x$; $\text{none} = 1 - 3(\text{only}) - 3(\text{exactly two}) - x$.
- Know-or-guess: $\mathbb{P}(K \mid C) = kN/(kN + g(n - N))$ with $k = \mathbb{P}(C \mid K)$, $g = \mathbb{P}(C \mid K^c)$; cross-multiply and solve for the integer N .
- Venn: $\mathbb{P}(A \mid B) = (\text{regions in } A \cap B)/(\text{regions in } B)$; “only” regions by inclusion–exclusion.
- Tree: branch labels are conditionals; multiply along a path for the joint.
- Conditional complement: $\mathbb{P}(A^c \mid B) = 1 - \mathbb{P}(A \mid B)$.
- Conditional addition: $\mathbb{P}(A \cup C \mid B) = \mathbb{P}(A \mid B) + \mathbb{P}(C \mid B) - \mathbb{P}(A \cap C \mid B)$.
- Not a rule: $\mathbb{P}(A \mid B^c) \neq 1 - \mathbb{P}(A \mid B)$. Get it from $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$ divided by $1 - \mathbb{P}(B)$.
- Multiplication: $\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B) = \mathbb{P}(A)\mathbb{P}(B \mid A)$; chain: $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B \mid A)\mathbb{P}(C \mid A \cap B)$.
- Conditioning on an intersection: $\mathbb{P}(C \mid A \cap B) = \mathbb{P}(A \cap B \cap C)/\mathbb{P}(A \cap B)$.
- Reversing one condition: $\mathbb{P}(B \mid A) = \mathbb{P}(A \mid B)\mathbb{P}(B)/\mathbb{P}(A)$ when $\mathbb{P}(A)$ is given.
- Both given at least one: $\mathbb{P}(A \cap B)/\mathbb{P}(A \cup B)$ with $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- Independence: $\mathbb{P}(A \mid B) = \mathbb{P}(A) = \mathbb{P}(A \mid B^c)$.
- Subset: $A \subseteq B \Rightarrow \mathbb{P}(A \mid B) = \mathbb{P}(A)/\mathbb{P}(B)$; disjoint: $\mathbb{P}(A \mid B) = 0$.

7 Outcome (g): Law of total probability and Bayes' theorem

7.1 What the outcome asks

The exam partitions a population into classes (risk classes of drivers, smokers and non-smokers, suppliers of a part, diseased and healthy patients) and gives the proportion of the population in each class. These proportions are the *prior* probabilities. Each class has its own conditional rate of some outcome (an accident, a death, a defect, a positive test), and the question runs in one of two directions. The forward direction is the law of total probability: combine the class proportions and the class rates to find the overall rate of the outcome. The reverse direction is Bayes' theorem: given that the outcome occurred, find the probability that the individual came from a particular class. Almost every question on this outcome adds one twist: two years of experience that are independent only within a class, a class proportion left as an unknown to be solved from a given overall rate, or conditioning on the outcome *not* occurring.

7.2 Law of total probability

Events $B_1, B_2, \dots, B_n$ form a *partition* of the sample space if they are mutually exclusive ($B_i \cap B_j = \emptyset$ for $i \neq j$) and exhaustive ($B_1 \cup \dots \cup B_n = S$). Every outcome lies in exactly one B_i . The simplest partition is $\{B, B^c\}$.

Key fact: Law of total probability

If $B_1, \dots, B_n$ partition the sample space and $\mathbb{P}(B_i) > 0$ for every i , then for any event A ,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A | B_i) \mathbb{P}(B_i) = \mathbb{P}(A | B_1) \mathbb{P}(B_1) + \dots + \mathbb{P}(A | B_n) \mathbb{P}(B_n).$$

Two-event case:

$$\mathbb{P}(A) = \mathbb{P}(A | B) \mathbb{P}(B) + \mathbb{P}(A | B^c) \mathbb{P}(B^c).$$

In words: the overall rate of A is the weighted average of the class rates of A , the weights being the class proportions.

Proof. Because the B_i partition S , the event A decomposes into the disjoint pieces $A \cap B_1, \dots, A \cap B_n$, so $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i)$ by finite additivity. By the multiplication rule, $\mathbb{P}(A \cap B_i) = \mathbb{P}(A | B_i) \mathbb{P}(B_i)$. Substituting gives the formula. $\square$

Tree view. The formula is a two-level tree. The first level branches on the class, with branch probabilities $\mathbb{P}(B_i)$; the second level branches on whether A occurs, with branch probabilities $\mathbb{P}(A | B_i)$ and $\mathbb{P}(A^c | B_i) = 1 - \mathbb{P}(A | B_i)$. Multiply along each path to get a joint probability, then add the paths that end in A .

- B_1 with probability $\mathbb{P}(B_1)$
 - A : path probability $\mathbb{P}(B_1) \mathbb{P}(A | B_1)$
 - A^c : path probability $\mathbb{P}(B_1) (1 - \mathbb{P}(A | B_1))$
- B_2 with probability $\mathbb{P}(B_2)$
 - A : path probability $\mathbb{P}(B_2) \mathbb{P}(A | B_2)$
 - A^c : path probability $\mathbb{P}(B_2) (1 - \mathbb{P}(A | B_2))$
- $\vdots$
- B_n with probability $\mathbb{P}(B_n)$

- A : path probability $\mathbb{P}(B_n)\mathbb{P}(A | B_n)$
- A^c : path probability $\mathbb{P}(B_n)(1 - \mathbb{P}(A | B_n))$

The $2n$ path probabilities sum to 1. $\mathbb{P}(A)$ is the sum of the n paths ending in A ; $\mathbb{P}(A^c)$ is the sum of the n paths ending in A^c .

7.3 Bayes' theorem

Bayes' theorem reverses the direction of conditioning. The data give $\mathbb{P}(A | B_i)$ (the rate of the outcome within a class); the question asks for $\mathbb{P}(B_i | A)$ (the probability of the class given the outcome).

Key fact: Bayes' theorem

Two events, with $\mathbb{P}(A) > 0$ and $0 < \mathbb{P}(B) < 1$:

$$\mathbb{P}(B | A) = \frac{\mathbb{P}(A | B) \mathbb{P}(B)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A | B) \mathbb{P}(B)}{\mathbb{P}(A | B) \mathbb{P}(B) + \mathbb{P}(A | B^c) \mathbb{P}(B^c)}.$$

Partition $B_1, \dots, B_n$:

$$\mathbb{P}(B_k | A) = \frac{\mathbb{P}(A | B_k) \mathbb{P}(B_k)}{\sum_{i=1}^n \mathbb{P}(A | B_i) \mathbb{P}(B_i)}.$$

In words:

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{sum over all classes of (prior} \times \text{likelihood)}}.$$

Here the prior is $\mathbb{P}(B_k)$, the likelihood is $\mathbb{P}(A | B_k)$, and the denominator is $\mathbb{P}(A)$ from the law of total probability.

Derivation. By the definition of conditional probability, $\mathbb{P}(B_k | A) = \mathbb{P}(A \cap B_k) / \mathbb{P}(A)$. The numerator is $\mathbb{P}(A | B_k) \mathbb{P}(B_k)$ by the multiplication rule, and the denominator is $\sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i)$ by the law of total probability. $\square$

Two consequences are worth keeping in view. First, the posterior probabilities $\mathbb{P}(B_i | A)$ for $i = 1, \dots, n$ sum to 1, because the numerators sum to the denominator. Second, the ratio of two posteriors is

$$\frac{\mathbb{P}(B_j | A)}{\mathbb{P}(B_k | A)} = \frac{\mathbb{P}(B_j) \mathbb{P}(A | B_j)}{\mathbb{P}(B_k) \mathbb{P}(A | B_k)},$$

because the denominator cancels. The observation A multiplies each class's prior odds by that class's likelihood.

7.4 The table method

Every question on this outcome can be organised in one table with one row per class. The steps:

1. List the classes B_i (they must cover everyone and not overlap).
2. Write the prior $\mathbb{P}(B_i)$ in column 2. The column must sum to 1.
3. Write the likelihood $\mathbb{P}(A | B_i)$ in column 3, where A is the event actually observed. If the observed event is “no accident”, column 3 holds $1 - (\text{accident rate})$.
4. Multiply across: joint $\mathbb{P}(A \cap B_i) = \mathbb{P}(B_i) \mathbb{P}(A | B_i)$ in column 4.
5. Sum column 4. The sum is $\mathbb{P}(A)$ (law of total probability).

6. Divide each joint by the column sum: posterior $\mathbb{P}(B_i | A)$ in column 5 (Bayes' theorem). Column 5 must sum to 1.

Worked table. A health insurer classifies its policyholders as smokers (20%), former smokers (30%), and never-smokers (50%). The probability of a claim in a year is 0.30 for a smoker, 0.20 for a former smoker, and 0.10 for a never-smoker. A randomly chosen policyholder files a claim. Find the probability that the policyholder is a smoker.

Class B_i	Prior $\mathbb{P}(B_i)$	Likelihood $\mathbb{P}(A B_i)$	Joint $\mathbb{P}(A \cap B_i)$	Posterior $\mathbb{P}(B_i A)$
Smoker	0.20	0.30	$0.20 \times 0.30 = 0.06$	$0.06/0.17 = 0.3529$
Former smoker	0.30	0.20	$0.30 \times 0.20 = 0.06$	$0.06/0.17 = 0.3529$
Never-smoker	0.50	0.10	$0.50 \times 0.10 = 0.05$	$0.05/0.17 = 0.2941$
Total	1.00		$\mathbb{P}(A) = 0.17$	1.0000

The overall claim rate is $\mathbb{P}(A) = 0.17$, and $\mathbb{P}(\text{smoker} | \text{claim}) = 0.06/0.17 = 0.3529$. Note that the prior for smokers was 0.20; observing a claim raised it to 0.35.

Natural frequencies. When the priors are proportions of a population, the same computation can be done by counting in a hypothetical population. Take 1000 policyholders.

- Smokers: $0.20 \times 1000 = 200$ people, of whom $0.30 \times 200 = 60$ file a claim.
- Former smokers: $0.30 \times 1000 = 300$ people, of whom $0.20 \times 300 = 60$ file a claim.
- Never-smokers: $0.50 \times 1000 = 500$ people, of whom $0.10 \times 500 = 50$ file a claim.

Total claimants: $60 + 60 + 50 = 170$. Of the 170 claimants, 60 are smokers, so $\mathbb{P}(\text{smoker} | \text{claim}) = 60/170 = 0.3529$. The counts are the joint column multiplied by 1000; the arithmetic is the same, but the counts are harder to get wrong under time pressure. Use 10 000 or 100 000 when the rates are small (a prevalence of 0.001, say) so that every count is a whole number.

7.5 Worked examples

Example 7.1: Three risk classes of drivers

An auto insurer classifies its drivers as standard, preferred, or ultra-preferred. Of the insurer's drivers, 60% are standard, 30% are preferred, and 10% are ultra-preferred. In a given year, the probability of an accident is 0.10 for a standard driver, 0.05 for a preferred driver, and 0.01 for an ultra-preferred driver. A driver chosen at random has an accident during the year. Calculate the probability that the driver is ultra-preferred.

(A) 0.001 (B) 0.010 (C) 0.013 (D) 0.076 (E) 0.132

Solution

Let A be the event of an accident. Build the table.

Class	Prior	Likelihood	Joint
Standard	0.60	0.10	0.060
Preferred	0.30	0.05	0.015
Ultra-preferred	0.10	0.01	0.001
Total	1.00		$\mathbb{P}(A) = 0.076$

$$\mathbb{P}(\text{ultra} | A) = \frac{0.001}{0.076} = 0.0132.$$

Answer (C). Choice (A) is the joint probability, not the posterior; choice (D) is $\mathbb{P}(A)$. Note the base-rate effect: ultra-preferred drivers are 10% of the population but only 1.3% of the drivers who have accidents.

Example 7.2: Medical test with low prevalence

A blood test for a disease has sensitivity 0.95 (the probability of a positive result given that the disease is present) and specificity 0.90 (the probability of a negative result given that the disease is absent). The disease is present in 1% of the population. A person chosen at random tests positive. Calculate the probability that the person has the disease.

(A) 0.009 (B) 0.088 (C) 0.095 (D) 0.475 (E) 0.950

Solution

Let D be disease present and T a positive test. The likelihood of a positive test in the healthy class is the false-positive rate $1 - \text{specificity} = 0.10$.

Class	Prior	$\mathbb{P}(T \mid \cdot)$	Joint
Disease	0.01	0.95	0.0095
No disease	0.99	0.10	0.0990
Total	1.00		$\mathbb{P}(T) = 0.1085$

$$\mathbb{P}(D \mid T) = \frac{0.0095}{0.1085} = 0.0876.$$

Answer (B). In natural frequencies with 10 000 people: 100 have the disease and 95 of them test positive; 9900 do not and 990 of them test positive; $95/(95 + 990) = 0.0876$. A test that is 95% sensitive and 90% specific still leaves a positive result with under a 9% chance of disease, because the healthy class is 99 times larger than the diseased class and its false positives dominate. Choice (E) confuses $\mathbb{P}(T \mid D)$ with $\mathbb{P}(D \mid T)$.

Example 7.3: Claim in year two given no claim in year one

An insurer has two classes of policyholders. Low-risk policyholders make up 70% of the insured population and have probability 0.10 of a claim in any given year. High-risk policyholders make up the remaining 30% and have probability 0.40 of a claim in any given year. For each policyholder, claims in different years are independent given the policyholder's class. A policyholder chosen at random has no claim in the first year. Calculate the probability that the policyholder has a claim in the second year.

(A) 0.135 (B) 0.167 (C) 0.190 (D) 0.210 (E) 0.250

Solution

Let N_1 be no claim in year 1 and C_2 a claim in year 2. Then $\mathbb{P}(C_2 \mid N_1) = \mathbb{P}(N_1 \cap C_2) / \mathbb{P}(N_1)$, and both numerator and denominator are computed by total probability over the classes. Within a class the years are independent, so $\mathbb{P}(N_1 \cap C_2 \mid \text{class}) = \mathbb{P}(N_1 \mid \text{class}) \mathbb{P}(C_2 \mid \text{class})$.

Class	Prior	$\mathbb{P}(N_1 \mid \cdot)$	Joint $\mathbb{P}(N_1 \cap \cdot)$	$\mathbb{P}(N_1 \cap C_2 \cap \cdot)$
Low	0.70	0.90	0.630	$0.630 \times 0.10 = 0.063$
High	0.30	0.60	0.180	$0.180 \times 0.40 = 0.072$
Total	1.00		$\mathbb{P}(N_1) = 0.810$	$\mathbb{P}(N_1 \cap C_2) = 0.135$

$$\mathbb{P}(C_2 \mid N_1) = \frac{0.135}{0.810} = 0.1667.$$

Answer (B). An equivalent route is to update the class probabilities first: $\mathbb{P}(\text{low} \mid N_1) = 0.630/0.810 = 7/9$ and $\mathbb{P}(\text{high} \mid N_1) = 2/9$; then $\mathbb{P}(C_2 \mid N_1) = (7/9)(0.10) + (2/9)(0.40) = 0.1667$. Choice (C) is the unconditional one-year claim rate $0.7(0.10) + 0.3(0.40) = 0.19$; the clean first year shifts weight toward the low-risk class and lowers the second-year rate. The years are *not* unconditionally independent, so $\mathbb{P}(C_2 \mid N_1) \neq \mathbb{P}(C_2)$.

Example 7.4: Unknown class proportion

An insurer issues policies to two types of policyholders. Type A policyholders have probability 0.05 of a claim in a year; type B policyholders have probability 0.20. The insurer's overall claim rate is 0.11 per policyholder per year. A policyholder chosen at random files a claim. Calculate the probability that the policyholder is type B.

(A) 0.40 (B) 0.60 (C) 0.67 (D) 0.73 (E) 0.80

Solution

Let $x = \mathbb{P}(A)$, so $\mathbb{P}(B) = 1 - x$. The law of total probability gives

$$0.11 = 0.05x + 0.20(1 - x) = 0.20 - 0.15x \implies x = \frac{0.09}{0.15} = 0.60.$$

So 60% are type A and 40% are type B. Now apply Bayes' theorem; the denominator is the given overall rate 0.11, so there is nothing further to add up:

$$\mathbb{P}(B \mid \text{claim}) = \frac{0.40 \times 0.20}{0.11} = \frac{0.08}{0.11} = 0.7273.$$

Answer (D). Choice (A) is the prior for type B; choice (B) is the proportion of type A.

Example 7.5: Classes given as a ratio

A company has twice as many male employees as female employees. In a given year, 6% of male employees and 9% of female employees file a disability claim. An employee chosen at random files a disability claim. Calculate the probability that the employee is female.

(A) 0.333 (B) 0.400 (C) 0.429 (D) 0.500 (E) 0.600

Solution

“Twice as many male as female” means the ratio male : female is 2 : 1, so the priors are $\mathbb{P}(M) = 2/3$ and $\mathbb{P}(F) = 1/3$. (The same information stated as odds “2 to 1 in favour of male” converts the same way: odds $a : b$ means probabilities $a/(a + b)$ and $b/(a + b)$.)

Class	Prior	Likelihood	Joint
Male	2/3	0.06	0.04
Female	1/3	0.09	0.03
Total	1		0.07

$$\mathbb{P}(F \mid \text{claim}) = \frac{0.03}{0.07} = 0.4286.$$

Answer (C). Using natural frequencies with 300 employees: 200 males with 12 claims, 100 females with 9 claims; $9/21 = 0.4286$. Choice (A) is the prior. Choice (E) comes from using the wrong ratio (treating female as the larger class).

Example 7.6: Which supplier

A manufacturer buys a component from three suppliers. Supplier A provides 50% of the components, supplier B provides 30%, and supplier C provides 20%. The proportion of defective components is 2% for supplier A, 3% for supplier B, and 5% for supplier C. A component chosen at random from inventory is found to be defective. Calculate the probability that it came from supplier C.

(A) 0.200 (B) 0.310 (C) 0.345 (D) 0.400 (E) 0.500

Solution

Supplier	Prior	Likelihood	Joint	Posterior
A	0.50	0.02	0.010	$0.010/0.029 = 0.345$
B	0.30	0.03	0.009	$0.009/0.029 = 0.310$
C	0.20	0.05	0.010	$0.010/0.029 = 0.345$
Total	1.00		0.029	1.000

$\mathbb{P}(C \mid \text{defective}) = 0.010/0.029 = 0.3448$. Answer (C). Choice (B) is the posterior for supplier B, a distractor for a candidate who reads the wrong row. In natural frequencies with 1000 components: A supplies 500 with 10 defects, B supplies 300 with 9, C supplies 200 with 10; $10/29$.

Example 7.7: Conditioning on the complement

An insurer’s drivers fall into three age groups: 30% are young, 50% are middle-aged, and 20% are senior. The probability of an accident in a year is 0.20 for a young driver, 0.08 for a middle-aged driver, and 0.12 for a senior driver. A driver chosen at random has no accident during the year. Calculate the probability that the driver is young.

(A) 0.240 (B) 0.274 (C) 0.300 (D) 0.484 (E) 0.516

Solution

The observed event is A^c , no accident. The likelihood in each row is $\mathbb{P}(A^c \mid \text{class}) = 1 - \mathbb{P}(A \mid \text{class})$, in every row.

Class	Prior	$\mathbb{P}(A^c \mid \cdot)$	Joint
Young	0.30	0.80	0.240
Middle	0.50	0.92	0.460
Senior	0.20	0.88	0.176
Total	1.00		$\mathbb{P}(A^c) = 0.876$

$$\mathbb{P}(\text{young} \mid A^c) = \frac{0.240}{0.876} = 0.2740.$$

Answer (B). Choice (D), 0.484, is $\mathbb{P}(\text{young} \mid A) = 0.060/0.124$, the answer to the opposite question; choice (E) is $1 - 0.484$, which is $\mathbb{P}(\text{not young} \mid A)$ and not $\mathbb{P}(\text{young} \mid A^c)$. A clean year lowers the probability of the young class from 0.30 to 0.27, a smaller shift than an accident produces, because a clean year is common in every class.

Example 7.8: Observation spanning two years

An insurer classifies 80% of its policyholders as good risks and 20% as bad risks. In each year a good risk has probability 0.05 of at least one claim and a bad risk has probability 0.25. Given the class, claim activity in different years is independent. A policyholder chosen at random has at least one claim during a two-year period. Calculate the probability that the policyholder is a bad risk.

(A) 0.200 (B) 0.400 (C) 0.529 (D) 0.556 (E) 0.714

Solution

Let E be the event of at least one claim in two years. Within a class, $\mathbb{P}(E \mid \text{class}) = 1 - (1 - p)^2$ where p is the class's one-year rate, using conditional independence.

Class	Prior	$\mathbb{P}(E \mid \cdot)$	Joint
Good	0.80	$1 - 0.95^2 = 0.0975$	0.0780
Bad	0.20	$1 - 0.75^2 = 0.4375$	0.0875
Total	1.00		$\mathbb{P}(E) = 0.1655$

$$\mathbb{P}(\text{bad} \mid E) = \frac{0.0875}{0.1655} = 0.5287.$$

Answer (C). Choice (D), 0.556, is $\mathbb{P}(\text{bad} \mid \text{claim in a single year}) = 0.05/0.09$; the likelihood must match the event actually observed, which here covers two years.

7.6 Variants of the likelihood and the denominator

The eight examples above use a likelihood that is a single given rate and a denominator that sums over every class. The sample exam changes one or both. The likelihood may be given only as a ratio between classes, may be a product of two stages, may be a binomial count, may be a formula in the class index, or may be a power because the same experiment is repeated. The denominator may sum over a subset of the classes because the observation says the item came from one of those classes. The table method is unchanged; what changes is what goes into the likelihood column and which rows are in the table.

Key fact: Shapes the likelihood can take

In every case the likelihood column holds $\mathbb{P}(\text{observed event} \mid B_i)$, computed inside class B_i as if the class were known.

- **Ratios only.** If the data say “class 2 is twice as likely as class 1” and “class 3 is half as likely as class 2”, set the class-1 rate to an unknown r and write the others as $2r$ and r . Every joint probability is a multiple of r , so r cancels in the posterior. Setting $r = 1$ (or any convenient value) gives the same posterior.
- **Two stages.** If the observed event needs two things to happen inside the class (an accident, then a claim above the deductible), the likelihood is the product $\mathbb{P}(A \mid B_i) \mathbb{P}(E \mid A \cap B_i)$. With class counts, multiply the count by each stage in turn.
- **A count out of n .** If n items are drawn from a large lot with defect rate p_i and exactly k are defective, the likelihood is the binomial probability $\binom{n}{k} p_i^k (1 - p_i)^{n-k}$; for $k = 1$ it is $np_i(1 - p_i)^{n-1}$; for $k = 0$ it is $(1 - p_i)^n$. “At most one” adds the $k = 0$ and $k = 1$ terms. The binomial coefficient is the same in every row and cancels, but the $(1 - p_i)^{n-k}$ factor differs between rows and must be kept.
- **A formula in the class index.** If the class is a size m with prior $\mathbb{P}(M = m)$ given by a formula and the observation is uniform on $\{1, \dots, m\}$, the likelihood of observing k is $1/m$ for $m \geq k$ and 0 for $m < k$. The denominator is a finite sum over the classes consistent with the observation.
- **Repeated observations.** If the same experiment is repeated n times and the repetitions are independent given the class, the likelihood of a run of n identical outcomes is p_i^n . The forward probability of the run is $\sum_i \mathbb{P}(B_i) p_i^n$, the posterior is $\mathbb{P}(B_i) p_i^n$ divided by that sum, and the probability that the next repetition gives the same outcome is $\sum_i \mathbb{P}(B_i \mid \text{run}) p_i$, which equals $\sum_i \mathbb{P}(B_i) p_i^{n+1} / \sum_i \mathbb{P}(B_i) p_i^n$.

Key fact: Restricted denominator

If the observation states that the item came from one of the classes in a subset $J \subset \{1, \dots, n\}$ and that A occurred, the observed event is $A \cap (\bigcup_{j \in J} B_j)$, and for $k \in J$

$$\mathbb{P}(B_k \mid A, \bigcup_{j \in J} B_j) = \frac{\mathbb{P}(A \mid B_k) \mathbb{P}(B_k)}{\sum_{j \in J} \mathbb{P}(A \mid B_j) \mathbb{P}(B_j)}.$$

The rows for classes outside J are dropped from the table. Renormalising the priors to sum to 1 over J first gives the same answer, because the common factor $\mathbb{P}(\bigcup_{j \in J} B_j)$ cancels.

Example 7.9: Likelihoods given only as ratios

A life insurer classifies its policyholders as sedentary (35%), moderately active (45%), and highly active (20%). A moderately active policyholder is half as likely as a sedentary policyholder to be hospitalised during the year, and a highly active policyholder is one-third as likely as a moderately active policyholder. A policyholder chosen at random is hospitalised during the year. Calculate the probability that the policyholder is sedentary.

- (A) 0.055 (B) 0.225 (C) 0.350 (D) 0.370 (E) 0.575

Solution

No hospitalisation rate is given, only ratios. Let $r = \mathbb{P}(H \mid \text{sedentary})$. Then $\mathbb{P}(H \mid \text{moderate}) = r/2$ and $\mathbb{P}(H \mid \text{active}) = \frac{1}{3}(r/2) = r/6$. Note that the second ratio is stated relative to the moderate class, not the sedentary class.

Class	Prior	Likelihood	Joint
Sedentary	0.35	r	$0.3500r$
Moderate	0.45	$r/2$	$0.2250r$
Active	0.20	$r/6$	$0.0333r$
Total	1.00		$0.6083r$

$$\mathbb{P}(\text{sedentary} \mid H) = \frac{0.3500r}{0.6083r} = 0.5753.$$

Answer (E). The unknown r cancels; the posterior depends only on the ratios. The same arithmetic with $r = 6$ uses whole-number rates 6, 3, 1: joints 2.10, 1.35, 0.20 and $2.10/3.65 = 0.5753$. Choice (C) is the prior; (D) is the moderate-class posterior $0.225/0.6083$; (B) is the moderate joint with $r = 1$. The cancellation works only for the event the ratios describe. The probability that a policyholder who was *not* hospitalised is sedentary needs $1 - r$, $1 - r/2$, and $1 - r/6$, which do not share a common factor, so that question cannot be answered without r .

Example 7.10: Item known to come from a subset of the classes

A property insurer writes policies in four regions.

Region	Proportion of policies	Probability of a claim in a year
Coastal	0.25	0.12
Urban	0.30	0.06
Suburban	0.35	0.04
Rural	0.10	0.05

A claim is filed during the year on a policy that is known to be from the coastal, suburban, or rural region. Calculate the probability that the policy is from the coastal or rural region.

- (A) 0.350 (B) 0.500 (C) 0.522 (D) 0.612 (E) 0.714

Solution

The observed event is “claim, and region is coastal, suburban, or rural”. The urban row is not consistent with the observation and leaves the table.

Region	Prior	Likelihood	Joint
Coastal	0.25	0.12	0.030
Suburban	0.35	0.04	0.014
Rural	0.10	0.05	0.005
Total (three regions)	0.70		0.049

$$\mathbb{P}(\text{coastal or rural} \mid \text{claim, one of the three regions}) = \frac{0.030 + 0.005}{0.049} = 0.7143.$$

Answer (E). Renormalising the priors first (0.25/0.70, 0.35/0.70, 0.10/0.70) gives joints 0.030/0.70, 0.014/0.70, 0.005/0.70 and the same ratio. Choice (C) keeps the urban row in the denominator, 0.035/0.067; (B) is the restricted prior 0.35/0.70, which ignores the claim; (D) is the coastal region alone, 0.030/0.049; (A) is the unrestricted prior. The prior column of a restricted table does not sum to 1; that is expected, and the posteriors over the three remaining rows still sum to 1.

Example 7.11: Likelihood is a binomial count

A reinsurer audits batches of claim files prepared by third-party administrators. Administrator X prepares 30% of the batches and other administrators prepare the remaining 70%. Each batch contains a very large number of files. In batches from X, 20% of the files contain a coding error; in batches from other administrators, 5% do. An auditor examines 25 randomly selected files from a batch and finds that exactly two contain a coding error. Calculate the probability that the batch was prepared by X.

(A) 0.116 (B) 0.300 (C) 0.632 (D) 0.872 (E) 0.884

Solution

Because the batch is very large, the 25 files are independent with the batch’s error rate p , and the observed event is a binomial count: $\mathbb{P}(\text{exactly 2 of 25} \mid p) = \binom{25}{2} p^2 (1-p)^{23} = 300 p^2 (1-p)^{23}$.

$$\begin{aligned}\mathbb{P}(2 \text{ of } 25 \mid X) &= 300(0.20)^2(0.80)^{23} = 12 \times 0.005903 = 0.07084, \\ \mathbb{P}(2 \text{ of } 25 \mid \text{other}) &= 300(0.05)^2(0.95)^{23} = 0.75 \times 0.30736 = 0.23052.\end{aligned}$$

Class	Prior	Likelihood	Joint
X	0.30	0.07084	0.02125
Other	0.70	0.23052	0.16136
Total	1.00		0.18261

$$\mathbb{P}(X \mid 2 \text{ of } 25) = \frac{0.02125}{0.18261} = 0.1164.$$

Answer (A). Two errors in 25 files is an 8% sample error rate, closer to 5% than to 20%, so the observation lowers X's probability from 0.30 to 0.12. Choice (C) uses p alone as the likelihood, $0.06/0.095$; (D) uses p^2 alone and drops the $(1 - p)^{23}$ factor, which is where the evidence against X lives; (E) is $1 - 0.116$. The coefficient 300 is common to both rows and could have been omitted, but the powers of $1 - p$ cannot.

Example 7.12: Two-stage likelihood with class counts

A commercial property insurer covers 250 restaurants, 500 retail shops, and 250 offices. In a year the probability of a fire loss is 0.08 for a restaurant, 0.04 for a shop, and 0.02 for an office. Given a fire loss, the probability that the resulting claim exceeds the deductible is 0.75 for a restaurant, 0.90 for a shop, and 0.60 for an office. The insurer picks a claim at random from all claims that exceed the deductible. Calculate the probability that the claim is on a restaurant.

- (A) 0.015 (B) 0.238 (C) 0.250 (D) 0.417 (E) 0.444

Solution

The observed event E (claim exceeds the deductible) requires a loss and then an excess claim, so within each class $\mathbb{P}(E \mid \text{class}) = \mathbb{P}(\text{loss} \mid \text{class}) \mathbb{P}(E \mid \text{loss, class})$. With counts, multiply through the stages.

Class	Policies	Losses	Claims above deductible	Posterior
Restaurants	250	$250 \times 0.08 = 20$	$20 \times 0.75 = 15$	$15/36 = 0.417$
Shops	500	$500 \times 0.04 = 20$	$20 \times 0.90 = 18$	$18/36 = 0.500$
Offices	250	$250 \times 0.02 = 5$	$5 \times 0.60 = 3$	$3/36 = 0.083$
Total	1000	45	36	1.000

$\mathbb{P}(\text{restaurant} \mid E) = 15/36 = 0.4167$. Answer (D). In probabilities, the likelihoods are $0.08 \times 0.75 = 0.060$, $0.04 \times 0.90 = 0.036$, and $0.02 \times 0.60 = 0.012$, and the posterior is $0.25(0.060)/[0.25(0.060) + 0.50(0.036) + 0.25(0.012)] = 0.015/0.036$. Choice (E) stops after the first stage, $20/45$; (B) uses only the second stage, $250(0.75)/[250(0.75) + 500(0.90) + 250(0.60)]$; (C) is the prior; (A) is the joint probability $15/1000$.

Example 7.13: Prior given by a formula, likelihood $1/(n + 1)$

A dental plan covers households of size $n = 1, 2, \dots, 6$, and the fraction of covered households of size n is $n/21$. For a household of size n , the number of members who file a claim during the year is uniformly distributed on $\{0, 1, \dots, n\}$. A household chosen at random has exactly three members who file a claim. Calculate the probability that the household has four members.

- (A) 0.182 (B) 0.190 (C) 0.200 (D) 0.247 (E) 0.264

Solution

The classes are the sizes n , with prior $\mathbb{P}(N = n) = n/21$; the priors sum to $(1+2+\cdots+6)/21 = 1$. Let K be the number of claimants. Given $N = n$, K takes each of the $n + 1$ values $0, 1, \dots, n$ with probability $1/(n + 1)$, so $\mathbb{P}(K = 3 \mid N = n) = 1/(n + 1)$ for $n \geq 3$ and 0 for $n = 1, 2$. Only $n = 3, 4, 5, 6$ remain in the table.

n	Prior $n/21$	Likelihood $1/(n + 1)$	Joint $\times 21$
3	3/21	1/4	3/4 = 0.7500
4	4/21	1/5	4/5 = 0.8000
5	5/21	1/6	5/6 = 0.8333
6	6/21	1/7	6/7 = 0.8571
Total			3.2405

$$\mathbb{P}(N = 4 \mid K = 3) = \frac{0.8000}{3.2405} = 0.2469.$$

Answer (D). The common factor $1/21$ cancels, so the joint column can be carried as $n/(n+1)$. Choice (A) keeps all six sizes in the denominator with likelihood $1/(n + 1)$ in every row, $0.8/4.4071$, which ignores that a household of size 1 or 2 cannot have three claimants; (B) is the prior $4/21$; (C) is the likelihood $1/5$; (E) is the posterior for $n = 6$.

Example 7.14: Repeated tosses of one coin drawn from a box

A box contains six coins. Three are fair, two land heads with probability 0.25, and one has heads on both faces. A coin is drawn at random from the box and tossed three times, and all three tosses are heads. Calculate the probability that the coin is one of the two that land heads with probability 0.25.

(A) 0.022 (B) 0.234 (C) 0.267 (D) 0.333 (E) 0.711

Solution

Given the coin, the tosses are independent, so the likelihood of three heads is p^3 where p is that coin's heads probability. The priors are the coin counts over 6.

Coin	Prior	p^3	Joint	Posterior
Fair	3/6	1/8	1/16 = 12/192	12/45 = 0.2667
$p = 0.25$	2/6	1/64	1/192	1/45 = 0.0222
Two-headed	1/6	1	1/6 = 32/192	32/45 = 0.7111
Total	1		45/192 = 0.2344	1

$\mathbb{P}(p = 0.25 \mid HHH) = 1/45 = 0.0222$. Answer (A). The column total $45/192 = 0.2344$ is the forward answer, the unconditional probability of three heads, which is choice (B). Choice (C) is the fair-coin posterior, (E) the two-headed posterior, and (D) the prior. The posteriors also give the probability that a fourth toss is heads: $(12/45)(1/2) + (1/45)(1/4) + (32/45)(1) = 38.25/45 = 0.85$, against an unconditional single-toss heads probability of $(1/2)(1/2) + (1/3)(1/4) + (1/6)(1) = 0.50$.

7.7 Traps

Trap: Answering $\mathbb{P}(A | B)$ when $\mathbb{P}(B | A)$ is asked

The data always give the rate of the outcome within each class, $\mathbb{P}(A | B_i)$. The question almost always asks for the class given the outcome, $\mathbb{P}(B_i | A)$. These are different numbers: in the medical-test example, $\mathbb{P}(T | D) = 0.95$ while $\mathbb{P}(D | T) = 0.088$. Before computing, write down in symbols which event is observed (goes to the right of the bar) and which is asked for (goes to the left). The answer choices routinely include the likelihood as a distractor.

Trap: Forgetting a class in the denominator

The denominator of Bayes' theorem is $\mathbb{P}(A)$, the sum of prior times likelihood over *every* class. Omitting a class (typically the third class in a three-class problem, or the “no disease” class in a test problem) makes the denominator too small and the posterior too large. Check that the prior column sums to 1 before summing the joint column. If the priors given in the problem do not sum to 1, the remaining mass belongs to a class the problem describes in words (“the remaining policyholders”) and that class must be in the table.

Trap: Using the overall rate as a likelihood

The likelihood column holds class-specific rates $\mathbb{P}(A | B_i)$. The overall rate $\mathbb{P}(A)$ belongs in the total row, not in any class row. When a problem gives the overall rate together with all but one of the class rates or priors, the overall rate is there so that the missing quantity can be solved from the law of total probability, as in the unknown-proportion example. Once solved, the given overall rate is the Bayes denominator; do not recompute it from the table and get a different number through rounding.

Trap: Conditional independence is given the class, not unconditional

In multi-year problems, “claims in different years are independent” means independent *within each class*: $\mathbb{P}(C_1 \cap C_2 | B_i) = \mathbb{P}(C_1 | B_i)\mathbb{P}(C_2 | B_i)$. It does not mean $\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2)$. Because the population mixes classes with different rates, a claim in year one raises the probability of the high-risk class and therefore raises the probability of a claim in year two: $\mathbb{P}(C_2 | C_1) > \mathbb{P}(C_2)$. A candidate who treats the years as unconditionally independent answers with the marginal one-year rate, which is always among the choices. Multiply the two years' probabilities inside each row of the table, never across the total row.

Trap: Conditioning on A^c requires $1 - \text{likelihood}$ in every row

When the observed event is “no accident”, “tested negative”, or “survived”, the likelihood in every row is $1 - \mathbb{P}(A | B_i)$, and the denominator is $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. Two wrong shortcuts appear in the choices: using the accident rates unchanged (which gives $\mathbb{P}(B_i | A)$), and computing $1 - \mathbb{P}(B_i | A)$ (which is $\mathbb{P}(B_i^c | A)$, the complement on the wrong side of the bar). Neither equals $\mathbb{P}(B_i | A^c)$. For a negative medical test the likelihoods are $1 - \text{sensitivity}$ in the disease row and the specificity in the healthy row.

Trap: Normalising by the wrong total

The posterior is the joint probability divided by the sum of the joint column, not by the prior, not by the likelihood, and not by 1. Dividing the joint by the class prior returns the likelihood; dividing by the likelihood returns the prior; leaving it undivided gives $\mathbb{P}(A \cap B_i)$, which the choices usually include (0.001 in the driver example, 0.018 in a test example). The posterior column must sum to 1; if it does not, the divisor is wrong.

Trap: Ratios cancel only for the event they describe

When the class rates are given only as ratios (“twice as likely to die”), the unknown base rate r cancels in $\mathbb{P}(B_i \mid \text{die})$ because every joint probability is a multiple of r . It does not cancel in $\mathbb{P}(B_i \mid \text{survive})$, because $1 - r$, $1 - 2r$, and $1 - r/2$ have no common factor. If a question with ratio-only rates asks about the complement, a base rate must be given somewhere in the stem; look for it. Also read which class each ratio is relative to: “light smokers are twice as likely as nonsmokers, but half as likely as heavy smokers” makes the heavy rate four times the nonsmoker rate, not twice.

Trap: Restricting the denominator, or failing to

If the stem says the item “is from one of” a named subset of classes, the classes outside the subset are not in the denominator; including them makes the posterior too small. If the stem says nothing of the kind, every class is in the denominator, including a class described only as “other” or “the rest”; excluding it makes the posterior too large. The rule is that the table holds exactly the classes consistent with everything the observation says. The same rule handles a formula prior over sizes m with a $1/m$ likelihood: sizes smaller than the observed count have likelihood 0 and drop out.

7.8 Practice problems

Practice 7.1

An insurer classifies its policyholders as low risk (45%), medium risk (35%), and high risk (20%). The probability of a claim in a year is 0.02 for low risk, 0.05 for medium risk, and 0.10 for high risk. A policyholder chosen at random files a claim. Calculate the probability that the policyholder is high risk.

- (A) 0.200 (B) 0.387 (C) 0.430 (D) 0.465 (E) 0.500

Practice 7.2

A screening test for a condition has sensitivity 0.90 and specificity 0.95. The condition is present in 2% of the population screened. A person screened at random tests positive. Calculate the probability that the person has the condition.

- (A) 0.018 (B) 0.067 (C) 0.269 (D) 0.474 (E) 0.900

Practice 7.3

An insurer's policyholders are 60% class A, with annual claim probability 0.10, and 40% class B, with annual claim probability 0.30. Given the class, claims in different years are independent. A policyholder chosen at random has a claim in year one. Calculate the probability that the policyholder has a claim in year two.

- (A) 0.180 (B) 0.200 (C) 0.233 (D) 0.250 (E) 0.300

Practice 7.4

A factory produces a part on two machines. Parts from machine 1 are defective with probability 0.02 and parts from machine 2 are defective with probability 0.05. The proportion of all parts that are defective is 0.032. A part chosen at random is defective. Calculate the probability that it was produced on machine 2.

- (A) 0.400 (B) 0.500 (C) 0.600 (D) 0.625 (E) 0.750

Practice 7.5

An auto insurer has three times as many preferred drivers as standard drivers. In a year, a standard driver has an accident with probability 0.12 and a preferred driver with probability 0.04. A driver chosen at random has an accident. Calculate the probability that the driver is standard.

- (A) 0.250 (B) 0.333 (C) 0.500 (D) 0.667 (E) 0.750

Practice 7.6

Of a life insurer's policyholders, 25% are smokers and 75% are non-smokers. The probability of dying within the year is 0.05 for a smoker and 0.01 for a non-smoker. A policyholder chosen at random survives the year. Calculate the probability that the policyholder is a smoker.

- (A) 0.238 (B) 0.242 (C) 0.250 (D) 0.375 (E) 0.625

Practice 7.7

A distributor stocks a component from four suppliers. Supplier A provides 40% of the stock with a 1% defect rate, supplier B provides 30% with a 2% defect rate, supplier C provides 20% with a 3% defect rate, and supplier D provides 10% with a 5% defect rate. A component chosen at random is defective. Calculate the probability that it came from supplier A.

- (A) 0.100 (B) 0.190 (C) 0.286 (D) 0.400 (E) 0.476

Practice 7.8

An insurer's policyholders are 30% young and 70% not young. The probability that a young policyholder files a claim in a year is 0.15. The overall probability that a policyholder files a claim in a year is 0.09. A policyholder chosen at random files no claim during the year. Calculate the probability that the policyholder is young.

- (A) 0.064 (B) 0.255 (C) 0.280 (D) 0.300 (E) 0.500

Practice 7.9

A health plan classifies its members as non-smokers, former smokers, light smokers, and heavy smokers. Non-smokers are 55% of the members. There are twice as many former smokers as light smokers and three times as many light smokers as heavy smokers. A former smoker is 1.5 times as likely as a non-smoker to be hospitalised during the year, a light smoker is twice as likely as a former smoker, and a heavy smoker is twice as likely as a light smoker. A member chosen at random is hospitalised during the year. Calculate the probability that the member is a heavy smoker.

- (A) 0.045 (B) 0.166 (C) 0.248 (D) 0.270 (E) 0.337

Practice 7.10

An auto insurer's drivers fall into four age groups.

Age group	Proportion of drivers	Probability of an accident in a year
Under 25	0.15	0.14
25–44	0.35	0.06
45–64	0.30	0.04
65 and over	0.20	0.08

A driver who is known not to be in the 25–44 group has no accident during the year. Calculate the probability that the driver is in the 45–64 group.

- (A) 0.245 (B) 0.300 (C) 0.310 (D) 0.462 (E) 0.479

Practice 7.11

A clinic receives 50% of its vaccine shipments from supplier A, 30% from supplier B, and 20% from supplier C. Each shipment contains a very large number of vials. The proportion of ineffective vials is 1% for supplier A, 4% for supplier B, and 10% for supplier C. The clinic tests 20 randomly selected vials from a shipment and finds at most one ineffective vial. Calculate the probability that the shipment came from supplier C.

- (A) 0.096 (B) 0.200 (C) 0.219 (D) 0.299 (E) 0.392

Practice 7.12

An insurer covers 200 sports cars, 500 sedans, and 300 trucks. The probability of an accident in a year is 0.15 for a sports car, 0.06 for a sedan, and 0.08 for a truck. Given an accident, the probability that the claim exceeds the deductible is 0.80 for a sports car, 0.70 for a sedan, and 0.50 for a truck. The insurer picks a claim at random from all claims that do not exceed the deductible. Calculate the probability that the claim is on a truck.

- (A) 0.211 (B) 0.286 (C) 0.333 (D) 0.444 (E) 0.500

Practice 7.13

A tour operator studies the groups that book a tour. The fraction of groups of size m is $m/36$ for $m = 1, 2, \dots, 8$. For a group of size m , the number of members who buy travel

insurance is uniformly distributed on $\{1, \dots, m\}$. A group is chosen at random, and at least five of its members buy travel insurance. Calculate the probability that the group has seven members.

- (A) 0.194 (B) 0.250 (C) 0.269 (D) 0.300 (E) 0.429

Practice 7.14

A box contains five spinners. Two land on red with probability 0.5, two land on red with probability 0.2, and one lands on red with probability 1. A spinner is drawn at random and spun twice, and it lands on red both times. Calculate the probability that a third spin of the same spinner lands on red.

- (A) 0.316 (B) 0.480 (C) 0.500 (D) 0.633 (E) 0.801

Practice 7.15

A drawer contains three coins that land heads with probability 0.5, 0.6, and 0.9 respectively. A coin is chosen at random, each coin being equally likely, and tossed four times. Calculate the probability of exactly three heads.

- (A) 0.296 (B) 0.346 (C) 0.395 (D) 0.729 (E) 0.887

7.9 Answers to practice problems

- (C). Joint probabilities $0.45(0.02) = 0.0090$, $0.35(0.05) = 0.0175$, $0.20(0.10) = 0.0200$; total 0.0465. $\mathbb{P}(\text{high} \mid \text{claim}) = 0.0200/0.0465 = 0.430$.
- (C). Joint: $0.02(0.90) = 0.018$ and $0.98(0.05) = 0.049$; total 0.067. $\mathbb{P}(D \mid T) = 0.018/0.067 = 0.269$. Choice (A) is the joint, (B) is $\mathbb{P}(T)$.
- (C). $\mathbb{P}(C_1) = 0.6(0.10) + 0.4(0.30) = 0.18$; $\mathbb{P}(C_1 \cap C_2) = 0.6(0.10)^2 + 0.4(0.30)^2 = 0.006 + 0.036 = 0.042$; $\mathbb{P}(C_2 \mid C_1) = 0.042/0.18 = 0.233$. Choice (A) is the unconditional rate.
- (D). With $x = \mathbb{P}(\text{machine 1})$: $0.02x + 0.05(1-x) = 0.032$ gives $x = 0.60$. $\mathbb{P}(\text{machine 2} \mid \text{def}) = 0.40(0.05)/0.032 = 0.020/0.032 = 0.625$.
- (C). Priors $\mathbb{P}(\text{std}) = 1/4$, $\mathbb{P}(\text{pref}) = 3/4$. Joint: $0.25(0.12) = 0.03$ and $0.75(0.04) = 0.03$; total 0.06. $\mathbb{P}(\text{std} \mid A) = 0.03/0.06 = 0.500$.
- (B). Survival likelihoods 0.95 and 0.99. Joint: $0.25(0.95) = 0.2375$ and $0.75(0.99) = 0.7425$; total 0.98. $\mathbb{P}(\text{smoker} \mid \text{survive}) = 0.2375/0.98 = 0.242$. Choice (E) is $\mathbb{P}(\text{smoker} \mid \text{die}) = 0.0125/0.02$; choice (D) is its complement.
- (B). Joint: 0.004, 0.006, 0.006, 0.005; total 0.021. $\mathbb{P}(A \mid \text{def}) = 0.004/0.021 = 0.190$. Choice (C) is the posterior for B or C.
- (C). First solve for the not-young rate y : $0.3(0.15) + 0.7y = 0.09$ gives $y = 0.045/0.7 = 0.0643$. Then $\mathbb{P}(\text{no claim}) = 1 - 0.09 = 0.91$ and $\mathbb{P}(\text{young} \mid \text{no claim}) = 0.3(0.85)/0.91 = 0.255/0.91 = 0.280$. Choice (A) is y ; choice (B) is the joint probability.
- (B). Let L be the light-smoker proportion; former $= 2L$, heavy $= L/3$, and $2L + L + L/3 = 0.45$ gives $L = 0.135$, so the priors are 0.55, 0.27, 0.135, 0.045. With non-smoker rate r the rates are $r, 1.5r, 3r, 6r$. Joints in units of r : 0.55, 0.405, 0.405, 0.27; total 1.63. $\mathbb{P}(\text{heavy} \mid H) = 0.27/1.63 = 0.166$. Choice (A) is the prior, (C) the former or light posterior, (D) the unnormalised joint, (E) the non-smoker posterior.
- (E). Drop the 25–44 row; the likelihood is 1– accident rate. The joints $0.15(0.86) = 0.129$,

- $0.30(0.96) = 0.288$, and $0.20(0.92) = 0.184$ sum to 0.601. $\mathbb{P}(45-64 \mid \text{no accident, not } 25-44) = 0.288/0.601 = 0.479$. Choice (D) is the restricted prior $0.30/0.65$; (C) keeps all four rows, $0.288/0.930$; (A) conditions on an accident instead, $0.012/0.049$.
11. (A). $\mathbb{P}(\text{at most 1 of 20} \mid p) = (1-p)^{20} + 20p(1-p)^{19}$: A: $0.8179 + 0.1652 = 0.9831$; B: $0.4420 + 0.3683 = 0.8103$; C: $0.1216 + 0.2702 = 0.3917$. Joints 0.4916, 0.2431, 0.0783; total 0.8130. $\mathbb{P}(C \mid \cdot) = 0.0783/0.8130 = 0.096$. Choice (C) uses the exactly-one term only; (D) is the posterior for B; (E) is C's likelihood.
 12. (D). Accidents: 30, 30, 24. Claims not exceeding the deductible: $30(0.20) = 6$, $30(0.30) = 9$, $24(0.50) = 12$; total 27. $\mathbb{P}(\text{truck} \mid \text{not exceed}) = 12/27 = 0.444$. Choice (B) stops after the accident stage, $24/84$; (A) conditions on exceeding instead, $12/57$; (C) is the sedan posterior.
 13. (D). $\mathbb{P}(\text{at least 5} \mid m) = (m-4)/m$ for $m \geq 5$ and 0 otherwise. Joint = $(m/36)(m-4)/m = (m-4)/36$ for $m = 5, 6, 7, 8$: $1/36, 2/36, 3/36, 4/36$; total $10/36$. $\mathbb{P}(M = 7 \mid \cdot) = 3/10 = 0.300$. Choice (A) is the prior $7/36$; (E) the likelihood $3/7$; (C) the restricted prior $7/26$.
 14. (E). Joint = prior $\times p^2$: $0.4(0.25) = 0.100$, $0.4(0.04) = 0.016$, $0.2(1) = 0.200$; total 0.316. Posteriors 0.316, 0.051, 0.633. $\mathbb{P}(\text{third red} \mid RR) = 0.316(0.5) + 0.051(0.2) + 0.633(1) = 0.801$; equivalently $\mathbb{P}(RRR)/\mathbb{P}(RR) = 0.2532/0.316$. Choice (A) is $\mathbb{P}(RR)$; (B) is the unconditional $\mathbb{P}(\text{red}) = 0.48$; (D) is the always-red posterior.
 15. (A). $\mathbb{P}(3 \text{ of } 4 \mid p) = 4p^3(1-p)$: 0.2500, 0.3456, 0.2916. Average with weights $1/3$: $0.8872/3 = 0.296$. Choice (C) averages p first, $4(2/3)^3(1/3) = 0.395$; (E) omits the $1/3$ weights; (B) is the second coin alone.

7.10 Sample exam cross-reference

- Bank 1, Q121: total probability with a likelihood p_i^2 for two rolls that are independent given the die; forward probability of the run.
- Bank 2, Q18: four-class table, posterior of one class given an accident.
- Bank 2, Q19: three-class table with small death rates; posterior of the smallest class.
- Bank 2, Q20: conditioning on the complement (survived); likelihood $1 - \text{rate in every row}$.
- Bank 2, Q21: three classes with death rates given only as chained ratios; base rate cancels.
- Bank 2, Q22: four-class table, posterior given at least one collision.
- Bank 2, Q24: medical test with sensitivity and false-positive rate; posterior given positive.
- Bank 2, Q25: two classes with the smoking rate given as a ratio; unknown rate cancels.
- Bank 2, Q26: denominator restricted to the three named model years; “other” row dropped.
- Bank 2, Q27: likelihood is a count, $30p(1-p)^{29}$ for exactly one of 30 vials.
- Bank 2, Q133: two-stage likelihood (accident, then claim above deductible) with class counts.
- Bank 2, Q150: prior given by a formula over sizes m , likelihood $1/m$, denominator summed over $m \geq 5$.

7.11 One-page summary

- A partition $B_1, \dots, B_n$: mutually exclusive, exhaustive, priors sum to 1. The two-class case is $\{B, B^c\}$.
- Law of total probability: $\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)$. Two classes: $\mathbb{P}(A) = \mathbb{P}(A \mid B) \mathbb{P}(B) + \mathbb{P}(A \mid B^c) \mathbb{P}(B^c)$. The overall rate is the prior-weighted average of the class rates.
- Bayes' theorem: $\mathbb{P}(B_k \mid A) = \frac{\mathbb{P}(A \mid B_k) \mathbb{P}(B_k)}{\sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)}$. Posterior = prior $\times$ likelihood, divided by the sum of prior $\times$ likelihood over all classes. Posteriors sum to 1.
- Table method: (1) one row per class; (2) prior column, sums to 1; (3) likelihood column

- $\mathbb{P}(\text{observed event} \mid \text{class})$; (4) joint = prior $\times$ likelihood; (5) sum the joint column to get $\mathbb{P}(\text{observed event})$; (6) posterior = joint / column sum.
- Natural frequencies: take 1000 or 10 000 people, count each class, count the outcome within each class, and divide the count in the target class by the total count with the outcome.
 - Observed event on the right of the bar, asked-for event on the left. The data give $\mathbb{P}(A \mid B_i)$; the question asks $\mathbb{P}(B_i \mid A)$.
 - Conditioning on A^c (no claim, negative test, survived): likelihood $1 - \mathbb{P}(A \mid B_i)$ in every row; denominator $1 - \mathbb{P}(A)$. Negative test: disease row $1 - \text{sensitivity}$, healthy row = specificity.
 - Sensitivity = $\mathbb{P}(T \mid D)$; specificity = $\mathbb{P}(T^c \mid D^c)$; false-positive rate = $1 - \text{specificity}$. Low prevalence makes $\mathbb{P}(D \mid T)$ small even for an accurate test.
 - Two years: independence holds within a class only. Multiply year probabilities inside each row: $\mathbb{P}(C_1 \cap C_2 \mid B_i) = p_i^2$, $\mathbb{P}(N_1 \cap C_2 \mid B_i) = (1 - p_i)p_i$, $\mathbb{P}(\text{at least one claim} \mid B_i) = 1 - (1 - p_i)^2$. Then $\mathbb{P}(C_2 \mid C_1) = \mathbb{P}(C_1 \cap C_2) / \mathbb{P}(C_1)$, each side by total probability. Equivalent: update the class posteriors after year one, then average the year-two rates.
 - Unknown proportion: set $\mathbb{P}(B_1) = x$, write the given overall rate as $\sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)$, solve for x , then use the given overall rate as the Bayes denominator.
 - Ratios and odds: “ k times as many in class 1 as class 2” or odds $k : 1$ means priors $k/(k + 1)$ and $1/(k + 1)$.
 - Rates given only as ratios: set one rate to r , write the others as multiples of r , and r cancels in the posterior for the event the ratios describe. It does not cancel for the complement. Read which class each ratio is relative to.
 - Restricted denominator: if the item is known to be from a subset of classes, the table holds only those rows; the prior column need not sum to 1. Without such a statement, every class stays in, including “other”.
 - Count likelihood: exactly k of n from a large lot with rate p_i is $\binom{n}{k} p_i^k (1 - p_i)^{n-k}$; exactly one is $n p_i (1 - p_i)^{n-1}$; at most one is $(1 - p_i)^n + n p_i (1 - p_i)^{n-1}$. The coefficient cancels, the $(1 - p_i)$ powers do not.
 - Two-stage likelihood: $\mathbb{P}(E \mid B_i) = \mathbb{P}(A \mid B_i) \mathbb{P}(E \mid A \cap B_i)$; with counts, multiply the class count through each stage. Conditioning on the second stage failing uses $1 - \mathbb{P}(E \mid A \cap B_i)$ in the second factor only.
 - Formula prior over sizes m with uniform likelihood: $\mathbb{P}(K = k \mid m) = 1/m$ on $\{1, \dots, m\}$ (or $1/(m + 1)$ on $\{0, \dots, m\}$), zero for $m < k$; the denominator is a finite sum over $m \geq k$. A tail observation “at least k ” has likelihood $(m - k + 1)/m$.
 - Repeated observations: likelihood p_i^n for a run of n ; forward $\mathbb{P}(\text{run}) = \sum_i \mathbb{P}(B_i) p_i^n$; posterior $\mathbb{P}(B_i) p_i^n / \mathbb{P}(\text{run})$; next repetition $\sum_i \mathbb{P}(B_i \mid \text{run}) p_i = \sum_i \mathbb{P}(B_i) p_i^{n+1} / \sum_i \mathbb{P}(B_i) p_i^n$.
 - Distractors to recognise: the likelihood, the joint probability, the overall rate, the prior, the posterior of a neighbouring class, and the complement on the wrong side of the bar.